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1.0 Theoretical Aspects

1 Version spaces and the consistency problem

Author(s): H. Hirsh (**Affiliation:** Rutgers Univ., USA), N. Mishra, L. Pitt

Journal: *Artif. Intell. (Netherlands)*, vol.156, no.2, p.115-38 (July 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0004-3702, [Full text](#)

Document type: Journal article

Abstract: A version space is a collection of concepts consistent with a given set of positive and negative examples. Mitchell [Artificial Intelligence 18 (1982) 203-226] proposed representing a version space by its boundary sets: the maximally general (C) and maximally specific consistent concepts (S). For many simple concept classes, the size of G and S is known to grow exponentially in the number of positive and negative examples. This paper argues that previous work on alternative representations of version spaces has disguised the real question underlying version space reasoning. We instead show that tractable reasoning with version spaces turns out to depend on the consistency problem, i.e., determining if there is any concept consistent with a set of positive and negative examples. Indeed, we show that tractable version space reasoning is possible if and only if there is an efficient algorithm for the consistency problem. Our observations give rise to new concept classes for which tractable version space reasoning is now possible, e.g., 1-decision lists, monotone depth two formulas, and halfspaces (30 refs.)

Inspec No.: 8200624

2 Theory revision with queries: Horn, read-once, and parity formulas

Author(s): J. Goldsmith (**Affiliation:** Dept. of Comput. Sci., Kentucky Univ., USA), R.H. Sloan, B. Szorenyi, G. Turan

Journal: *Artif. Intell. (Netherlands)*, vol.156, no.2, p.139-76 (July 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0004-3702, [Full text](#)

Document type: Journal article

Abstract: A theory, in this context, is a Boolean formula; it is used to classify instances, or truth assignments. Theories can model real-world phenomena, and can do so more or less correctly. The theory revision, or concept revision, problem is to correct a given, roughly correct concept. This problem is considered here in the model of learning with equivalence and membership queries. A revision algorithm is considered efficient if the number of queries it makes is polynomial in the revision distance between the initial theory and the target theory, and polylogarithmic in the number of variables and the size of the initial theory. The revision distance is the minimal number of syntactic revision operations, such as the deletion or addition of literals, needed to obtain the target theory from the initial theory. Efficient revision algorithms are given for Horn formulas and read-once formulas, where revision operators are restricted to deletions of variables or clauses, and for parity formulas, where revision operators include both deletions and additions of variables. We also show that the query complexity of the read-once revision algorithm is near-optimal (59 refs.)

Inspec No.: 8200625

3 A differential semantics for jointree algorithms

Author(s): J.D. Park, A. Darwiche (**Affiliation:** Dept. of Comput. Sci., California Univ., USA)

Journal: *Artif. Intell. (Netherlands)*, vol.156, no.2, p.197-216 (July 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0004-3702, [Full text](#)

Document type: Journal article

Abstract: A new approach to inference in belief networks has been recently proposed, which is based on an algebraic representation of belief networks using multi-linear functions. According to this approach, belief network inference reduces to a simple process of evaluating and differentiating multi-linear functions. We show here that mainstream inference algorithms based on jointrees are a special case of the approach based on multi-linear functions, in a very precise sense. We use this result to prove new properties of jointree algorithms. We also discuss some practical and theoretical implications of this new finding (15 refs.)

Inspec No.: 8200627

4 Modelling for fault detection of electric motors

Author(s): A. Rakar, D. Juricic (**Affiliation:** Dept. of Syst. & Control, Jozef Stefan Inst., Ljubljana, Slovenia)

Journal: *Stroj. Vestn. (Slovenia)*, vol.50, no.5, p.267-76 (2004) **Publisher:** Strojniški Vestnik, Slovenia **Language:** Slovenian **ISSN:** 0039-2480

Document type: Journal article

Abstract: A semi-physical model aimed at detection of incipient faults in electric motors is presented. In order to gain high sensitivity to faults a physical model is combined with a black-box model based on an adaptive-network-based fuzzy inference system (ANFIS) as a corrective term. The method is applied to vacuum-cleaner motors. The architecture and hybrid learning procedure is presented. In the first step, the parameters of the physical model are identified by a simple least-squares method. Then, the modelling error is compensated using an adaptive-network learning procedure. In this way, the meaning of the physical parameters can be preserved. Next, the detection of the electrical faults of the motor - sparking of the brushes, changes in electrical parameters, etc. - are presented, where there is the most significant physical modelling error. The diagnostic results show a higher sensitivity to faults, which enables reliable fault detection. Consequently, the false and missed alarm ratio is reduced as well (9 refs.)

Inspec No.: 8201254

5 Representation dependence in probabilistic inference

Author(s): J.Y. Halpern (**Affiliation:** Dept. of Comput. Sci., Cornell Univ., Ithaca, NY, USA), D. Koller

Journal: *J. Artif. Intell. Res. (USA)*, vol.21, p.319-56 (2004) **Publisher:** AI Access Foundation & Morgan Kaufmann Publishers, USA **Language:** English **ISSN:** 1076-9757

Document type: Journal article

Abstract: Non-deductive reasoning systems are often representation dependent representing the same situation in two different ways may cause such a system to return two different answers. Some have viewed this as a significant problem. For example, the principle of maximum entropy has been subjected to much criticism due to its representation dependence. There has, however, been almost no work investigating representation dependence. In this paper, we formalize this notion and show that it is not a problem specific to maximum entropy. In fact, we show that any representation-

independent probabilistic inference procedure that ignores irrelevant information is essentially entailment, in a precise sense. Moreover, we show that representation independence is incompatible with even a weak default assumption of independence. We then show that invariance under a restricted class of representation changes can form a reasonable compromise between representation independence and other desiderata, and provide a construction of a family of inference procedures that provides such restricted representation independence, using relative entropy (27 refs.)

Inspec No.: 8201759

6 Dynamic extensions of arrow logic

Author(s): P. Balbiani (**Affiliation:** Inst. de Recherche en Inf. de Toulouse, Univ. Paul Sabatier, Toulouse, France), D. Vakarelov

Journal: *Ann. Pure Appl. Log. (Netherlands)*, vol.127, no.1-3, p.1-15 (June 2004) **Publisher:** Elsevier, Netherlands

Language: English **ISSN:** 0168-0072, [Full text](#)

Document type: Journal article

Abstract: This paper is devoted to the complete axiomatization of dynamic extensions of arrow logic based on a restriction of prepositional dynamic logic with intersection. Our deductive systems contain an unorthodox inference rule: the inference rule of intersection. The proof of the completeness of our deductive systems uses the technique of the canonical model (19 refs.)

Inspec No.: 8201977

7 Modular and incremental proofs of AC-termination

Author(s): C. Marche, X. Urbain (**Affiliation:** Univ. Paris-Sud, Orsay, France)

Journal: *J. Symb. Comput. (UK)*, vol.38, no.1, p.873-97 (July 2004) **Publisher:** Academic Press, UK **Language:** English

ISSN: 0747-7171, [Full text](#)

Document type: Journal article

Abstract: Termination is a non-modular property of rewriting systems, thus it is a difficult task to discover termination proofs for rewriting systems of a large number of rules. Recently, new modular and incremental termination criteria, suitable for automation, were proposed, using an approach based on notions of termination under non-deterministic collapse and dependency pairs, which apply to hierarchical combinations of rewriting systems. We extend this approach and corresponding results to the important case of rewriting modulo associativity and commutativity (46 refs.)

Inspec No.: 8203585

8 Modelling the effects of semantic ambiguity in word recognition

Author(s): J.M. Rodd (**Affiliation:** Dept. of Exp. Psychol., Cambridge Univ., UK), M.G. Gaskell, W.D. Marslen-Wilson

Journal: *Cogn. Sci. (USA)*, vol.28, no.1, p.89-104 (Jan.-Feb. 2004) **Publisher:** Elsevier, USA **Language:** English **ISSN:** 0364-0213, [Full text](#)

Document type: Journal article

Abstract: Most words in English are ambiguous between different interpretations; words can mean different things in different contexts. We investigate the implications of different types of semantic ambiguity for connectionist models of word recognition. We present a model in which there is competition to activate distributed semantic representations. The model

performs well on the task of retrieving the different meanings of ambiguous words, and is able to simulate data reported by Rodd, Gaskell, and Marslen-Wilson [*J. Mem. Lang.* 46 (2002) 245] on how semantic ambiguity affects lexical decision performance. In particular, the network shows a disadvantage for words with multiple unrelated meanings (e.g., bark) that coexists with a benefit for words with multiple related word senses (e.g., twist). The ambiguity disadvantage arises because of interference between the different meanings, while the sense benefit arises because of differences in the structure of the attractor basins formed during learning. Words with few senses develop deep, narrow attractor basins, while words with many senses develop shallow, broad basins. We conclude that the mental representations of word meanings can be modelled as stable states within a high-dimensional semantic space, and that variations in the meanings of words shape the landscape of this space (29 refs.)

Inspec No.: 8203661

9 Failure of motor learning for large initial errors

Author(s): T.D. Sanger (**Affiliation:** Dept. of Neurology & Neurological Sci., Stanford Univ., CA, USA)

Journal: *Neural Comput. (USA)*, vol.16, no.9, p.1873-86 (Sept. 2004) **Publisher:** MIT Press, USA **Language:** English **ISSN:** 0899-7667, [Full text](#)

Document type: Journal article

Abstract: For certain complex motor tasks, humans may experience the frustration of a lack of improvement despite repeated practice. We investigate a computational basis for failure of motor learning when there is no prior information about the system to be controlled and when it is not practical to perform a thorough random exploration of the set of possible commands. In this case, if the desired movement has never yet been performed, then it may not be possible to learn the correct motor commands since there is no appropriate training examples. We derive the mathematical basis for this phenomenon when the controller can be modeled as a linear combination of nonlinear basis functions trained using a gradient descent learning rule on the observed commands and their results. We show that there are two failure modes for which continued training examples are not going to lead to improvement in performance. We suggest that this may provide a model for the lack of improvement in human skills that can occur despite repeated practice of a complex task (18 refs.)

Inspec No.: 8204288

Abstract: For certain complex motor tasks, humans may experience the frustration of a lack of improvement despite repeated practice. We investigate a computational basis for failure of motor learning when there is no prior information about the system to be controlled and when it is not practical to perform a thorough random exploration of the set of possible commands. In this case, if the desired movement has never yet been performed, then it may not be possible to learn the correct motor commands since there is no appropriate training examples. We derive the mathematical basis for this phenomenon when the controller can be modeled as a linear combination of nonlinear basis functions trained using a gradient descent learning rule on the observed commands and their results. We show that there are two failure modes for which continued training examples are not going to lead to improvement in performance. We suggest that this may provide a model for the lack of improvement in human skills that can occur despite repeated practice of a complex task (18 refs.)

Inspec No.: 8204288

10 Rules and refutation rules for the logic of finite n-ary trees

Author(s): Tom Skura (**Affiliation:** Dept. of Logic, Wroclaw Univ., Poland)

Journal: *J. Logic Comput. (UK)*, vol.14, no.3, p.429-35 (June 2004) **Publisher:** Oxford University Press, UK **Language:** English **ISSN:** 0955-792X

Document type: Journal article

Abstract: In this paper we give a new syntactic decision procedure for the logic of finite n-ary trees (axiomatized by Gabbay and de Jongh by means of semantic methods). In the decision procedure we use the simple concept of a rule both for proving valid formulas and for refuting invalid ones. Refutation rules with certain normal forms are introduced, and it is shown in a constructive way that every normal form is either provable or refutable (6 refs.)

Abstract: In this paper we give a new syntactic decision procedure for the logic of finite n-ary trees (axiomatized by Gabbay and de Jongh by means of semantic methods). In the decision procedure we use the simple concept of a rule both for proving valid formulas and for refuting invalid ones. Refutation rules with certain normal forms are introduced, and it is shown in a constructive way that every normal form is either provable or refutable (6 refs.)

Inspec No.: 8204447

11 Advances in support vector machines

Author(s): Xu Jian-hua (**Affiliation:** Sch. of Math. & Comput. Sci., Nanjing Normal Univ., China), Zhang Xue-gong, Li Yan-da

Journal: *Control Decis. (China)*, vol.19, no.5, p.481-95 (May 2004) **Publisher:** Northeastern Univ, China **Language:** Chinese **ISSN:** 1001-0920

Document type: Journal article

Abstract: Vapnik and his collaborators proposed a useful algorithm: support vector machines, which can implement the structural risk minimization principle in statistical learning theory. This novel algorithm handles the classification problems successfully. Since then more attention have been paid to statistical learning theory and support vector machines. The attractive research areas are to improve or modify support vector machines by optimization techniques, and to design the novel non-linear machine learning algorithms based on statistical learning theory and some ideas in support vector machines, etc. The advances in such algorithm studies in the last ten years are reviewed (26 refs.)

Inspec No.: 8204458

12 Parallel system methods for management and control of complex systems

Author(s): Wang Fei-yue (**Affiliation:** Key Lab. of Complex Syst. & Intelligence Sci., Chinese Acad. of Sci., China)

Journal: *Control Decis. (China)*, vol.19, no.5, p.485-514 (May 2004) **Publisher:** Northeastern Univ, China **Language:** English **ISSN:** 1001-0920

Document type: Journal article

Abstract: The basic idea, concepts, and framework of management and control for complex systems based on parallel systems methods are presented. The relationship of control systems and parallel systems is discussed. Parallel system methods are a further development and natural extension of control systems and computer-based modeling and simulation, resulting from new and recent advances in computing structure, algorithms and technology. Parallel system methods can provide effective tools for control and management of complex systems that cannot be modeled precisely or experimented repeatedly (22 refs.)

Inspec No.: 8204459

13 Modeling method based on support vector machines within the Bayesian evidence framework

Author(s): Yan Wei-wu, Chang Jun-lin, Shao Hui-he (**Affiliation:** Dept. of Autom., Shanghai Jiaotong Univ., China)

Journal: *Control Decis. (China)*, vol.19, no.5, p.525-33 (May 2004) **Publisher:** Northeastern Univ, China **Language:** Chinese **ISSN:** 1001-0920

Document type: Journal article

Abstract: Regression algorithms and modeling method based on standard support vector machine (standard SVM) and least squares support vector machine (LS SVM) within the Bayesian evidence framework are discussed. A parameter selecting and tuning method within the Bayesian evidence framework is proposed for SVM regression with Gauss kernel. SVM modeling method within the Bayesian evidence framework is applied to nonlinear system identification, and effective results are obtained (7 refs.)

Inspec No.: 8204467

14 Beyond the Floquet theorem: generalized Floquet formalisms and quasienergy methods for atomic and molecular multiphoton processes in intense laser fields

Author(s): S.-I. Chu (**Affiliation:** Dept. of Chem., Kansas Univ., Lawrence, KS, USA), D.A. Telnov

Journal: *Phys. Rep. (Netherlands)*, vol.390, no.1-2, p.1-131 (Feb. 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0370-1573, [Full text](#)

Document type: Journal article

Abstract: The advancement of high-power and short-pulse laser technology in the past two decades has generated considerable interest in the study of multiphoton and very high-order nonlinear optical processes of atomic and molecular systems in intense and superintense laser fields, leading to the discovery of a host of novel strong-field phenomena which cannot be understood by the conventional perturbation theory. The Floquet theorem and the time-independent Floquet Hamiltonian method are powerful theoretical framework for the study of bound-bound multiphoton transitions driven by periodically time-dependent fields. However, there are a number of significant strong-field processes which cannot be directly treated by the conventional Floquet methods. In this review article, we discuss several recent developments of generalized Floquet theorems, formalisms, and quasienergy methods, beyond the conventional Floquet theorem, for accurate nonperturbative treatment of a broad range of strong-field atomic and molecular processes and phenomena of current interests. Topics covered include (a) artificial intelligence (AI)-most-probable-path approach (MPPA) for effective treatment of ultralarge Floquet matrix problem; (b) nonHermitian Floquet formalisms and complex quasienergy methods for nonperturbative treatment of bound-free and free-free processes such as multiphoton ionization (MPI) and above-threshold ionization (ATI) of atoms and molecules, multiphoton dissociation (MPD) and above-threshold dissociation (ATD) of molecules, chemical bond softening and hardening, charge-resonance enhanced ionization (CREI) of molecular ions, and multiple high-order harmonic generation (HHG), etc.; (c) many-mode Floquet theorem (MMFT) for exact treatment of multiphoton processes in multicolor laser fields with nonperiodic time-dependent Hamiltonian; (d) Floquet-Liouville supermatrix (FLSM) formalism for exact nonperturbative treatment of time-dependent Liouville equation (allowing for relaxations and dephasing mechanisms) and high-order nonlinear optical processes (such as intensity-dependent nonlinear optical susceptibilities and multiphoton resonance fluorescence, etc.); (e) generalized Floquet approaches for the treatment of nonadiabatic and complex geometric phases involving multiphoton transitions; (f) generalized Floquet techniques for the treatment of multiphoton processes in intense laser pulse fields with nonperiodic time-dependent Hamiltonians; (g) Floquet formulations of time-dependent density functional theory (DFT) and time-dependent current DFT for nonperturbative treatment of multiphoton processes of many-electron quantum systems in periodic or polychromatic (quasiperiodic) laser fields. For each generalized Floquet approach, we present also the corresponding development of new computational techniques for facilitating the study of strong-field processes and phenomena. The advancement of these generalized Floquet formalisms and quasienergy methods provides powerful new theoretical frameworks and accurate computational methods for nonperturbative and ab initio treatment of a wide range of interesting and challenging laser-induced chemical and physical processes and insightful

exploration of strong-field atomic and molecular physics (273 refs.)

Inspec No.: 8204837

15 Toward a theory of intelligence

Author(s): P. Kugel (**Affiliation:** Dept. of Comput. Sci., Boston Coll., Chestnut Hill, MA, USA)

Journal: *Theor. Comput. Sci. (Netherlands)*, vol.317, no.1-3, p.13-30 (4 June 2004) **Publisher:** Elsevier, Netherlands

Language: English **ISSN:** 0304-3975, [Full text](#)

Document type: Journal article

Abstract: In 1950, Turing suggested that intelligent behavior might require "a departure from the completely disciplined behavior involved in computation", but nothing that a digital computer could not do. In this paper, the author wants to explore Turing's suggestion by asking what it is, beyond computation, that intelligence might require, why it might require it and what knowing the answers to the first two questions might do to help us understand artificial and natural intelligence (26 refs.)

Inspec No.: 8205064

16 Hypercomputation with quantum adiabatic processes

Author(s): T.D. Kieu (**Affiliation:** Centre for Atom Opt. & Ultrafast Spectrosc., Swinburne Univ. of Technol., Hawthorn, Vic., Australia)

Journal: *Theor. Comput. Sci. (Netherlands)*, vol.317, no.1-3, p.93-104 (4 June 2004) **Publisher:** Elsevier, Netherlands

Language: English **ISSN:** 0304-3975, [Full text](#)

Document type: Journal article

Abstract: Despite the recursive non-computability of Hilbert's tenth problem, we outline and argue for a quantum algorithm that is based on the quantum adiabatic theorem. It is explained how this algorithm can solve Hilbert's tenth problem. The algorithm is then considered in the context of several "no-go" arguments against such hypercomputation. Logical arguments are usually based on Cantor's diagonal technique used for proving noncomputability of the Turing halting problem, which is related to Hilbert's tenth problem. Physical arguments are related to the limited computability of a class of quantum computation based on qubits and dimensionally finite quantum logical gates (19 refs.)

Inspec No.: 8205066

17 The modal argument for hypercomputing minds

Author(s): S. Bringsjord, K. Arkoudas (**Affiliation:** Dept. of Comput. Sci., Rensselaer Polytech. Inst., Troy, NY, USA)

Journal: *Theor. Comput. Sci. (Netherlands)*, vol.317, no.1-3, p.167-90 (4 June 2004) **Publisher:** Elsevier, Netherlands

Language: English **ISSN:** 0304-3975, [Full text](#)

Document type: Journal article

Abstract: We now know both that hypercomputation (or super-recursive computation) is mathematically well-understood, and that it provides a theory that according to some accounts for some real-life computation (e.g., operating systems that, unlike Turing machines, never simply output an answer and halt) better than the standard theory of computation at and below the "Turing limit". But one of the things we do not know is whether the human mind hypercomputes, or merely computes-this despite informal arguments from Godel, Lucas, Penrose and others for the view that, in light of incompleteness theorems, the human mind has powers

exceeding those of TMs and their equivalents. All these arguments fail; their fatal flaws have been repeatedly exposed in the literature. However, we give herein a novel, formal modal argument showing that since it's mathematically possible that human minds are hypercomputers, such minds are in fact hypercomputers. We take considerable pains to anticipate and rebut objections to this argument (80 refs.)

Inspec No.: 8205068

18 An adaptive algorithm for efficient message diffusion in unreliable environments

Author(s): B. Garbinato (**Affiliation:** Univ. de Lausanne, Switzerland), F. Pedone, R. Schmidt

Conference: 2004 International Conference on Dependable Systems and Networks , Florence, Italy **Conference Date:** 28 June-1 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Fault-Tolerant Comput.; IFIP WG 10.4 on Dependable Comput. and Fault Tolerance **Publication:** 2004 International Conference on Dependable Systems and Networks **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2052 9 **Page:** 507-16

Document type: Conference paper

Abstract: In this paper, we propose a novel approach for solving the reliable broadcast problem in a probabilistic unreliable model. Our approach consists in first defining the optimality of probabilistic reliable broadcast algorithms and the adaptiveness of algorithms that aim at converging toward such optimality. Then, we propose an algorithm that precisely converges toward the optimal behavior, thanks to an adaptive strategy based on Bayesian statistical inference. We compare the performance of our algorithm with that of a typical gossip algorithm through simulation. Our results show, for example, that our adaptive algorithm quickly converges toward such exact knowledge (13 refs.)

Inspec No.: 8205322

19 Reductions between disjoint NP-pairs

Author(s): C. Glaber (**Affiliation:** Lehrstuhl fur Informatik IV, Univ. Wurzburg, Germany), A.L. Selman

Conference: Proceedings. 19th IEEE Annual Conference on Computational Complexity, Amherst, MA, USA **Conference Date:** 21-24 June 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Found. of Comput. (TCMF) **Publication:** Proceedings. 19th IEEE Annual Conference on Computational Complexity **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2120 7 **Page:** 42-53

Document type: Conference paper

Abstract: Razborov (1994) proved that existence of an optimal proof system implies existence of a many-one complete disjoint NP-pair. Kobler, Messner, and Toran (2003) defined a stronger form of many-one reduction and claimed to improve Razborov's result by showing under the same assumption that there is a strongly many-one complete disjoint NP-pair. Here we show that the two results are equivalent. More generally, we prove that all of the following assertions are equivalent: There is a many-one complete disjoint NP-pair; there is a strongly many-one complete disjoint NP-pair; there is a Turing complete disjoint NP-pair such that all reductions are smart reductions; there is a complete disjoint NP-pair for one-to-one, invertible reductions; the class of all disjoint NP-pairs is uniformly enumerable. Let A, B, C, and D be nonempty sets belonging to NP. A smart reduction between the disjoint NP-pairs (A,B) and (C,D) is a Turing reduction with the additional property that if the input belongs to $A \cup B$, then all queries

belong to $C \cup D$. We prove under the reasonable assumption $UP \cap co-UP$ has a P-bi-immune set that there exist disjoint NP-pairs (A,B) and (C,D) such that (A,B) is truth-table reducible to (C,D), but there is no smart reduction between them. This paper contains several additional separations of reductions between disjoint NP-pairs. We exhibit an oracle relative to which DisjNP has a truth-table-complete disjoint NP-pair, but has no many-one-complete disjoint NP-pair (22 refs.)

Inspec No.: 8205803

20 A clausal resolution method for branching-time logic ECTL

Author(s): A. Bolotov, A. Basukoski (**Affiliation:** Harrow Sch. of Comput. Sci., Westminster Univ., UK)

Editor(s): C. Combi, G. Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 140-7

Document type: Conference paper

Abstract: We expand the applicability of the clausal resolution technique to the branching-time temporal logic ECTL⁺. ECTL⁺ is strictly more expressive than the basic computation tree logic CTL and its extension, ECTL, as it allows Boolean combinations of fairness and single temporal operators. We show that any ECTL⁺ formula can be translated to a normal form the structure of which was initially defined for CTL and then applied to ECTL. This enables us to apply to ECTL⁺ a resolution technique defined over the set of clauses. Our correctness argument also bridges the gap in the correctness proof for ECTL: we show that the transformation procedure for ECTL preserves unsatisfiability (12 refs.)

Inspec No.: 8205914

21 SVM training time reduction using vector quantization

Author(s): G. Lebrun, C. Charrier, H. Cardot (**Affiliation:** Groupe Vision et Anal. d'Image, LUSAC, Saint-Lo, France)

Conference: Proceedings of the 17th International Conference on Pattern Recognition, Cambridge, UK **Conference Date:** 23-26 Aug. 2004 **Publication:** Proceedings of the 17th International Conference on Pattern Recognition **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2128 2 **Page:** 160-3 Vol.1

Document type: Conference paper

Abstract: In this paper, we describe a new method for training SVM on large data sets. Vector quantization is applied to reduce a large data set by replacing examples by prototypes. Training time for choosing optimal parameters is greatly reduced. Some experimental results yields to demonstrate that this method can reduce training time by a factor of 100, while preserving classification rate. Moreover this method allows to find a decision function with a low complexity when the training data set includes noisy or error examples (10 refs.)

Inspec No.: 8206270

22 Comparing optimal bounding ellipsoid and support vector machine active learning

Author(s): I. Gokcen, D. Joachim (**Affiliation:** Dept. of Electr. Eng. & Comput. Sci., Tulane Univ., New Orleans, LA, USA), J. R. Deller

Conference: Proceedings of the 17th International Conference on Pattern Recognition, Cambridge, UK **Conference Date:** 23-26 Aug. 2004 **Publication:** Proceedings of the 17th International Conference on Pattern Recognition **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2128 2 **Page:** 172-5 Vol.1

Document type: Conference paper

Abstract: In this paper we propose two active learning algorithms combining statistical active learning methods based on SVM and optimal bounding algorithms (OBE) of adaptive system identification. We unify SVM and OBE by demonstrating the similarities and representing SVM in the OBE interpretation. Samples are judiciously selected based on a volume measure provided by OBE using both simple heuristic and greedy optimal strategies. Preliminary experiments illustrate the effectiveness of the proposed algorithms as compared to similar methods (12 refs.)

Inspec No.: 8206271

23 Divide-and-conquer algorithm for creating neighborhood graph for clustering

Author(s): O. Virmajoki, P. Franti (**Affiliation:** Dept. of Comput. Sci., Joensuu Univ., Finland)

Conference: Proceedings of the 17th International Conference on Pattern Recognition, Cambridge, UK **Conference Date:** 23-26 Aug. 2004 **Publication:** Proceedings of the 17th International Conference on Pattern Recognition **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2128 2 **Page:** 264-7 Vol.1

Document type: Conference paper

Abstract: K-nearest neighbor graph has been used for reducing the number of distance calculations in PNN-based clustering. The bottleneck of the approach is the creation of the graph. In this paper, we develop a fast divide-and-conquer method for graph creation based on the algorithm previously used in the closest pair problem. The proposed algorithm is then applied to agglomerative clustering, in which it outperforms previous projection-based algorithm for high dimensional spatial data sets (9 refs.)

Inspec No.: 8206276

24 A powerful finite mixture model based on the generalized Dirichlet distribution: unsupervised learning and applications

Author(s): N. Bouguila, D. Ziou (**Affiliation:** Sherbrooke Univ., Que., Canada)

Conference: Proceedings of the 17th International Conference on Pattern Recognition, Cambridge, UK **Conference Date:** 23-26 Aug. 2004 **Publication:** Proceedings of the 17th International Conference on Pattern Recognition **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2128 2 **Page:** 280-3 Vol.1

Document type: Conference paper

Abstract: This paper presents a new finite mixture model based on a generalization of the Dirichlet distribution. For the estimation of the parameters of this mixture we use a GEM (generalized expectation maximization) algorithm based on a Newton-Raphson step. The experimental results involve the

comparison of the performance of Gaussian and generalized Dirichlet mixtures in the classification of several pattern-recognition data sets (8 refs.)

Inspec No.: 8206278

25 Proof reuse with extended inductive types

Author(s): O. Boite (**Affiliation:** Lab. CEDRIC, Inst. d'Informatique d'Entreprise, Evry, France)

Editor(s): K.Slind, A.Bunker, G.Gopalakrishnan

Conference: Theorem Proving in Higher Order Logics. 17th International Conference, TPHOLs 2004. Proceedings, Park City, UT, USA **Conference Date:** 14-17 Sept. 2004
Publication: Theorem Proving in Higher Order Logics. 17th International Conference, TPHOLs 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3223) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23017 3 **Page:** 50-65

Document type: Conference paper

Abstract: Proof assistants based on type theory such as COQ or LEGO, put emphasis on inductive specifications and proofs by featuring expressive inductive types. It is frequent to modify an existing specification or proof by adding or modifying constructors in inductive types. In the context of language design, adding constructors is a common practice to make progress step-by-step. In this article, we propose a mechanism to extend and parameter inductive types, and a proof reuse mechanism founded on proof terms reuse (27 refs.)

Inspec No.: 8206329

26 A decision procedure for geometry in Coq

Author(s): J. Narboux (**Affiliation:** LIX, Ecole Polytechnique, France)

Editor(s): K.Slind, A.Bunker, G.Gopalakrishnan

Conference: Theorem Proving in Higher Order Logics. 17th International Conference, TPHOLs 2004. Proceedings, Park City, UT, USA **Conference Date:** 14-17 Sept. 2004
Publication: Theorem Proving in Higher Order Logics. 17th International Conference, TPHOLs 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3223) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23017 3 **Page:** 225-40

Document type: Conference paper

Abstract: We present in this paper the development of a decision procedure for affine plane geometry in the Coq proof assistant. Among the existing decision methods, we have chosen to implement one based on the area method developed by Chou, Gao and Zhang, which provides short and "readable" proofs for geometry theorems. The idea of the method is to express the goal to be proved using three geometric quantities and eliminate points in the reverse order of their construction thanks to some elimination lemmas (18 refs.)

Inspec No.: 8206332

27 Classification-based behavior model for detection of abnormal states in systems

Author(s): G. Vachkov, K. Komatsu (**Affiliation:** Dept. of Reliability-Based Inf. Syst. Eng., Kagawa Univ., Kagawa-Ken, Japan), S. Fujii

Conference: Proceedings. 2004 International Conference on Intelligent Mechatronics and Automation, Chengdu, China **Conference Date:** 26-31 Aug. 2004 **Publication:** Proceedings.

2004 International Conference on Intelligent Mechatronics and Automation (IEEE Cat. No.04EX952) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8748 1 **Page:** 611-16

Document type: Conference paper

Abstract: In this paper, a special model in the form of lookup table is proposed for detecting different abnormal states in mechanical and other industrial systems. The model is generated by repetitive use of a classification procedure that uses a self-organized map and neural gas algorithm for competitive unsupervised learning. This classification procedure makes a kind of mapping from the measured "Parameter-Space" to the preliminary defined "Operating Mode-Space". The computed modes recognition vectors show the relative weight of each operating mode in the entire set of operating modes. They represent the first part of the system behavior model, while the second part consists of the vectors of the deviated (abnormal) parameters. These are preliminary generated in a systematical, combinatorial way. Thus produced classification-based behavior model (CBB Model) is further used for solving the general problem of the fault diagnosis. Here the most plausible deviated parameters vector has to be found which produces as close as possible the computed modes recognition vector. Simulations with real data from a hydraulic excavator are used in order to prove the applicability and the merits of the proposed method for abnormality detection and fault diagnosis (9 refs.)

Inspec No.: 8206390

28 Constructing decision trees with multiple response variables

Author(s): Seong-Jun Kim (**Affiliation:** Dept. of Ind. Eng., Kangnung Nat. Univ., South Korea), Kang Bae Lee

Journal: *Int. J. Mange. Dec. Mak. (Switzerland)*, vol.4, no.4, p.337-53 (2003) **Publisher:** Inderscience Enterprises, Switzerland **Language:** English **ISSN:** 1462-4621

Document type: Journal article

Abstract: Data mining is a process of discovering meaningful patterns in large data sets that are useful for decision making and has recently received an amount of attention in a wide range of business and engineering fields. Decision tree, also known as recursive partitioning or rule induction, is one of the most frequently used methods for data mining. A decision tree, on a divide-and-conquer basis, provides a set of rules for classifying samples in the learning data set. Most of works on decision tree have been conducted for the case of single response variable. However situations where multiple response variables should be considered arise from many applications, for example, manufacturing process monitoring, customer management, and clinical and health analysis. This article concerns constructing decision trees when there are two or more response variables in the data set. In this article, we investigate node homogeneity criteria such as entropy and Gini index and then present three approaches to constructing decision trees with multiple response variables. To do so, we first describe extensions of entropy and a Gini index to the case in which multiple response variables are of concern. A weighting method for node splitting is also explained. Next, we present a decision tree minimising an expected loss due to misclassifications. To illustrate the procedures, numerical examples are given with discussions (9 refs.)

Inspec No.: 8204843

2.0 Systems and Techniques

29 Probabilistic temporal interval networks

Author(s): V. Ryabov (**Affiliation:** Jyvaskyla Univ., Finland), A. Trudel

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004
Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 64-7

Document type: Conference paper

Abstract: A probabilistic temporal interval network is a constraint satisfaction problem where the nodes are temporal intervals and the edges are uncertain interval relations. We attach a probability to each of Allen's basic interval relations. An uncertain relation between two temporal intervals is represented as a disjunction of Allen's probabilistic basic relations. Using the operations of inversion, composition, and addition, defined for this probabilistic representation, we present a path consistency algorithm (10 refs.)

Inspec No.: 8205904

30 Fuzzy calendar algebra and its applications to data mining

Author(s): Wan-Jui Lee, Shie-Jue Lee (**Affiliation:** Dept. of Electr. Eng., Nat. Sun Yat-Sen Univ., Kaohsiung, Taiwan)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004
Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 71-8

Document type: Conference paper

Abstract: Temporal expressions are widely used in our daily life. Calendar algebra has been studied for years to provide a formal specification for constructing temporal expressions. However, temporal requirements specified by human beings tend to be ill-defined or uncertain. To deal with such kind of uncertain information, we propose the fuzzy calendar algebra which allows users to describe desired temporal expressions easily and naturally. The operations provided reflect the way in which people reason about temporal requirements in daily life. By using the fuzzy calendar algebra, users can define complicated calendars with multiple time granularities in which different time intervals can have different weights according to their matching degrees to the specified calendar. This can help users to discover the knowledge in the time intervals that are of interest to them. We show the usefulness of the algebra by incorporating it with an incremental data miner to mine fuzzy temporal association rules from temporal databases (20 refs.)

Inspec No.: 8205905

31 Induction in time-varying domains: motivation, origins, and encouragements

Author(s): M. Kubat (**Affiliation:** Dept. of Electr. & Comput. Eng., Miami Univ., Coral Gables, FL, USA)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 316-21

Document type: Conference paper

Abstract: Automated acquisition of knowledge and skills by way of machine-learning techniques is a new promising technology for embedded systems. The paper briefly summarizes the ambitions of this technology, and then discusses in some detail the work that has been done in one of its subfields: induction in time-varying domains (39 refs.)

Inspec No.: 8205240

32 Features with fuzzy probability

Author(s): A. Pieczyrski (**Affiliation:** Inst. of Control & Comput. Eng., University of Zielona Gora, Poland), S. Robak, A. Walaszek-Babiszewska

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 323-8

Document type: Conference paper

Abstract: This paper proposes an expert system description based on a feature diagram tree, annotated with weighted variant features in the software family context. The principle of how some variant features may be described on the basis of fuzzy logic is presented and fuzzy probability of market conditions is introduced. The proposed description integrates the external cross-tree constraints, and the resulting homogeneity of the logic then allows its usage to further applications, e.g., generators, or to the purposes of constraint based reasoning itself. The appliance of the proposed approach is demonstrated by the example of a mobile phone tariff description. In the example descriptions of the decision tree for customers have been prepared (14 refs.)

Inspec No.: 8205241

33 The many ways of facing temporal uncertainty in planning and scheduling

Author(s): T. Vidal (**Affiliation:** LGP/ENIT, Tarbes, France)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004
Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning

(TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004
Language: English **ISBN:** 0 7695 2155 X **Page:** 9-10

Document type: Conference paper

Abstract: Generating off line a plan of activities before executing it on line is no longer realistic when one expects disturbances from an uncertain environment. Focusing on temporal plans and uncertain temporal constraints, we show that there are three main ways of tackling the issue: the reactive way, the proactive way and the progressive way, which are illustrated through existing models and systems. In realistic applications, one usually needs all of them, requiring a unified model. Such a model are introduced together with the first results of a modular system developed in the ILOG programming environment (9 refs.)

Inspec No.: 8205896

34 The SOL time theory: a formalization of structured temporal objects and repetition

Author(s): D.R. Cukierman, J.P. Delgrande (**Affiliation:** Simon Fraser Univ., Burnaby, BC, Canada)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004
Language: English **ISBN:** 0 7695 2155 X **Page:** 28-35

Document type: Conference paper

Abstract: We propose to formally represent time with structured temporal objects. Structured temporal objects denote related time intervals (and recursively, related temporal objects) which are conceived as structured objects, rather than relations among such intervals. The major emphasis in this approach is on temporal repetition. To that effect, a new temporal object, the time loop, is defined. The intent of a time loop is to capture a structured notion of repetition. We propose a first order theory formalizing these objects. The building blocks of this formalism are time intervals and Allen's qualitative interval relations. We prove a number of key desirable results including the consistency of the theory, and extensively compare expressions in this theory with previous related work. We argue that this theory presents temporality and temporal repetition in a simple, commonsense manner. Furthermore, we argue that it presents an alternative, succinct and more general view than previous proposals to represent temporal repetition (28 refs.)

Inspec No.: 8205899

35 Recursive representation of periodicity and temporal reasoning

Author(s): L. Anselma (**Affiliation:** Dipt. di Informatica, Universitd di Torino, Italy)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International

Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004

Language: English **ISBN:** 0 7695 2155 X **Page:** 52-9

Document type: Conference paper

Abstract: Representing and reasoning with repeated and periodic events is important in many real-world domains, such as protocol and guideline management. In this set, it is important to give support to complex periodicities, that can involve non-symmetric repetitions, imprecision, variability, pauses between repetitions, and nested time intervals. Also, in these domains it can be useful to give support to composite events, as well as classes of events (i.e. types of actions) and instances of events (i.e. specific actions). In this paper, we propose a general-purpose domain-independent knowledge server dealing with all these issues. In particular, we describe a compact and (hopefully) user-friendly formalism for representing repetition/periodicity temporal constraints that supports arbitrarily nested repetitions as well as possibly imprecise and variable delays between repetitions. Moreover, we define two algorithms for performing consistency checking on knowledge bases of (possibly repeated/periodic) classes and instances of events retaining the efficiency of less expressive approaches (19 refs.)

Inspec No.: 8205902

36 From research to operations: integrating components with an aspect-oriented framework and ontology

Author(s): D.P. Fletcher (**Affiliation:** NASA Ames Res. Center, Moffet Field, CA, USA), F. Akkawi, D.P. Duncavage, R. Alena

Conference: 2004 IEEE Aerospace Conference Proceedings, Big Sky, MT, USA **Conference Date:** 6-13 March 2004
Sponsor(s): Aerosp. and Electron. Syst. Soc **Publication:** 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No.04TH8720) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8155 6 **Page:** 3064-78 Vol.5

Document type: Conference paper

Abstract: Bringing technology from the research world into an operational environment poses many challenges. Typically, software systems having their origins in low technical readiness level research projects have few, if any, formal requirements associated with them. This paucity of formal requirements coupled with the challenges associated with coordinating multiple, distributed research-oriented software projects makes it difficult to design and build software systems that will ultimately be useful in an operational environment. Targeted for current and next-generation space vehicles, the diagnostic applications that compose the advanced diagnostic system (ADS) under development in our lab at NASA-Ames Research Center are realizations of research projects associated with multiple organizations and generally are not designed according to stringent requirements nor with integration into the ADS environment in mind. The core functionality of a Diagnostic Client Application, usually having its basis in artificial intelligence research, is the primary (and perhaps sole) consideration of the application developer. Research funds generally are not available for implementing aspects such as logging and security, both of which are critical in aerospace diagnostic systems such as the ADS. Aspect-oriented programming (AOP) is a new software development methodology that complements object-oriented programming and addresses the complexity of software systems by achieving a separation of functional and interaction components (aspects). Aspects such as logging and security are defined as properties that cut across groups of functional components (diagnostic applications). Aspects can be thought

about and analyzed separately from each other and from the core functionality of the software system. AOP provides the modular separation of crosscutting concerns, where the aspect code is scattered/tangled throughout the software system. An AOP framework takes advantage of what AOP provides and enables us to build software systems that can be extended and adapted during runtime. In this paper we present an AOP framework for integrating software components into an advanced (artificial intelligence-based) diagnostic system and introduce a basic ontology for sharing knowledge between a community of diagnostic applications (agents) (13 refs.)

Inspec No.: 8206468

37 Uninterrupted learning agent using finite state machine

Author(s): Tapanan Yeophantong, Peerapol Moemeng (**Affiliation:** Autonomous Syst. Res. Lab., Assumption Univ., Bangkok, Thailand)

Conference: 2004 IEEE Aerospace Conference Proceedings, Big Sky, MT, USA **Conference Date:** 6-13 March 2004 **Sponsor(s):** Aerosp. and Electron. Syst. Soc **Publication:** 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No.04TH8720) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8155 6 **Page:** 3090-7 Vol.5

Document type: Conference paper

Abstract: In a multi-agent system (MAS), an issue arises as agent may possibly encounter a new event, an event that has never been perceived by the agent or was never anticipated by a human developer during the agent implementation. The ability to recognise an event as being on a new class, without having to shut the system down, would significantly enhance the agent's learning potentials. This paper presents the means for allowing an agent to recognise new events without having to interrupt its services. Our scheme is based on a dynamic construction of finite-state machines (FSM) from the general regularities implicitly embedded in the data. We have extended additional functionality in FSM transition function to enhance the recognition. Our experiment has shown that our extended FSM can enhance the learning process and allows the system to continue working without system interruption (8 refs.)

Inspec No.: 8206469

38 Complexity results and approximation strategies for MAP explanations

Author(s): J.D. Park, A. Darwiche (**Affiliation:** Dept. of Comput. Sci., California Univ., Los Angeles, CA, USA)

Journal: *J. Artif. Intell. Res. (USA)*, vol.21, p.101-33 (2004) **Publisher:** AI Access Foundation & Morgan Kaufmann Publishers, USA **Language:** English **ISSN:** 1076-9757

Document type: Journal article

Abstract: MAP is the problem of finding a most probable instantiation of a set of variables given evidence. MAP has always been perceived to be significantly harder than the related problems of computing the probability of a variable instantiation (Pr), or the problem of computing the most probable explanation (MPE). This paper investigates the complexity of MAP in Bayesian networks. Specifically, we show that MAP is complete for NP and provide further negative complexity results for algorithms based on variable elimination. We also show that MAP remains hard even when MPE and Pr become easy. For example, we show that MAP is NP-complete when the networks are restricted to polytrees,

and even then can not be effectively approximated. Given the difficulty of computing MAP exactly, and the difficulty of approximating MAP while providing useful guarantees on the resulting approximation, we investigate best effort approximations. We introduce a generic MAP approximation framework. We provide two instantiations of the framework; one for networks which are amenable to exact inference (Pr), and one for networks for which even exact inference is too hard. This allows MAP approximation on networks that are too complex to even exactly solve the easier problems, Pr and MPE. Experimental results indicate that using these approximation algorithms provides much better solutions than standard techniques, and provide accurate MAP estimates in many cases (25 refs.)

Inspec No.: 8201758

39 PHA*: finding the shortest path with A* in an unknown physical environment

Author(s): A. Felner (**Affiliation:** Dept. of Inf. Syst. Eng., Ben-Gurion Univ. of the Negev, Israel), R. Stern, A. Ben-Yair, S. Kraus, N. Netanyahu

Journal: *J. Artif. Intell. Res. (USA)*, vol.21, p.631-70 (2004) **Publisher:** AI Access Foundation & Morgan Kaufmann Publishers, USA **Language:** English **ISSN:** 1076-9757

Document type: Journal article

Abstract: We address the problem of finding the shortest path between two points in an unknown real physical environment, where a traveling agent must move around in the environment to explore unknown territory. We introduce the physical-A* algorithm (PHA*) for solving this problem. PHA* expands all the mandatory nodes that A* would expand and returns the shortest path between the two points. However, due to the physical nature of the problem, the complexity of the algorithm is measured by the traveling effort of the moving agent and not by the number of generated nodes, as in standard A*. PHA* is presented as a two-level algorithm, such that its high level, A*, chooses the next node to be expanded and its low level directs the agent to that node in order to explore it. We present a number of variations for both the high-level and low-level procedures and evaluate their performance theoretically and experimentally. We show that the travel cost of our best variation is fairly close to the optimal travel cost, assuming that the mandatory nodes of A* are known in advance. We then generalize our algorithm to the multiagent case, where a number of cooperative agents are designed to solve the problem. Specifically, we provide an experimental implementation for such a system. It should be noted that the problem addressed here is not a navigation problem, but rather a problem of finding the shortest path between two points for future usage (32 refs.)

Inspec No.: 8201762

40 Multi-agent systems specification and certification: a situation and state calculus approach

Author(s): P. Gouveia, J. Ramos (**Affiliation:** Dep. de Matematica, IST, Lisboa, Portugal)

Journal: *Ann. Math. Artif. Intell. (Netherlands)*, vol.41, no.2-4, p.301-38 (Aug. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1012-2443, [Full text](#)

Document type: Journal article

Abstract: We address the topic of specifying multi-agent systems using the situation and state calculus (SSC). SSC has been proposed as an extension of the situation calculus to overcome some limitations of the usual notion of state. The

envisaged multi-agent system specification framework allows the uniform treatment of both local and global properties, providing also techniques for reasoning about such specifications. When a certain intended property is not inferred from a specification, we cannot always just add to it the corresponding formula. Indeed, it is often the case that specification axioms are required to be formulae of a certain kind. The task of identifying the new axioms that should be added to the specification in order to ensure the intended property has an abductive nature. Herein, we develop abductive reasoning techniques to tackle this problem (28 refs.)

Inspec No.: 8201890

41 Improving performance of heterogeneous agents

Author(s): F. Ozcan (**Affiliation:** IBM Almaden Res. Center, San Jose, CA, USA), V.S. Subrahmanian, J. Dix

Journal: *Ann. Math. Artif. Intell. (Netherlands)*, vol.41, no.2-4, p.339-95 (Aug. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1012-2443, [Full text](#)

Document type: Journal article

Abstract: Agents provide services not only to humans users but also to agents in one or more multiagent systems. When agents are confronted with multiple tasks to perform (or requests to satisfy), the agent can reduce load on itself by attempting to take advantage of commonalities between the tasks that need to be performed. In this paper, we develop a logical theory by which such "heavily loaded" agents can merge commonalities amongst such tasks. In our framework, agents can be built on top of legacy codebases. We propose a logical formalism called invariants using which agent developers may specify known commonalities between tasks - after this, we propose a sound and complete mechanism to derive all possible derived commonalities. An obvious A*-based algorithm may be used to merge a set of tasks in a way that minimised expected execution cost. Unfortunately the execution time of this algorithm is prohibitive, even when only 10 tasks need to be merged, thus making it unusable in practice. We develop heuristic algorithms for this problem that take much less time to execute and produce almost as good ways of merging tasks (50 refs.)

Inspec No.: 8201891

42 Extracting fuzzy classification rules from partially labeled data

Author(s): A. Klose (**Affiliation:** Inst. for Knowledge & Language Eng., Otto-von-Guericke Univ. Magdeburg, Germany)

Journal: *Soft Comput. (Germany)*, vol.8, no.6, p.417-27 (May 2004) **Publisher:** Springer-Verlag, Germany **Language:** English **ISSN:** 1432-7643, [Full text](#)

Document type: Journal article

Abstract: The interpretability and flexibility of fuzzy if-then rules make them a popular basis for classifiers. It is common to extract them from a database of examples. However, the data available in many practical applications are often unlabeled, and must be labeled manually by the user or by expensive analyses. The idea of semi-supervised learning is to use as much labeled data as available and try to additionally exploit the information in the unlabeled data. In this paper, we describe an approach to learn fuzzy classification rules from partially labeled datasets (51 refs.)

Inspec No.: 8202638

43 Belief, plausibility, and probability measures on interval-valued type 2 fuzzy sets

Author(s): I.B. Turkmen (**Affiliation:** Dept. of Mech. & Ind. Eng., Toronto Univ., Canada)

Journal: *Int. J. Intell. Syst. (USA)*, vol.19, no.7, p.681-99 (July 2004) **Publisher:** Wiley, USA **Language:** English **ISSN:** 0884-8173, [Full text](#)

Document type: Journal article

Abstract: Belief (Bel), plausibility (Pl), and probability (P) measures can be formulated on interval-valued type 2 fuzzy sets with their representation in terms of α -level (crisp) sets. Recently, it was shown that interval-valued type 2 fuzzy sets naturally arise with modified and restricted multivalued maps of Dempster. An analogy to Dempster's upper and lower probabilities, upper and lower beliefs, and Pl and P measures can be determined over interval-valued type 2 fuzzy sets. It is shown that the application of appropriate t-norms over α -values of α -level crisp sets in combination with the axiom of Bel measure entails appropriate weights for the overlapping α -level (crisp) set conjunctions where Bel measure is applicable (17 refs.)

Inspec No.: 8203778

44 NTM-agent: text mining agent for net auction

Author(s): Y. Kusumura, Y. Hijikata, S. Nishida (**Affiliation:** Graduate Sch. of Eng. Sci., Osaka Univ., Toyonaka, Japan)

Journal: *IEICE Trans. Inf. Syst. (Japan)*, vol.E87-D, no.6, p.1386-96 (June 2004) **Publisher:** Inst. Electron. Inf. & Commun. Eng, Japan **Language:** English **ISSN:** 0916-8532

Document type: Journal article

Abstract: Net auctions have been widely utilized with the recent development of the Internet. However, it is a problem that there are too many items for bidders to select the most suitable one. We aim at supporting the bidders on net auctions by automatically generating a table which contains the features of several items for comparison. We construct a system called NTM-agent (Net auction text mining agent). The system collects Web pages of items and extracts the items' features from the pages. After that, it generates a table which contains the extracted features. This research focuses on two problems in the process. The first problem is that if the system collects items automatically, the results contain the items which is different from the items of the user's target. The second problem is that the descriptions in net auctions are not uniform (There are different formats such as sentences, items and tables. The subjects of some sentences are omitted.). Therefore, it is difficult to extract the information from the descriptions by conventional methods of information extraction. This research proposes methods to solve the problems. For the first problem, NTM-agent filters the items by correlation rules about the keywords in the titles and the item descriptions. These rules are created semi-automatically by a support tool. For the second problem, NTM-agent extracts the information by distinguishing the formats. It also learns the feature values from plain examples for the future extraction (16 refs.)

Inspec No.: 8204881

45 A lattice of classes of user-defined symbolic periodicities

Author(s): L. Egidi, P. Terenziani (**Affiliation:** Dipt. di Informatica, Universita del Piemonte Orientale, Alessandria, Italy)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004
Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 13-20

Document type: Conference paper

Abstract: User-defined calendars and periodicities are gaining an increasing relevance in AI and DB theory and applications. Thus several representation languages have been introduced to model them. Even focusing the attention on symbolic languages only, several different proposals can be chosen, and the comparisons between them are not trivial at all. In this paper, we propose a lattice of properties about periodicity and use it in order to classify different symbolic approaches in the literature. We then propose a language which covers all the properties in the lattice (15 refs.)

Inspec No.: 8205897

46 A possibility theory-based approach to the handling of uncertain relations between temporal points

Author(s): A. Hadjali, D. Dubois, H. Prade (**Affiliation:** IRISA/ENSSAT, Lannion, France)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004
Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 36-43

Document type: Conference paper

Abstract: Uncertain relations between temporal points are represented by means of possibility distributions over the three basic relations "smaller than", "equal to", and "greater than". Operations for computing inverse relations, for composing relations, for combining relations coming from different sources and pertaining to the same temporal points, or for representing negative information, are defined. An illustrative example of representing and reasoning with uncertain temporal relations is given. This paper shows how possibilistic temporal uncertainty can be handled in the setting of point algebra. Moreover, the paper emphasizes the advantages of the possibilistic approach over a probabilistic approach previously proposed. This work does for the temporal point algebra what the authors previously did for the temporal interval algebra (20 refs.)

Inspec No.: 8205900

47 The role of labelled partitionings for modelling periodic temporal notions

Author(s): H.J. Ohlbach (**Affiliation:** Inst. fur Informatik, Univ. Munchen, Germany)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 60-3

Document type: Conference paper

Abstract: The key notion for modelling calendar systems as well as many periodic events, for example the seasons, is the notion of a partitioning of the real numbers. A partitioning of $\mathbb{R}$ splits the time axis into a finite or infinite sequence of intervals. Basic time units like seconds, minutes, hours, days, weeks, months, years etc. can all be represented by finite partitionings of $\mathbb{R}$. There are a lot of other temporal notions which can be modelled as partitions either: the seasons, the ecclesiastical calendars, financial years, semesters at universities, the sequence of sunrises and sunsets, the sequence of the tides, the sequence of school holidays etc. In this paper, a formalization of periodic temporal notions by means of partitionings $\mathbb{R}$ is presented (11 refs.)

Inspec No.: 8205903

48 System framework and key enabling techniques for plant e-maintenance

Author(s): Liu Jian, Yu Dejie (**Affiliation:** Hunan Univ., China), Li Degang, Li Rong

Journal: *China Mech. Eng. (China)*, vol.15, no.13, p.1171-6 (2004) **Publisher:** China Mech. Eng, China **Language:** Chinese **ISSN:** 1004-132X

Document type: Journal article

Abstract: On the basis of extending tele-maintenance with e-service, the definition and connotation of e-maintenance were expatriated to answer changes confronted by plant maintenance. Considering its characteristics of alliance, intelligence, online and business, the system framework of e-maintenance was designed from four hierarchies as workshops, enterprise, alliance and logic. Key enabling techniques of e-maintenance, like the equipment e-encapsulation, work flow management, the construction, cooperation and operation of alliance and quantitative efficiency evaluation, were systematically expounded. Above ideas provide guidance to implementation of the e-maintenance (10 refs.)

Inspec No.: 8201645

49 Effective dimensions of hierarchical latent class models

Author(s): N.L. Zhang (**Affiliation:** Dept. of Comput. Sci., Hong Kong Univ. of Sci. & Technol., China), T. Kocka

Journal: *J. Artif. Intell. Res. (USA)*, vol.21, p.1-17 (2004) **Publisher:** AI Access Foundation & Morgan Kaufmann Publishers, USA **Language:** English **ISSN:** 1076-9757

Document type: Journal article

Abstract: Hierarchical latent class (HLC) models are tree-structured Bayesian networks where leaf nodes are observed while internal nodes are latent. There are no theoretically well justified model selection criteria for HLC models in particular and Bayesian networks with latent nodes in general. Nonetheless, empirical studies suggest that the BIC score is a reasonable criterion to use in practice for learning HLC models. Empirical studies also suggest that sometimes model selection can be improved if standard model dimension is replaced with effective model dimension in the penalty term of the BIC score. Effective dimensions are difficult to compute. In

this paper, we prove a theorem that relates the effective dimension of an HLC model to the effective dimensions of a number of latent class models. The theorem makes it computationally feasible to compute the effective dimensions of large HLC models. The theorem can also be used to compute the effective dimensions of general tree models (18 refs.)

Inspec No.: 8201757

50 On the semantics of deliberation in IndiGolog: from theory to implementation

Author(s): S. Sardina (**Affiliation:** Dept. of Comput. Sci., Toronto Univ., Ont., Canada), G. De Giacomo, Y. Lesperance, H.J. Levesque

Journal: *Ann. Math. Artif. Intell. (Netherlands)*, vol.41, no.2-4, p.259-99 (Aug. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1012-2443, [Full text](#)

Document type: Journal article

Abstract: We develop an account of the kind of deliberation that an agent that is doing planning or executing high-level programs under incomplete information must be able to perform. The deliberator's job is to produce a kind of plan that does not itself require deliberation to interpret. We characterize these as epistemically feasible programs: programs for which the executing agent, at every stage of execution, by virtue of what it knew initially and the subsequent readings of its sensors, always knows what step to take next towards the goal of completing the entire program. We formalize this notion and characterize deliberation in the situation calculus based IndiGolog agent programming language in terms of it. We also show that for certain classes of problems, which correspond to those with bounded solutions and those with solutions without sensing, the search for epistemically feasible programs can be limited to programs of a simple syntactic form. Finally, we discuss implementation issues and execution monitoring and replanning too (33 refs.)

Inspec No.: 8201889

51 Organization of the information flow in the perception-action loop of evolved agents

Author(s): A.S. Klyubin, D. Polani, C.L. Nehaniv (**Affiliation:** Sch. of Comput. Sci., Hertfordshire Univ., Hatfield Herts, UK)

Conference: 2004 NASA/DoD Conference on Evolvable Hardware, Seattle, WA, USA **Conference Date:** 24-26 June 2004 **Sponsor(s):** Nat. Aeronautics and Space Administration (NASA); Dept. of Defense (DoD) **Publication:** 2004 NASA/DoD Conference on Evolvable Hardware **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2145 2 **Page:** 177-80

Document type: Conference paper

Abstract: Sensor evolution in nature aims at improving the acquisition of information from the environment and is intimately related with selection pressure towards adaptivity and robustness. Recent work in the area aims at studying the perception-action loop in a formalized information-theoretic manner. This paves the way towards a principled and general understanding of the mechanisms guiding the evolution of sensors in nature and provides insights into the design of mechanisms of artificial sensor evolution. In our paper we study the perception-action loop of agents. We evolve finite-state automata as agent controllers to solve an information acquisition task in a simple virtual world and study how the information flow is organized by evolution. Our analysis of the

evolved automata and the information flow provides insight into how evolution organizes sensoric information acquisition, memory, processing and action selection. In addition, the results are compared to ideal information extraction schemes following from the Information Bottleneck principle (8 refs.)

Inspec No.: 8205850

52 The architecture of the symbol engine in a programmable autonomous business

Author(s): N.V. Flor (**Affiliation:** New Mexico Univ., Albuquerque, NM, USA)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 119-26

Document type: Conference paper

Abstract: A programmable autonomous business is a kind of online software agent whose function is not to collect or synthesize information for people, but rather to gather and generate customers and income for entrepreneurs. Functionally and compositionally autonomous businesses are indistinguishable from conventional ones. However, an autonomous business is built entirely in software and capable of operating profitably without any employees or managers guiding it. This paper analyses an instance of a successful autonomous business "covering the processes driving the production of its symbols. A general architecture for symbol engines is induced from these processes. The paper discusses how this architecture applies to other successful autonomous businesses, and concludes with a set of design requirements for symbol engines in autonomous businesses (17 refs.)

Inspec No.: 8206157

53 IPSI-PF: a business process matchmaking engine

Author(s): A. Wombacher, B. Mahleko, E. Neuhold (**Affiliation:** Integrated Publication & Inf. Syst. Inst., Fraunhofer Gesellschaft, Darmstadt, Germany)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 137-45

Document type: Conference paper

Abstract: Success of Web services mainly depends on the availability of tools facilitating usage of technology within the addressed B2B integration problems. One severe problem in loosely coupled systems is service discovery including a sufficient matchmaking definition. The concept for service discovery in Web service architecture is UDDI providing limited querying functionality and not being capable to deal with the multiple dimensions of a service, like for example semantic, workflow, or quality of service aspects. The IPSI process finder (IPSI-PF) provides a matchmaking definition and an engine focusing on process aspects. In particular, the matchmaking engine realizes service discovery by extending the capabilities of UDDI to ease the integration with additional

UDDI extensions addressing other service description dimensions (28 refs.)

Inspec No.: 8206159

3.0 Applications

54 Belief change and pre-orders: a brief overview

Author(s): S. Benferhat (**Affiliation:** CRIL, Univ. d'Artois, France), S. Lagrue, O. Papini

Journal: *Int. J. Geogr. Inf. Sci. (UK)*, vol.18, no.4, p.353-88 (June 2004) **Publisher:** Taylor & Francis, UK **Language:** English **ISSN:** 1365-8816, [Full text](#)

Document type: Journal article

Abstract: We deal with iterated belief change. Epistemic states are suitable to represent an intelligent agent's knowledge. Most of the time, when dealing with geographic information, the agent faces incomplete, uncertain, or inaccurate information and needs a mechanism for either revision, update, or fusion to handle their change in beliefs in the presence of new items of information. After some preliminaries on the representation of epistemic states by means of pre-orders, We present three approaches to belief change: revision, update, and fusion. For each of these approaches, the background, postulates, representation by pre-orders, and semantic and syntactic requirements are given. In each case, the approaches are illustrated by examples (47 refs.)

Inspec No.: 8202090

55 The beginnings of a logical semantics framework for the integration of thematic map data

Author(s): N. Wilson (**Affiliation:** Dept. of Comput. Sci., Keele Univ., UK)

Journal: *Int. J. Geogr. Inf. Sci. (UK)*, vol.18, no.4, p.389-415 (June 2004) **Publisher:** Taylor & Francis, UK **Language:** English **ISSN:** 1365-8816, [Full text](#)

Document type: Journal article

Abstract: The integration of spatial datasets from different sources is becoming an increasingly important issue. It is very desirable to have a rigorous approach to integration, as an ad hoc approach can easily lead to incorrect inferences. We take a formal approach, giving the beginnings of a logical semantics framework which allows meaning to be defined mathematically for spatial datasets which represent certain types of thematic map data. The basic idea is to interpret the datasets as summarizations of spatial variables, which are in turn interpreted as sets of possible worlds (ways the world could be). This semantics approach can give a formal meaning to pairs (or sets) of such datasets, which can then be used to determine the valid inferences from an integrated dataset (10 refs.)

Inspec No.: 8202091

56 Le fauteuil intelligent VAHM-3: architecture, commande et premiers resultats

Author(s): A. Pruski, Y. Morere, M. Ennaji (**Affiliation:** Lab. d'Autom. des Syst. Cooperatifs, Metz Univ., France)

Journal: *APII-JESA J. Eur. Syst. Autom. (France)*, vol.37, no.7-8, p.911-27 (2003) **Publisher:** Editions Hermes, France **Language:** French **ISSN:** 1269-6935

Document type: Journal article

Abstract: This article has for object to give some results obtained by the application of a symbiotic architecture on the intelligent wheelchair VAHM in its third release. This architecture aims to realize a strong interaction between the person and the wheelchair. Every entity, the man or the machine, brings to the other its contribution allowing the couple to achieve a goal with a minimum effort for the person while allowing him to remain the master of the action (11 refs.)

Inspec No.: 8202487

57 Strategic knowledge for sustainable development: the need for reflexivity and learning at the interface between science and society

Author(s): A. Grunwald (**Affiliation:** Inst. for Technol. Assessment & Syst. Anal., Forschungszentrum Karlsruhe, Germany)

Journal: *Int. J. Foresight Innov. Policy (Switzerland)*, vol.1, no.1-2, p.150-67 (2004) **Publisher:** Inderscience Enterprises, Switzerland **Language:** English **ISSN:** 1740-2816

Document type: Journal article

Abstract: Redirecting society onto paths of more sustainable development is inconceivable without the support of scientific knowledge. Scientific contributions to sustainable development, however, don't often follow the classical routes to cognition, and don't often correspond to the traditional concept of science. Our hypothesis is that strategic knowledge for sustainable development extends far beyond explanatory and observational cognisance. It much rather consists of a problem-oriented combination of explanatory, orientation-, and action-guiding knowledge. For the generation and application of this strategic knowledge, new trails, which are only rudimentarily discernible, have to be trod. Above all, reflexivity and making societal learning possible are important requirements. This has consequences, not only for the self-concept of the sciences, but also for the relationship between science, politics, and other societal areas (31 refs.)

Inspec No.: 8203065

58 Multimodal story-based communication: integrating a movie and a conversational agent

Author(s): Y.I. Nakano, T. Murayama (**Affiliation:** Res. Inst. of Sci. & Technol. for Soc., Tokyo, Japan), T. Nishida

Journal: *IEICE Trans. Inf. Syst. (Japan)*, vol.E87-D, no.6, p.1338-46 (June 2004) **Publisher:** Inst. Electron. Inf. & Commun. Eng, Japan **Language:** English **ISSN:** 0916-8532

Document type: Journal article

Abstract: In story-based communication, where a message is conveyed in story form, it is important to embody the story with expressive materials. However, it is quite difficult for users to create rich multimedia contents using multimedia editing tools. This paper proposes a Web-based multimedia environment, SPOC (stream-oriented public opinion channel), aiming at helping nonskilful people to convert their stories into TV-like programs very easily. The system can produce a digital camera work for graphics and video clips as well as generate an agent animation automatically according to a narration text. Findings in evaluation experiments showed that SPOC is easy-to-use and easy-to-learn for novice users. Given a short instruction, the subjects not only mastered the operations of the software, but also succeeded in creating highly original programs. In subjective evaluation, the subjects answered that they enjoyed using the software without feeling difficulty. These results suggest that this system reduces user's cost in

making a program, and encourages communication in a network community (25 refs.)

Inspec No.: 8204875

59 Net-centric human-robotics operations: integration of business/military process management, grid computing, and robotics

Author(s): J.T. Durham, W.C. Torrez (**Affiliation:** Space & Naval Warfare Syst. Center, San Diego, CA, USA), J.E. Manley

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 810-11

Document type: Conference paper

Abstract: An effort is underway to more fully integrate business/military process management (BPM) and grid computing with distributed robotics and automation capabilities. This work is a logical extension of BPM whereby process and workflow management are extended to include the configuration and reengineering of both human and software agents that collectively synchronize and coordinate both human and robotics driven activities (36 refs.)

Inspec No.: 8205496

60 Validating simulation models

Author(s): K.G. Troitzsch (**Affiliation:** Univ. Koblenz-Landau, Germany)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004, Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 265-70

Document type: Conference paper

Abstract: This paper discusses aspects of validating simulation models designed to describe, explain and predict real-world phenomena. It starts with a short review of arguments used in the simsoc mailing list discussion on theory, simulation and explanation a few months ago, deals with the use of quantitative and qualitative computational models to make quantitative and qualitative predictions or rather to draw conclusions from complex antecedents, and then discusses different types of explanation and prediction (and the relation between these two). It closes with an overview of topics in validity and validation from the point of view of the structuralist programme in the philosophy of science (19 refs.)

Inspec No.: 8205741

61 Agent-based modelling of human acting, deciding and behaviour - the reference model PECS

Author(s): B. Schmidt, B. Schneider (**Affiliation:** Passau Univ., Germany)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004,

Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 378-87

Document type: Conference paper

Abstract: Agent-based modelling and simulation is a powerful tool in studying complex systems. For the purpose of designing related simulation models, it is helpful to utilize already existing problem-related design patterns: reference models. The PECS reference model provides capabilities for object-oriented model specification. Its application area is settled in the field of agent-based simulation, especially for models deriving from studies in sociology, economy and psychology. Major aspect is the modelling of human acting and deciding, interaction of human beings with each other and with their environment, where a human being is considered a psychosomatic unit with cognitive capacities embedded in a social environment. Accordingly, the PECS architecture takes into account physical conditions, emotional states, cognitive capabilities and social status that are represented each by specialised components: Physics, Emotion, Cognition and Social Status from which the name of the PECS reference model derive. The presented case study is supposed to explain the interplay between the PECS components in order to control and to coordinate the available reactive, deliberative and reflective behavioural patterns that enable a PECS agent to show the most complex form of human behaviour: emotional intelligent behaviour (15 refs.)

Inspec No.: 8205746

3.1 Business and Finance

62 Personalized trust management in decision making: a dynamic clustering approach

Author(s): G.P. Papamichail (**Affiliation:** Dept. of Manage. Sci. & Technol., Athens Univ. of Econ. & Bus., Greece), D.P. Papamichail

Journal: *J. Comput. Inf. Technol. CIT (Croatia)*, vol.12, no.1, p.47-53 (March 2004) **Publisher:** Univ. Comput. Centre, Zagreb, Croatia **Language:** English **ISSN:** 1330-1136

Document type: Journal article

Abstract: This paper presents a personalized approach for distributed trust management by employing the k-means range algorithm, a combination of the partitionial k-means clustering algorithm with orthogonal range search concepts. The aim of this approach is to aid the human or computer agent in organizing information from multiple sources into clusters according to its "trust features". Thus the agent can perform complicated trust assessments in real-time situations and cooperate with decision-making software to assist in purchasing activities. We conclude by discussing the implications and advantages of this approach in trust management in traditional and mobile e-commerce applications (15 refs.)

Inspec No.: 8200876

63 Concurrent auctions across the supply chain

Author(s): M. Babaioff, N. Nisan (**Affiliation:** Sch. of Comput. Sci. & Eng., Jerusalem Hebrew Univ., Israel)

Journal: *J. Artif. Intell. Res. (USA)*, vol.21, p.595-629 (2004) **Publisher:** AI Access Foundation & Morgan Kaufmann Publishers, USA **Language:** English **ISSN:** 1076-9757

Document type: Journal article

Abstract: With the recent technological feasibility of electronic commerce over the Internet, much attention has been given to the design of electronic markets for various types of electronically-tradable goods. Such markets, however, normally need to function in some relationship with markets for other related goods, usually those downstream or upstream in the supply chain. Thus, for example, an electronic market for rubber tires for trucks likely need to be strongly influenced by the rubber market as well as by the truck market. In this paper we design protocols for exchange of information between a sequence of markets along a single supply chain. These protocols allow each of these markets to function separately, while the information exchanged ensures efficient global behavior across the supply chain. Each market that forms a link in the supply chain operates as a double auction, where the bids on one side of the double auction come from bidders in the corresponding segment of the industry, and the bids on the other side are synthetically generated by the protocol to express the combined information from all other links in the chain. The double auctions in each of the markets can be of several types, and we study several variants of incentive compatible double auctions, comparing them in terms of their efficiency and of the market revenue (26 refs.)

Inspec No.: 8201761

64 Analytic applications and decision -centric BI

Author(s): H. Morris

Journal: *DM Rev. (USA)*, vol.14, no.6, p.44, 48, 50 (June 2004) **Publisher:** Faulkner & Gray, USA **Language:** English **ISSN:** 1521-2912

Document type: Journal article

Abstract: We describe about analytic applications and decision -centric BI (business intelligence). Research in psychology and decision sciences has challenged the assumption that apply rational principles in making decisions that work to optimize expected results. This descriptive, empirical research has catalogued many of the biases that enter into decision making. We discuss about two types of decisions - operational and strategic. Strategic decisions have broad organizational scope but are infrequent, while operational decisions have more restricted scope but repeat frequently. IDC research shows that progress is being made by capturing and automating decision-making processes for repeatable, operational decisions. Strategic decision are important. However, because they are infrequent, there is little opportunity to apply lessons learned on an ongoing basis and provide software-based automation to support such a process. That is not the case with operational decisions with are repeatable

Inspec No.: 8202105

65 Putting value of information theory into practice: a methodology for building sequential decision support systems

Author(s): S. Mussi (**Affiliation:** CILEA, Italy)

Journal: *Expert Syst. (UK)*, vol.21, no.2, p.92-103 (May 2004) **Publisher:** Blackwell Publishers, UK **Language:** English **ISSN:** 0266-4720, [Full text](#)

Document type: Journal article

Abstract: We present a methodology for building sequential decision support systems based on decision theory using value of information (for short, DT-VOI based SDSSs). DT-VOI based SDSSs support decision-makers in difficult

problems of sequential decision-making. In particular we consider the problem of building DT-VOI based SDSSs which are capable of supporting decisions in critical situations where (1) making a decision entails knowing the states of some critical hypotheses, and such knowledge is acquired by performing suitable tests; (2) test outcomes are uncertain; (3) performing a test entails, in general, some drawbacks, so that a trade-off exists between such drawbacks and the value of the information provided by the test; (4) performing a test has the side-effect that it changes the expected benefit from performing other tests; (5) exceptional situations alter probability and utility default values (29 refs.)

Inspec No.: 8202712

66 The impact of problem size on decision processes: an experimental investigation on very large choice problems with support of decision support systems

Author(s): H. Wang (**Affiliation:** Dept. of Manage. Sci. & Inf. Syst., Wright State Univ., Dayton, OH, USA), P.-C. Chu

Journal: *Expert Syst. (UK)*, vol.21, no.2, p.104-18 (May 2004) **Publisher:** Blackwell Publishers, UK **Language:** English **ISSN:** 0266-4720, [Full text](#)

Document type: Journal article

Abstract: Choice problems as a class of decision problems have attracted great attention for the last couple of decades. Among the frameworks and supporting theories used in their study, two have had the greatest impact: bounded rationality and cost-benefit. Both theories could find support from past empirical studies under different conditions or problem environments. In the past studies, problem size has been shown to play an important role in decision-making. As problem size increases, a decision process may be detoured and the decision outcome may be different. We investigate the impact of problem size on three important aspects of the computer-aided decision process - strategy selection, decision time/effort, and decision quality -through very large choice problems (25 refs.)

Inspec No.: 8202713

67 Intelligent aircraft maintenance support system using genetic algorithms and case-based reasoning

Author(s): C. Chiu, N.-H. Chiu (**Affiliation:** Dept. of Information Manage., Yuan-Ze Univ., Yuan Ze, Taiwan), C.-I. Hsu

Journal: *Int. J. Adv. Manuf. Technol. (UK)*, vol.24, no.5-6, p.440-6 (2004) **Publisher:** Springer-Verlag, UK **Language:** English **ISSN:** 0268-3768, [Full text](#)

Document type: Journal article

Abstract: The maintenance of aircraft components is crucial for avoiding aircraft accidents and aviation fatalities. To provide reliable and effective maintenance support, it is important for the airline companies to utilise previous repair experience with the aid of advanced decision support technology. Case-based reasoning (CBR) is a machine learning method that adapts previous similar cases to solve current problems. To effectively retrieve similar aircraft maintenance cases, this research proposes using a CBR system to aid electronic ballast fault diagnosis of Boeing 747-400 airplanes. By employing genetic algorithms (GA) to enhance dynamic weighting and the design of non-similarity functions, the proposed CBR system is able to achieve superior learning performance as compared to those with

either equal/varied weights or linear similarity functions (16 refs.)

Inspec No.: 8203598

68 IT and business process outsourcing: the knowledge potential

Author(s): L. Willcocks (**Affiliation:** Sch. of Bus., Warwick Univ., Coventry, UK), J. Hindle, D. Feeny, M. Lacity

Journal: *Inf. Syst. Manage. (USA)*, vol.21, no.3, p.7-15 (Summer 2004) **Publisher:** Auerbach Publications, USA **Language:** English **ISSN:** 1058-0530

Document type: Journal article

Abstract: Despite the widespread trends in IT and business process outsourcing, there has been too little focus on what happens to knowledge when an organization outsources. We present a framework for evaluating the knowledge potential within five different types of insourcing and outsourcing arrangements. A detailed example of an enterprise partnership relationship is described as a benchmark for how companies can leverage the knowledge potential from IT and BPO outsourcing (11 refs.)

Inspec No.: 8204848

69 Key issues for global IT sourcing: country and individual factors

Author(s): M.T. Rao (**Affiliation:** Albers Sch. of Bus., Seattle Univ., WA, USA)

Journal: *Inf. Syst. Manage. (USA)*, vol.21, no.3, p.16-21 (Summer 2004) **Publisher:** Auerbach Publications, USA **Language:** English **ISSN:** 1058-0530

Document type: Journal article

Abstract: Globalization trends, trade liberalization in developing countries, quantum improvements in worldwide telecommunications infrastructures, and cost-cutting pressures have together compelled many companies to examine global opportunities for sourcing some of their IT operations. Such global sourcing, whether it involves insourcing or outsourcing, requires IT project managers to deal with issues unique to operating in heterogeneous international environments. This article examines the influences of country-level and individual-level factors on the effective management of offshore IT sourcing relationships and offers a set of best practices that project managers should consider when entering into such global arrangements (13 refs.)

Inspec No.: 8204849

70 Software sales by IT departments: managing the opportunities and issues

Author(s): F.H. Grupe, S. Gesh, S. Jooste (**Affiliation:** Dept. of Inf. Syst., Nevada Univ., Reno, NV, USA)

Journal: *Inf. Syst. Manage. (USA)*, vol.21, no.3, p.65-9 (Summer 2004) **Publisher:** Auerbach Publications, USA **Language:** English **ISSN:** 1058-0530

Document type: Journal article

Abstract: Corporate managers are keenly aware of the need to implement cost reductions as well as new profit-making initiatives. These sometimes-conflicting demands can lead to innovative approaches. For example, some IT departments have become profit centers by selling their own software products. This article addresses some of the benefits associated with becoming a vendor, as well as the

management challenges that must be addressed for this type of initiative to be successful (9 refs.)

Inspec No.: 8204856

71 Agent architecture for smart adaptive systems

Author(s): L. Lhotska, O. Stepankova (**Affiliation:** Dept. of Cybern., Czech Tech. Univ., Prague, Czech Republic)

Journal: *Trans. Inst. Meas. Control (UK)*, vol.26, no.3, p.245-60 (2004) **Publisher:** Inst. Meas. Control, UK **Language:** English **ISSN:** 0142-3312, [Full text](#)

Document type: Journal article

Abstract: This paper discusses the possibility of utilizing a multiagent system as a platform for development of a smart adaptive system in medicine. A knowledge-based model of the agents' mutual awareness (social knowledge) is presented.

The tri-base acquaintance (3bA) model formalizing the agent's social behaviour knowledge and co-operation neighbourhood knowledge is enhanced to support attention focussing and to ensure a holistic complex perception of the problem. The paper focuses on an analysis of the content of 3bA knowledge bases for applications in medical decision support. An architecture for a multiagent system, ADAM, for medical decision support and monitoring is proposed (18 refs.)

Inspec No.: 8205121

72 IEEE International Conference on E-Commerce Technology

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7

Document type: Conference proceedings

Abstract: The following topics are dealt with: e-commerce; dynamic pricing; B2B commerce; e-catalogs; B2C decision support; Web service composition; business process management; service-oriented enterprise applications; mobile agents; economic incentives; e-payment; security; privacy; planning and interoperability

Inspec No.: 8206143

73 Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119)

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7

Document type: Conference proceedings

Abstract: The following topics are dealt with: mathematical knowledge management; mathematical statement retrieval; set theory; formal mathematics; mathematical service matching; differential equations; World Wide Web; Internet; mathematical knowledge repository; mathematical knowledge representation; predicate logic; digital library; theorem proving; proof verification; mathematics learning; matrices; proof assistant environments; mathematical formula parsing; and mathematical Web services

Inspec No.: 8206320

74 Web-service semantic enabled implementation of machine vs. machine business negotiation

Author(s): L. Vasiliu, M. Zaremba, M. Moran, C. Bussler
(Affiliation): Digital Enterprise Res. Inst., Galway, Ireland), J. Browne

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 106-15

Document type: Conference paper

Abstract: The ultimate target of business to business (B2B) integration and deployment is complete business process automation within and between enterprises with no human intervention. From a business point of view, negotiation is the main mechanism that modern enterprises use to achieve targets that maximise their profits. This paper proposes an automated B2B negotiation solution: the implementation of semantic-enabled machine versus machine business negotiation as a Web service. We argue that the shift from human vs. human to machine vs. machine business negotiation is facilitated by using semantic Web technology implemented with Web services. The negotiation process in this work is designed for multiple machines (a minimum of 3 computers), using a negotiation algorithm that emulates the behaviour of human business negotiations. Possible development directions are brought forward in the paper's conclusion (25 refs.)

Inspec No.: 8205405

75 Manners externalize semantics for on-demand composition of context-aware services

Author(s): A. Arsanjani **(Affiliation):** IBM Global Services, Maharishi Univ. of Manage., Fairfield, IA, USA), F. Curbera, N. Mukhi

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 583-90

Document type: Conference paper

Abstract: Architectural mismatch; the semantic and syntactic gap preventing component composition, can be partially alleviated through the ubiquity of the protocols for service composition (syntax). And yet, the semantics of service or component composition have remained elusive. The contribution of this paper builds on the notions introduced by grammar-oriented object design (GOOD) using the notion of manners to specify the semantics of services and drive their automated composition through dynamically reconfigure architectures enabled through GOOD. This leads to the use of manners for explicit service semantics representation in a re-configurable architectural style. We describe the base requirements for automated component and services composition and show how automated assembly of components and services can be accomplished through a dynamically reconfigurable (DyRec) architectural style. In addition, we show how this vision of dynamic reconfiguration is naturally supported by the Web services framework, because it derives from some of the core requirements of the service oriented architectures (SOA) style embodied by Web

services. Further, we discuss how such a DyRec architectural style can be implemented using grammar-oriented object design (GOOD) and describe a project that led to a tool called the Business Compiler, to accomplish dynamic composition and collaboration (28 refs.)

Inspec No.: 8205460

76 Mapping a business process model to a semantic Web service model

Author(s): Li Guo, Yun-Heh, C. Burger, D. Roberston
(Affiliation): CISA, Edinburgh Univ., UK)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 746-9

Document type: Conference paper

Abstract: The purpose of OWL-S is to support effective automation of various Web services related activities including service discovery, composition, execution and monitoring. For our work, we are interested in automated Web services composition for a business process model. OWL-S process ontology provides a standard language for describing the composition of Web services. Thus we can treat composed Web services as a process model. However, since OWL-S is an XML based syntax description which is used for automatic agent processing, it is difficult to read and understand for human reader, especially for those with little XML knowledge but from a process modelling background, for example, businessmen. Therefore it can not be easily and directly used for process modelling tasks. The Fundamental Business Process Modelling Language (FBPML) is a diagram based business process modelling language which merges IDEF3, RAD and PSL. It has a precise operational semantics and describes business processes in a conventional logic. Because it is diagram based, it can be easily accepted by processes modelers who normally are familiar with such diagrams. However, its problem is that it does not provide a direct specification for Web services based agents. This paper bridges the gap via conceptual mapping between two different process modelling languages, FBPML and OWL-S (9 refs.)

Inspec No.: 8205481

77 IRIS: a framework for mediator-based composition of service-oriented software

Author(s): U. Radetzki, A.B. Cremers **(Affiliation):** Dept. of Comput. Sci., Bonn Univ., Germany)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 752-6

Document type: Conference paper

Abstract: Due to a rapidly increasing volume of information and autonomous services, domain experts spend considerable time and effort in correlating and integrating these resources into meaningful business processes. In this paper we present IRIS, a framework that effectively supports users in creating service compositions with the special focus on interoperability. We illustrate our work by means of a

scientific business process from the area of life science informatics. For the sake of adaptability and reusability our approach relies on a component-based mediator model. The problem of mediator discovery and composition is addressed by an ontology-based registry approach (6 refs.)

Inspec No.: 8205482

78 Net-centric human-robotics operations: integration of business/military process management, grid computing, and robotics

Author(s): J.T. Durham, W.C. Torrez (**Affiliation:** Space & Naval Warfare Syst. Center, San Diego, CA, USA), J.E. Manley

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 810-11

Document type: Conference paper

Abstract: An effort is underway to more fully integrate business/military process management (BPM) and grid computing with distributed robotics and automation capabilities. This work is a logical extension of BPM whereby process and workflow management are extended to include the configuration and reengineering of both human and software agents that collectively synchronize and coordinate both human and robotics driven activities (36 refs.)

Inspec No.: 8205496

79 Semantic features and factual agents: a model to represent changeable situations

Author(s): T. Galinho, F. Serin (**Affiliation:** Comput. Sci. Lab., Le Havre Univ., France)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004, Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 351-6

Document type: Conference paper

Abstract: In this paper we present a part of a preventive monitoring information system. First, we describe the global architecture of the multiagent decision support system. Then, we focus on one organization of agents called factual agents whose aim is to represent semantic features. We also expose our ontology with its modeling. Our goal is to store and to use general and specific knowledge and information, in order to compare factual agents (8 refs.)

Inspec No.: 8205744

80 A Markov-decision-based price comparison problem for mobile agent-based Internet commerce system (MAGICS)

Author(s): P.P.Y. Lam, H.C.B. Chan (**Affiliation:** Dept. of Comput., Hong Kong Polytech. Univ., China)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July

2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 34-41

Document type: Conference paper

Abstract: Mobile agents are well suited for facilitating the consumer buying process on the Internet, particularly for comparing prices. In this paper, we present a mobile agent-based e-commerce system, called MAGICS, that addresses some issues concerning price comparisons. After providing an overview of the MAGICS architecture, we use the Markov decision theory to formulate a price comparison problem. The objective of the problem is to obtain an optimal decision policy in order to buy a product at the lowest expected cost, including the traveling costs. The problem was solved using the backward induction algorithm. Some analytical results are presented and discussed (17 refs.)

Inspec No.: 8206147

81 The architecture of the symbol engine in a programmable autonomous business

Author(s): N.V. Flor (**Affiliation:** New Mexico Univ., Albuquerque, NM, USA)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 119-26

Document type: Conference paper

Abstract: A programmable autonomous business is a kind of online software agent whose function is not to collect or synthesize information for people, but rather to gather and generate customers and income for entrepreneurs. Functionally and compositionally autonomous businesses are indistinguishable from conventional ones. However, an autonomous business is built entirely in software and capable of operating profitably without any employees or managers guiding it. This paper analyses an instance of a successful autonomous business "covering the processes driving the production of its symbols. A general architecture for symbol engines is induced from these processes. The paper discusses how this architecture applies to other successful autonomous businesses, and concludes with a set of design requirements for symbol engines in autonomous businesses (17 refs.)

Inspec No.: 8206157

82 IPSI-PF: a business process matchmaking engine

Author(s): A. Wombacher, B. Mahleko, E. Neuhold (**Affiliation:** Integrated Publication & Inf. Syst. Inst., Fraunhofer Gesellschaft, Darmstadt, Germany)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 137-45

Document type: Conference paper

Abstract: Success of Web services mainly depends on the availability of tools facilitating usage of technology within the addressed B2B integration problems. One severe problem in loosely coupled systems is service discovery including a sufficient matchmaking definition. The concept for service discovery in Web service architecture is UDDI providing limited querying functionality and not being capable to deal with the multiple dimensions of a service, like for example semantic, workflow, or quality of service aspects. The IPSI process finder (IPSI-PF) provides a matchmaking definition and an engine focusing on process aspects. In particular, the matchmaking engine realizes service discovery by extending the capabilities of UDDI to ease the integration with additional UDDI extensions addressing other service description dimensions (28 refs.)

Inspec No.: 8206159

83 Formal specification and verification of the SET/A protocol with an approach

Author(s): V.S.W. Lam, J. Padget (**Affiliation:** Dept. of Comput. Sci., Bath Univ., UK)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 229-35

Document type: Conference paper

Abstract: In this paper, we specify the SET/A protocol which is an agent-based payment protocol for credit card transactions in UML statechart diagrams. Then we translate them into the NuSMV language and analyze the payment protocol using the NuSMV model checker. Although the payment protocol satisfies desired properties including data integrity and deadlock freedom in the absence of an agent failure, a failure analysis reveals that not all desired properties are satisfied when the agent crashes. The agent may fail while it is travelling to a merchant's server or when it is running in the merchant's server. To ensure that a transaction is resilient to the agent failure, an extended SET/A protocol is proposed. Whenever a timeout occurs, the cardholder sends an inquiry request to the merchant directly for finding out the transaction result. Our work contributes the e-commerce field by using an integrated approach for modelling and analyzing the SET/A protocol (22 refs.)

Inspec No.: 8206170

84 LITESET/A++: a new agent-assisted secure payment protocol

Author(s): Yan Wang (**Affiliation:** Dept. of Comput., Macquarie Univ., Sydney, NSW, Australia), Tiejian Li

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 244-51

Document type: Conference paper

Abstract: In this paper, we proposed a new agent-assisted secure payment protocol LITESET/A++, which aims at

enabling the dispatched consumer-agent to choose the "best" one after negotiating with merchants to close a deal, autonomously sign contracts and make the payment on behalf of the cardholder without the possibility of disclosing any secret to any participant. This is realized by adopting the signature-share and signcryption-share schemes, and employing a trusted third party (TTP). In LITESET/A++, the principle that each participant knows what is strictly necessary for his/her role is followed as in SET payment protocol while the non-repudiation property is improved (31 refs.)

Inspec No.: 8206172

85 E-marketplace using artificial immune system as matchmaker

Author(s): Tung-Wan Cheng, Wan-Ling Wang (**Affiliation:** Inst. of Inf. Manage., Nat. Chiao Tung Univ., Hsinchu, Taiwan), An-Pin Chen

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 358-61

Document type: Conference paper

Abstract: Architecture immune dealing system (AIDS) is a system that uses the artificial immune algorithm that is implemented in the e-marketplace. It has similar architecture as the artificial immune system with agents acting to search for an appropriate match of buyer and seller. The AIDS represents the next generation of services in the e-marketplace, aiming to achieve an efficient and fair e-marketplace with less cost (11 refs.)

Inspec No.: 8206190

86 Promoting agent-based knowledge discovery in medical intensive care units

Author(s): A. Lourenco, O. Belo (**Affiliation:** Campus de Gualtar, Univ. do Minho, Braga, Portugal)

Journal: *WSEAS Trans. Comput. (Greece)*, vol.2, no.2, p.403-8 (April 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2750

Document type: Journal article

Abstract: Decision support has become vital in major scientific and organisational areas. Data analysis and knowledge acquisition are no longer means to an end, they have become the end itself. Knowledge discovery is a priority, constantly demanding for new, better suited efforts. Systems or tools capable of dealing with the steadily growing amount of data presented by information systems and digging through their structure, relations, contexts and ultimately, their contents, are in order. In this paper, an agent-based decision support system based in data warehousing systems contents is described, analysing its actual potential and skills in knowledge discovery as a whole and, in particular, in data preparation for data mining. Different requirements, algorithm dependent and nondependent, are pointed, analysing valid approaches. An algebraic model is used to specify the core of some of the embraced steps, as well as, to demarcate the data flow throughout the knowledge acquisition process. A case study is presented in order to demonstrate the application of the system to real world scenarios (10 refs.)

Inspec No.: 8203035

87 Mounting of auction agent under dynamic environment by Q-learning and SARSA learning

Author(s): T. Katou (**Affiliation:** Graduate Sch. Eng. Div, Hosei Univ., Koganei, Japan), K. Nagasaka

Editor(s): N.Callaos, M.Margenstern, J.Zhang, O.Castillo, E.-E.Doberkat

Conference: SCI 2003. 7th World Multiconference on Systemics, Cybernetics and Informatics Proceedings, Orlando, FL, USA **Conference Date:** 27-30 July 2003 **Sponsor (s):** WOSC: World Organization on Systemics and Cybernetics; Centre for Syst. Studies; Syst. Soc. of Poland; Soc. Applied Syst. Res.; Slovenian Artificial Intelligence Soc.; Simon Bolivar Univ.; Polish Syst. Soc.; Italian Soc. of Systemics; ISSS; ISI; IFSR; Cybernetics and Human Knowing; CUST; Concurrency and Architecture Group, the Telematics Eng. Department of the Univ. of Las Palmas of Gran Canaria; Tunisian Sci. Soc.; ANS; Lab. of Res. of Computational Intelligence/Department of Informatic/San Luis Nat. Univ.; American Soc. of Cybernetics; Wolfram Res. Inc **Publication:** SCI 2003. 7th World Multiconference on Systemics, Cybernetics and Informatics Proceedings **Publisher:** IIS, USA, 2003 **Language:** English **ISBN:** 980 6560 01 9 **Page:** 472-5 Vol.5

Document type: Conference paper

Abstract: In recent years, electronic commerce is fairly popular with the development of the Internet network system, the number of participants of Internet auctions increases very rapidly, and the prepared automatic bid agents system makes the auction procedures easier and faster without restrictions of time or place. However the policy of such an automatic agents' bids is very simple; that is, the agent presents new bids, step by step by adding small amount of money until the maximum value which is determined beforehand, and the unsucceeded maximum value decides the farewell from the auction. Hence the performance of this type of automatic agents system is not flexible to various situations. A student of Tokyo University, Leo Ohtake started to introduce Q-learning method as only one learning method of one agent and other agent's learning method are conventional. From his simulation study, agents with Q-learning obtain better results than those of conventional agents. In the virtual auction system, we introduce our agents with SARSA learning system. This system contains agents with Q-learning and most of agents are conventional. SARSA learning is a variant of Q-learning. The objective our research is the comparison of the performances among agents in our virtual auction system (8 refs.)

Inspec No.: 8205526

3.2 Industries

88 Application of neural networks and orthogonal experimental design to blank design in metal forming processes

Author(s): Han Lifen, Li Guangyao, Zhong Zhihua, Gan Nianfei (**Affiliation:** Hunan Univ., China)

Journal: *China Mech. Eng. (China)*, vol.15, no.14, p.1266-71 (2004) **Publisher:** China Mech. Eng, China **Language:** Chinese **ISSN:** 1004-132X

Document type: Journal article

Abstract: A neural network for blank design of quadrature cup with flange was designed to simulate the nonlinear mapping relations among six process parameters and blank shape, in which the application of BP neural network and orthogonal experimental design was discussed. By means of the

application of orthogonal experimental design to the determination of training samples for the neural network training and structural parameters of the BP neural network, the required quantity of training samples and the number of calculation for optimal structural parameters of the BP network are decreased largely. Therefore, the design period can be shortened (10 refs.)

Inspec No.: 8201658

89 Intelligent aircraft maintenance support system using genetic algorithms and case-based reasoning

Author(s): C. Chiu, N.-H. Chiu (**Affiliation:** Dept. of Information Manage., Yuan-Ze Univ., Yuan Ze, Taiwan), C.-I. Hsu

Journal: *Int. J. Adv. Manuf. Technol. (UK)*, vol.24, no.5-6, p.440-6 (2004) **Publisher:** Springer-Verlag, UK **Language:** English **ISSN:** 0268-3768, [Full text](#)

Document type: Journal article

Abstract: The maintenance of aircraft components is crucial for avoiding aircraft accidents and aviation fatalities. To provide reliable and effective maintenance support, it is important for the airline companies to utilise previous repair experience with the aid of advanced decision support technology. Case-based reasoning (CBR) is a machine learning method that adapts previous similar cases to solve current problems. To effectively retrieve similar aircraft maintenance cases, this research proposes using a CBR system to aid electronic ballast fault diagnosis of Boeing 747-400 airplanes. By employing genetic algorithms (GA) to enhance dynamic weighting and the design of non-similarity functions, the proposed CBR system is able to achieve superior learning performance as compared to those with either equal/varied weights or linear similarity functions (16 refs.)

Inspec No.: 8203598

90 An agent-based dynamic information network for supply chain management

Author(s): H.J. Ahn (**Affiliation:** Sch. of Manage., KAIST, South Korea), H. Lee

Journal: *BT Technol. J. (UK)*, vol.22, no.2, p.18-27 (April 2004) **Publisher:** British Telecommunications plc, UK **Language:** English **ISSN:** 1358-3948, [Full text](#)

Document type: Journal article

Abstract: One of the main research issues in supply chain management is to improve the global efficiency of supply chains. However, the improvement efforts often fail because supply chains are complex, are subject to frequent changes, and collaboration and information sharing in the supply chains are often unfeasible. This paper presents a practical collaboration framework for supply chain management wherein multi-agent systems form dynamic information networks and co-ordinate their production and order planning according to synchronised estimation of market demands. In the framework, agents employ an iterative relaxation contract net protocol to find the most desirable suppliers by using data envelopment analysis. Furthermore, the chain of buyers and suppliers, from the end markets to raw material suppliers, form dynamic information networks for synchronised planning. This paper presents an agent-based dynamic information network for supply chain management and discusses the associated pros and cons (19 refs.)

Inspec No.: 8204601

91 Designing Web services with Tropos

Author(s): D. Lau, J. Mylopoulos (**Affiliation:** Dept. of Comput. Sci., Toronto Univ., Ont., Canada)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 306-13

Document type: Conference paper

Abstract: We propose a methodology for designing Web services. The methodology is founded on Tropos (Perini et al., 2001 and Castro, et al., 2002), an agent-oriented software development technique, and supports early and late requirements analysis, as well as architectural and detailed design. An online retailer example is used for illustration of the proposed methodology. We also compare the generated design with a sample design presented in [BPT01] (9 refs.)

Inspec No.: 8205428

92 An efficient parallel collision detection algorithm for virtual prototype environments

Author(s): M. Figueiredo, T. Fernando (**Affiliation:** Centre for Virtual Environ., Salford Univ., UK)

Editor(s): N.-F.Tzeng

Conference: Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004), Newport Beach, CA, USA **Conference Date:** 7-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Parallel Processing; IEEE Comput. Soc. Tech. Committee on Distributed Processing **Publication:** Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004) **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2152 5 **Page:** 249-56

Document type: Conference paper

Abstract: The automatic recognition of geometric constraints in virtual assembly and maintenance operations relies in the determination of intersecting surfaces between virtual prototypes. This is a key challenge in many virtual prototype applications, where it is necessary to find collisions precisely and interactively. This paper presents an algorithm to determine intersecting surfaces at interactive speed in a virtual prototyping environment. The proposed algorithm is based on the overlapping axis-aligned bounding box (OAABB). The OAABB concept is used effectively to eliminate the number of surfaces that cannot intersect and improve performance. The algorithm also facilitates the implementation using parallel computing methods. OpenMP is used, taking advantage of shared memory multiple processors and reducing the overall time complexity of the collision detection algorithm. To achieve an efficient parallel simulation, it is necessary to provide an efficient load balancing scheme. Our experiences in parallelising the code to achieve a better work distribution are also described. Results show that the proposed collision detection achieves interactive rates in real industrial applications as desired (10 refs.)

Inspec No.: 8206077

93 Assisting seller pricing strategy selection for electronic auctions

Author(s): Jia Zhang (**Affiliation:** Dept. of Comput. Sci., Northern Illinois Univ., DeKalb, IL, USA), Ning Zhang, Jen-Yao Chung

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 27-33

Document type: Conference paper

Abstract: This research examines sellers' pricing strategy oriented to bidders' risk preferences, within the context of an electronic auction. A business model is then proposed to predict the probability of potential bidders' risk preference, which sellers can consider to set pricing strategy to maximize their profits. A tailored intelligent auction registry engine is constructed based upon an initial version of our business model. Because the extensible markup language (XML)-based simple object access protocol (SOAP) is utilized to support interactions with other resources, our engine is decoupled from the underlying technology choices; therefore, it can be deployed to different platforms (18 refs.)

Inspec No.: 8206146

94 A Markov-decision-based price comparison problem for mobile agent-based Internet commerce system (MAGICS)

Author(s): P.P.Y. Lam, H.C.B. Chan (**Affiliation:** Dept. of Comput., Hong Kong Polytech. Univ., China)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 34-41

Document type: Conference paper

Abstract: Mobile agents are well suited for facilitating the consumer buying process on the Internet, particularly for comparing prices. In this paper, we present a mobile agent-based e-commerce system, called MAGICS, that addresses some issues concerning price comparisons. After providing an overview of the MAGICS architecture, we use the Markov decision theory to formulate a price comparison problem. The objective of the problem is to obtain an optimal decision policy in order to buy a product at the lowest expected cost, including the traveling costs. The problem was solved using the backward induction algorithm. Some analytical results are presented and discussed (17 refs.)

Inspec No.: 8206147

95 COSIMA^{B2B} - sales automation for e-procurement

Author(s): W. Kiessling, S. Fischer, S. Doring (**Affiliation:** Augsburg Univ., Germany)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 59-68

Document type: Conference paper

Abstract: We present a fully automated electronic sales agent for e-procurement portals. The key technologies for this

breakthrough are based on preferences modeled as strict partial orders, enabling a deep personalization of the B2B sales process. The interplay of several novel middleware components achieves to automate skills that so far could be executed only by a human vendor. As personalized search engine for XML based e-catalogs, we use Preference XPath; the Preference Presenter implements a sales psychology based presentation of search results, supporting various human sales strategies; the Preference Repository provides the management of situated long-term preferences; the flexible Personalized Price Offer and the multi-objective Preference Bargainer provide a personalized price fixing and the opportunity to bargain about the price of an entire product bundle, applying up/cross and down selling techniques. Our prototype COSIMA^{B2B}, supported by industrial partners, has been successfully demonstrated at a large computer fair (16 refs.)

Inspec No.: 8206150

96 The e-salesman system

Author(s): M.P. Ting, V. Seth, J. Gao (**Affiliation:** San Jose State Univ., CA, USA)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 353-7

Document type: Conference paper

Abstract: Current e-commerce Websites employ designs that are elaborate displays of product catalogs. The drawback is the presence of complex navigation and keyword searches to find products. Customers rely solely on their own decision to purchase a product. In addition, it's difficult to distinguish online retailers from one another based on the Web-front; the identity of the Web business and the portrayal of genuineness are lost. The human and personal element seen in traditional shopping is most often not present when shopping online. This paper discusses and proposes a solution to these problems. The e-salesman system (eSS) aims to simulate the human element of traditional shopping for online sales (8 refs.)

Inspec No.: 8206189

97 MACS: Multi-agent charter system

Author(s): M. Stula, D. Stipanicev, M. Bonkovic (**Affiliation:** Dept. for Modelling & Intelligent Syst., Split Univ., Croatia)

Journal: *WSEAS Trans. Comput. (Greece)*, vol.2, no.3, p.657-61 (July 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2750

Document type: Journal article

Abstract: Multi-agent systems (MAS) could be (and already are) applied in different areas. In LARIS-Laboratory for robotics and intelligent systems, FESB, University of Split, a number of experiments investigating multi-agent system capabilities are currently in progress. A yacht charter system can be realized as e-commerce system connecting charter providers and customers. Present-day market provides standard solutions in a form of a database with different querying tools (<http://www.croatia-yachtcharter.com>, <http://www.vip-yachting.com>, <http://www.kingcharters.com>). Multi-agent system lends

possibility for a charter system improvement and automation (10 refs.)

Inspec No.: 8203048

3.3 Education

98 Intelligent agents for distance learning

Author(s): A. Amandi, M. Campo, M. Armentano, L. Berdun (**Affiliation:** Fac. de Ciencias Exactas, UNICEN Campus Universitario, Buenos Aires, Argentina)

Journal: *Inf. Educ. (Lithuania)*, vol.2, no.2, p.161-80 (2003)

Publisher: Lithuanian Acad. Sci, Lithuania **Language:** English **ISSN:** 1648-5831

Document type: Journal article

Abstract: Distance learning involves a lot of work of human assistants. These assistants need to be connected for answering student doubts and questions. Intelligent agents can do part of this repetitive work because they can observe students interacting with educational courses, detect learning troubles of these students, and then suggest them some way for overcoming those troubles. However, a design problem appears with this promised possibility: how to connect educational applications with these agents. This paper presents a solution to this problem, in which both the capture of student's intentions and agent intervention for helping students are specified. These two architectural design points are defined as connection points. The first connection point is named student intentions. Student intentions define situations in which agents might help. This connection point depends on the user interface of the educational application that students are using; the agent needs to know the gestures that students could do for interpreting their intentions. The second connection point is named agent interventions. Agent interventions define the context in which agent might assist and the type of help that might give, like a suggestion or a warning. This solution is introduced in the context of one specific application for distance learning named SAVER, which is used for exemplifying each architectural design point (26 refs.)

Inspec No.: 8203796

99 Web-delivered supplemental instruction: dynamic customizing of search algorithms to enhance independent learning for developmental mathematics students

Author(s): I. Taksa (**Affiliation:** Dept. of Comput. Inf. Syst., New York City Univ., USA), R. Goldberg

Journal: *Math. Comput. Educ. (USA)*, vol.38, no.2, p.152-64 (Spring 2004) **Publisher:** MATYC Journal Inc, USA **Language:** English **ISSN:** 0730-8639

Document type: Journal article

Abstract: Traditional peer-to-peer supplemental instruction (SI) was introduced into higher education over a quarter of a century ago and promptly became an integral part of the developmental mathematics curricula in many senior and community colleges. Later, some colleges introduced video-based supplemental instruction (VSI) and, in recent years, Web-delivered supplemental instruction (WdSI), enhancing the delivery of SI. A major shortcoming of these new approaches is the fixed content of a prerecorded SI and its restricted availability. Furthermore, frequent changes in curricula and new or revised textbooks can quickly make SI incomplete. This research suggests a novel approach to a Web-delivered SI (WdSI), where the content and elucidation

level of the SI are based on the user-supplied topic description and are precisely selected from an enormous, continuously expanding and constantly available Internet collection. A meta search engine with dynamically adjustable search algorithms was designed and used to conduct experiments; the results demonstrated a high quality of on-demand delivery of relevant SI for any discipline or subject matter (16 refs.)

Inspec No.: 8201256

100 Native-speaker/non-native-speaker discourse in the MOO: topic negotiation and initiation in a synchronous text-based environment

Author(s): K. Schwienhorst (**Affiliation:** Trinity Coll. Dublin, Ireland)

Journal: *Comput. Assist. Lang. Learn. (Netherlands)*, vol.17, no.1, p.35-50 (Feb. 2004) **Publisher:** Swets & Zeitlinger, Netherlands **Language:** English **ISSN:** 0958-8221, [Full text](#)

Document type: Journal article

Abstract: A number of researchers in computer-mediated communication have pointed towards its potential to stimulate learner participation and engagement in the classroom. However, in many cases only anecdotal reports were provided. In addition, it is unclear whether the pedagogical set-up or the technology involved is responsible for changes in learner attitudes. We have conducted bilingual MOO (object-oriented multiple-user domain) projects based on tandem learning for the past three years, and collected extensive learner data on native-speaker/ nonnative-speaker discourse during that time. We look in detail at issues such as topic negotiation and initiation, both in the learner's L1 and L2. For instance, some researchers on native-speaker/nonnative-speaker discourse have claimed that native speakers initiate the majority of topics, and that nonnative speakers preferably initiate a topic by using questions. An analysis of learner data is compared to existing research on face-to-face native-speaker/ nonnative-speaker discourse. This comparison provides us with an empirical basis to assess whether a combination of tandem learning and a synchronous communication technology like the MOO can support learner participation and engagement in the language classroom. We also discuss whether the technology or the pedagogy is responsible for any differences in learner engagement (27 refs.)

Inspec No.: 8201665

101 Incorporating the use of e-mail into a language program

Author(s): E. Torii-Williams (**Affiliation:** Wellesley Coll., MA, USA)

Journal: *Comput. Assist. Lang. Learn. (Netherlands)*, vol.17, no.1, p.109-22 (Feb. 2004) **Publisher:** Swets & Zeitlinger, Netherlands **Language:** English **ISSN:** 0958-8221, [Full text](#)

Document type: Journal article

Abstract: Until recently, most foreign language education has been centered on the acquisition of grammar and vocabulary. While recognizing that grammar and vocabulary are essential tools for communication, the national standards for foreign language learning advocate "the acquisition of the ability to communicate in meaningful and appropriate ways with users of other languages" (1996) and identify five goals of today's foreign language classroom: communication, cultures, connections, comparisons, and communities. The present study attempts to investigate how the use of e-mail in the target language motivates and enhances the students'

learning of the language based on the five goals of the national standards. Students in a post secondary Japanese class were asked to communicate by e-mail in Japanese. At the end of the course, the students completed questionnaires regarding the project. The researcher examined their messages and the questionnaires. The study describes how an out of class activity with a focus on meaning can not only provide a great number of learning opportunities as described in the national standards, but also promote a deeper understanding of the Japanese language and culture. Moreover, it illustrates how a standards-based project with an emphasis on meaning can assist in the developmental and creative aspects of second language acquisition (15 refs.)

Inspec No.: 8201666

102 Instructional systems, learning sciences, design, technology: a more complete pathway

Author(s): A.A. Carr-Chellman (**Affiliation:** Dept. of Learning & Performance Syst., Pennsylvania State Univ., University Park, PA, USA)

Journal: *Educ. Technol. (USA)*, vol.44, no.3, p.40-4 (May-June 2004) **Publisher:** Educational Technology Publications, USA **Language:** English **ISSN:** 0013-1962

Document type: Journal article

Abstract: This article attempts to look first at those issues, which may distract us from a more fruitful pathway, more significant collaborative discourse. These issues include what a field is, what is not, reliance on theory and practice, the nature of collaboration and insularity, and the importance of research to both fields. The field of instructional design is beginning to realize that the cognitive models on which the learning sciences is built tend to be more robust than some of the more dated models. A first step toward a joint path may be a couple of stepping stones which recognize the differences in preparation and the ability for both communities to learn from one another's historical theories. The most significant discussion aimed at creating a more worn path between these two communities involves deeper understandings of the terms science, technology, and design (11 refs.)

Inspec No.: 8201881

103 The science of instruction and the technology of instructional design

Author(s): M.D. Merrill (**Affiliation:** Dept. of Instructional Technol., Utah State Univ., Logan, UT, USA)

Journal: *Educ. Technol. (USA)*, vol.44, no.3, p.45-6 (May-June 2004) **Publisher:** Educational Technology Publications, USA **Language:** English **ISSN:** 0013-1962

Document type: Journal article

Abstract: The technology of instructional design, like other technologies, is not a natural phenomenon. It is man made, designed to serve our needs. Design research involves inventing procedures and processes, which incorporate what we learn from instructional science. Instructional science is concerned with the discovery of the natural principles involved in instructional strategies; and instructional design is the use of these scientific principles to invent instructional design procedures and tools. This article suggests that the learning sciences are the study of learning (cognition) from instruction (design) in social contexts. Instructional design, on the other hand, is concerned with the efficient and effective design and development of instructional products and environments that accomplish specific learning goals (9 refs.)

Inspec No.: 8201882

104 Instructional technology and the learning sciences: multiple communities and political realities

Author(s): J.M. Spector (**Affiliation:** Syracuse Univ., NY, USA)
Journal: *Educ. Technol. (USA)*, vol.44, no.3, p.47-9 (May-June 2004) **Publisher:** Educational Technology Publications, USA
Language: English **ISSN:** 0013-1962
Document type: Journal article

Abstract: The article discusses differences between two professional communities: the learning sciences community and the instructional technology community. This discussion of increased articulation among instructional and learning researchers should continue. Issues include common interests (improving learning and instruction), multiple perspectives, and the political environment in schools, universities, and society. From a personal point of view, it is nearly always enriching and enlightening to engage in multiple-disciplinary work

Inspec No.: 8201883

105 A model of multitutor ontology-based learning environments

Author(s): A. Mitrovic (**Affiliation:** Dept. of Comput. Sci., Canterbury Univ., New Zealand), V. Devedzic
Journal: *Int. J. Contin. Eng. Educ. Life-Long Learn. (Switzerland)*, vol.14, no.3, p.229-45 (2004) **Publisher:** Inderscience Enterprises, Switzerland **Language:** English
ISSN: 0957-4344

Document type: Journal article

Abstract: The paper proposes the M-OBLIGE model for building multitutor ontology-based learning environments. The model is based on local ontologies, describing the domain of each individual tutor in the environment, and external ontologies, describing more general concepts. The ontologies are used by ontology processors to decide which tutors might benefit a student who needs to learn new concepts. The model allows domain expertise to be shared, and can be used as a framework for integrating multiple tutors on the Web. We show how the model can be applied to tutors in the database domain. We also illustrate the process of developing the ontologies for an existing system (31 refs.)

Inspec No.: 8202247

106 Using links to describe imprecise relationships in educational contents

Author(s): M.-A. Sicilia, E. Garcia (**Affiliation:** Comput. Sci. Dept., Univ. of Alcala, Spain), P. Diaz, I. Aedo
Journal: *Int. J. Contin. Eng. Educ. Life-Long Learn. (Switzerland)*, vol.14, no.3, p.260-75 (2004) **Publisher:** Inderscience Enterprises, Switzerland **Language:** English
ISSN: 0957-4344

Document type: Journal article

Abstract: Hypermedia links describe purposeful relationships between contents, and in consequence, their semantic intention can be modelled by the notion of link type. Ontologies can be used to extend that notion of link type, eventually resulting in shared conceptualisations useful in achieving consistent linking practices. In addition, the relationship embodied by the link is in many cases inherently imprecise, which has been overlooked in existing systems. In this work, we describe how ontologies of link types with support for expressing imprecision can be used to enhance educational Web contents. The computational semantics of the ontology, combined with fuzziness to specify graded relations between pairs of link sources and targets for specific

users or groups, enable the implementation of extended adaptive behaviours that may entail inferences and approximate reasoning. An implementation, based on an SCORM-compliant runtime environment, of such kind of extended links is also described, demonstrating their capabilities in the definition of learning-oriented adaptations (41 refs.)

Inspec No.: 8202249

107 A design environment to articulate design intention of learning contents

Author(s): Y. Hayashi, M. Ikeda (**Affiliation:** Sch. of Knowledge Sci., Japan Adv. Inst. of Sci. & Technol., Japan), R. Mizoguchi
Journal: *Int. J. Contin. Eng. Educ. Life-Long Learn. (Switzerland)*, vol.14, no.3, p.276-96 (2004) **Publisher:** Inderscience Enterprises, Switzerland **Language:** English
ISSN: 0957-4344

Document type: Journal article

Abstract: Design of learning contents is more difficult to support than design of a physical substance because its object is abstract and it is difficult to establish a framework to express it appropriately. To address this issue, we converted an abstract concept about learning content design to an ontology. Then, on the basis of that ontology, we developed iDesigner, a design support environment. iDesigner realises the following two points: (1) the designers are compelled to make the implicit results of their work explicit in order to deepen thinking about the design of an abstract matter (2) the designers are provided with basic information to verify content validity by simulating the change of understanding of a learner in the learning contents model at a conceptual level, which is the intermediate result of design (15 refs.)

Inspec No.: 8202250

108 A constructivist approach to online college learning

Author(s): A.P. Rovai (**Affiliation:** Sch. of Educ., Regent Univ., Virginia Beach, VA, USA)
Journal: *Internet High. Educ. (USA)*, vol.7, no.2, p.79-93 (2004) **Publisher:** Elsevier, USA **Language:** English **ISSN:** 1096-7516, [Full text](#)

Document type: Journal article

Abstract: The key elements of online course design and pedagogy suggested by research as promoting effective learning are discussed through the lens of constructivist epistemology. Presentation of content, instructor-student and student-student interactions, individual and group activities, and student assessment are each addressed, in turn. The focus is on learning and recognition that, from time-to-time, all students are teachers as they bring diverse expertise, experiences, and worldviews to the task of learning. Reflection on past experiences, interaction with other members of the learning community, immediate instructor behavior, authentic group activities, and diverse assessment tasks with timely and detailed feedback are underscored (47 refs.)

Inspec No.: 8202489

109 Blended learning: uncovering its transformative potential in higher education

Author(s): D.R. Garrison, H. Kanuka (**Affiliation:** Learning Commons, Calgary Univ., Alta., Canada)

Journal: *Internet High. Educ. (USA)*, vol.7, no.2, p.95-105 (2004) **Publisher:** Elsevier, USA **Language:** English **ISSN:** 1096-7516, [Full text](#)

Document type: Journal article

Abstract: The purpose of this paper is to provide a discussion of the transformative potential of blended learning in the context of the challenges facing higher education. Based upon a description of blended learning, its potential to support deep and meaningful learning is discussed. From here, a shift to the need to rethink and restructure the learning experience occurs and its transformative potential is analyzed. Finally, administrative and leadership issues are addressed and the outline of an action plan to implement blended learning approaches is presented. The conclusion is that blended learning is consistent with the values of traditional higher education institutions and has the proven potential to enhance both the effectiveness and efficiency of meaningful learning experiences (18 refs.)

Inspec No.: 8202490

110 Distance learning versus being there

Author(s): S. Reisman (**Affiliation:** Dept. of Inf. Syst. & Decision Sci., California State Univ., Fullerton, CA, USA)

Journal: *IT Prof. (USA)*, vol.6, no.5, p.8-10 (Sept.-Oct. 2004)

Publisher: IEEE, USA **Language:** English **ISSN:** 1520-9202, [Full text](#)

Document type: Journal article

Abstract: The Chronicle of Higher Education recently reported that "the boom in educational technology has not lived up to its promise of revolutionizing the classroom and making higher education more profitable" (Carnevale, 2004). The Chronicle later highlighted a study showing that "while the innovations have made courses more convenient, the spending [on IT in the classroom] has yet to have a large impact on learning" (Young, 2004). On the other hand, this same article states that "of the students surveyed, 76.1% said their experience using a course-management system was positive or very positive, with only 6.6% saying it was negative or very negative (17.3% were neutral on the issue)." So how effective is online versus traditional instruction? The author's personal observation is that his online students performed about as well as my traditional students on all their assignments, tests, and exercises. Different types of practice materials were available to the different classes, each appropriate to the teaching or learning situation in use. However, despite studies on instructional methods, there has never been a definitive answer to the question

Inspec No.: 8202590

111 The next generation of e-learning strategies for media rich online teaching and engaged learning

Author(s): Chye Seng Lee, Daniel Tiong Hok Tan, Wee Sen Goh (**Affiliation:** Nanyang Technol. Univ., Singapore, Singapore)

Journal: *Int. J. Distance Educ. Technol. (USA)*, vol.2, no.3, p.1-17 (Oct.-Dec. 2004) **Publisher:** Idea Group Publishing, USA **Language:** English **ISSN:** 1539-3100

Document type: Journal article

Abstract: In a short span of three years, the Nanyang Technological University (NTU) in Singapore witnessed significant growth in the adoption of e-learning. With the use of professors-friendly e-learning applications, NTU has been able to achieve critical mass buy-in by the academic staff when the e-learning take-up rate achieved 85% of the existing NTU

course curriculum. As NTU moves on to celebrate the third year of e-learning, measures were taken with the careful design considerations that aimed to "humanize" e-learning, that is, make e-learning interactive and engaging with active collaborations and student learning involvement. This includes the proliferation in the use of the video talking head format synchronized with the lecture presentation, live audio-video delivery, text chat and document annotations of a lecture presentation and delivery. This paper reviews the processes NTU adopted in adding the human touch to traditional e-learning projects and serves as a good case study for other institutions with a similar aim to achieve interactive and engaged online learning (13 refs.)

Inspec No.: 8204453

112 An e-learning system based on the top-down method and the cellular models

Author(s): N. Fujii, S. Yukita, N. Koike (**Affiliation:** Hosei Univ., Tokyo, Japan), T.L. Kunii

Journal: *Int. J. Distance Educ. Technol. (USA)*, vol.2, no.3, p.77-93 (Oct.-Dec. 2004) **Publisher:** Idea Group Publishing, USA **Language:** English **ISSN:** 1539-3100

Document type: Journal article

Abstract: As the broadband connectivity to the Internet becomes common, Web based e-learning and distance learning have come to play the central roles for self-learning, where learners are given much flexibility in choosing place and time to study. However, the learners still have to spend a lot of time before reaching the learning goal. This discourages the learners from continuing their studies and diminishes their motivation. To overcome this problem and to let the learners keep focusing on their primary interests, we propose a top-down e-learning system called TDeLS. The TDeLS can offer learners the learning materials based on the top-down (i.e., goal-oriented) method, according to the learners' demands and purposes. Moreover, the TDeLS can distribute them to the learners through the Internet, and manage the database for learning materials. In order to share learning materials among learners through the Web, these learning materials are wrapped in XML with a specially designed vocabulary for TDeLS. We employed the cellular models that ensure the consistency among design modules and support a top-down design methodology. In this paper, we present the TDeLS for hardware logic design courses based on the cellular models. The primary goal is to design complex logic circuits in Verilog HDL, which is an industrial-standard hardware description language. This paper also presents the basic XML vocabulary designed to describe hardware modules efficiently, and a brief introduction to the structure and functions of the proposed system that implements the TDeLS (15 refs.)

Inspec No.: 8204454

113 Network virtual laboratory for external devices programming

Author(s): R. Klempous, J. Nikodem, T. Walkowiak (**Affiliation:** Inst. of Eng. Cybern., Wroclaw Univ. of Technol., Poland), J. Rozenblit

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol. **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based

Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 293-8

Document type: Conference paper

Abstract: The paper describes an approach to replace traditional computer laboratories by Internet virtual ones. The purpose of this laboratory is to make available advanced learning contents in a network environment. Devices available for students are fully simulated by computers. Furthermore, remote access to programming environment is implemented, which gives the user an ability to write programs controlling a virtual device. The paper describes all elements of this system (12 refs.)

Inspec No.: 8205237

114 Interfaces that adapt like humans

Author(s): S. Alexander, A. Sarrafzadeh (**Affiliation:** Inst. of Inf. & Math. Sci., Massey Univ., New Zealand)

Editor(s): M.Masoodian, S.Jones, B.Rogers

Conference: Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings, Rotorua, New Zealand **Conference Date:** 29 June-2 July 2004 **Sponsor(s):** Dept. of Comput. Sci., Univ. of Waikato; Univ. of Waikato Usability Lab.; IFIP Tech. Committee No. 13 on Human-Comput. Interaction (IFIP TC.13); AMC SIGCHI, New Zealand Chapter (SIGCHI-NZ); COMPUTERWORLD; Human Interface Technol. Lab. New Zealand (HITLabNZ); Orbit Corporate Travel **Publication:** Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3101) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 22312 6 **Page:** 641-5

Document type: Conference paper

Abstract: Whenever people talk to each other, non-verbal behavior plays a very important role in regulating their interaction. However, almost all human-computer interactions take place using a keyboard or mouse - computers are completely oblivious to the non-verbal behaviour of their users. This paper outlines the plan for an interface that aims to adapt like a human to the non-verbal behaviour of users. An intelligent tutoring system (ITS) for counting and addition is being implemented in conjunction with the New Zealand Numeracy project. The system's interface detects the student's non-verbal behaviour using in-house image processing software, enabling it to adapt to the student's non-verbal behaviour in similar ways to a human tutor. We have conducted a video study of how human tutors interpret the non-verbal behaviour of students, which has laid the foundation for this research (10 refs.)

Inspec No.: 8205723

115 Understanding interaction experience in mobile learning

Author(s): F.H.A. Razak (**Affiliation:** Dept. of Comput., Lancaster Univ., UK)

Editor(s): M.Masoodian, S.Jones, B.Rogers

Conference: Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings, Rotorua, New Zealand **Conference Date:** 29 June-2 July 2004 **Sponsor(s):** Dept. of Comput. Sci., Univ. of Waikato; Univ. of Waikato Usability Lab.; IFIP Tech. Committee No. 13 on Human-Comput. Interaction (IFIP TC.13); AMC SIGCHI, New Zealand Chapter (SIGCHI-NZ); COMPUTERWORLD; Human Interface Technol. Lab. New Zealand (HITLabNZ); Orbit Corporate Travel **Publication:** Computer Human Interaction. 6th Asia

Pacific Conference, APCHI 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3101) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 22312 6 **Page:** 672-4

Document type: Conference paper

Abstract: The convergence of mobile technology and e-learning has generated considerable excitement among both practitioners and academics. Mass media continually promotes an idea about m-learning. Content developers also continue to deliver learning on wireless, mobile learning devices based on their often abstract conceptions of what the 'generalised' learners might want to learn. We are now seeing the adoption of e-learning into m-learning without a clear understanding of motivations and circumstances surrounding m-learning and m-learners. This research looks into interaction experience that is believed to play a significant role in the development of usable m-learning (3 refs.)

Inspec No.: 8205726

116 Adaptive access to a proof planner

Author(s): E. Melis, A. Meier (**Affiliation:** German Res. Inst. for Artificial Intelligence, Saarbrücken, Germany), M. Pollet

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Białowieża, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Białystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 251-64

Document type: Conference paper

Abstract: Mathematical tools such as computer algebra systems and interactive and automated theorem provers are complex systems and can perform difficult computations. Typically, such tools are used by a (small) group of particularly trained and skilled users to assist in mathematical problem solving. They can also be used as back-engines for interactive exercises in learning environments. This, however, suggests the adaptation of the choice of functionalities of the tool to the learner. This paper addresses the adaptive usage of the proof planner Multi for the learning environment ActiveMath. The proof planner is a back-engine for interactive proof exercises. We identify different dimensions in which the usage of such a service system can be adapted and investigate the architecture realizing the adaptive access to Multi (13 refs.)

Inspec No.: 8206326

117 The role of cooperation in creating a library online tutorial

Author(s): E.M. Diel, T.K. Flett (**Affiliation:** St. Charles Community Coll., St. Peters, MO, USA)

Journal: *Ref. Libr. (USA)*, no.83-84, p.175-82 (2003)

Publisher: Haworth Press, USA **Language:** English **ISSN:** 0276-3877, [Full text](#)

Document type: Journal article

Abstract: Reference librarians at St. Charles Community College discovered the benefits of cooperation when they began the process of creating an online library tutorial. In the fall of 1999, librarians realized that their walk-in library sessions were becoming ineffective due to poor attendance by students. The tours didn't fit into on-campus students' busy schedules, and failed to serve distance students at all. Two

reference librarians decided to work together to create a Web-based tutorial introducing students to library online resources. It was an informal process that started with information gathering on what tutorials exist, how they were developed and what type of software was used. After the librarians decided on the format and appropriate software, they brought together their creative and technical strengths to design an appealing and functional tutorial. To create a "virtual tour" of the library's physical layout, the librarians also collaborated with the Instructional Support Center, a group of educational technology specialists who are part of SCC's community college consortium. The tutorial was completed in only a year, partly because working in a small library allows for constant contact between the librarians, but also due to the efficiency of using cooperation. This successful collaborative project eventually won the Missouri Community College Association's 2001 Technology Innovation Award (7 refs.)

Inspec No.: 8201826

118 Librarians as researchers and the digital research library

Author(s): R. Wallach (**Affiliation:** Archit. & Fine Arts Libr., Southern California Univ., USA)

Journal: *New Rev. Inf. Libr. Res. (UK)*, vol.9, p.135-43 (2003)

Publisher: Taylor Graham Publishing, UK **Language:** English **ISSN:** 1361-455X, [Full text](#)

Document type: Journal article

Abstract: This paper discusses the changing relationship between academic institutions and their libraries, particularly in areas of knowledge management. It also discusses the role of the librarian/researcher, and posits the possibility that this role and the librarian's relationship to the larger institution may be changing. The paper uses as case study the Public Art in Los Angeles Web site, developed by the author. The project, which is ongoing, started in 1996 out of curiosity was there public art in Los Angeles, one of the world's largest cities, which grew exponentially in the 20th century, and was heavily car-oriented? This on-going project is driven by the interests of an individual librarian, with some support from the institution. Issues raised in the paper are: (1) Who owns the digital and the print information? (2) Who controls and provides access? (3) What is the usage of the collection? (7 refs.)

Inspec No.: 8202629

119 Seeking or skipping regularities? Novice tendencies and the role of invariants

Author(s): D. Ginat (**Affiliation:** Dept. of Sci. Educ., Tel-Aviv Univ., Israel)

Journal: *Inf. Educ. (Lithuania)*, vol.2, no.2, p.211-22 (2003)

Publisher: Lithuanian Acad. Sci, Lithuania **Language:** English **ISSN:** 1648-5831

Document type: Journal article

Abstract: Every repetitive process encapsulates a regularity pattern, which may be expressed as an invariant assertion. Invariants embody implicit, insightful properties that characterize the execution of programming statements. Due to their implicit nature, invariants may be less apparent to algorithmic problem solvers. Yet, invariants are essential for designing correct and efficient algorithms. This paper illustrates the essential role of invariants, and examines whether novices tend to look for invariant properties during their algorithmic problem solving. The paper presents a study in which two novel algorithmic challenges were displayed to a group of motivated, novice students. Student solutions to

these challenges demonstrate an operational reasoning approach, which does not capture the essence of the problems at hand, and yields non-satisfying results. Some solutions were incorrect, others were inefficient, and some had no convincing justification. These results, and the correct and efficient solutions to both challenges illuminate the importance of assertional reasoning and the fundamental role of invariants (11 refs.)

Inspec No.: 8203797

3.4 Medicine

120 Prediction of membrane protein types by using support vector machine

Author(s): Guo Zong-ming, Zhang Zhi-zhou, Pan Yu-xi, Huang Zhen-de, Feng Guo-yin, He Lin (**Affiliation:** Bio-X Life Sci. Res. Center, Shanghai Jiaotong Univ., China)

Journal: *J. Shanghai Jiaotong Univ. (China)*, vol.38, no.5, p.806-9 (May 2004) **Publisher:** Shanghai Jiaotong University Press, China **Language:** Chinese **ISSN:** 1006-2467

Document type: Journal article

Abstract: Support vector machine (SVM) based on statistical learning theory was applied for the classification of five membrane protein types by self-consistency, jackknife and independent dataset test methods based on the training dataset of 2 059 membrane proteins and the test dataset of 2 625 membrane proteins derived from the article according to their amino acid compositions. The results obtained by the three typical test methods of statistical prediction field are quite promising, indicating the current approach can play a complementary role to the other existing methods for predicting the types of membrane proteins (15 refs.)

Inspec No.: 8203791

121 Binocular disparity encoding cells generated through an infomax based learning algorithm

Author(s): K. Okajima (**Affiliation:** Syst. Device & Functional Res., NEC Corp., Tsukuba, Japan)

Journal: *Neural Netw. (UK)*, vol.17, no.7, p.953-62 (Sept. 2004) **Publisher:** Elsevier, UK **Language:** English **ISSN:** 0893-6080, [Full text](#)

Document type: Journal article

Abstract: A learning algorithm for a model binocular cell was derived according to an information maximization principle and by using a low signal-to-noise-ratio approximation. The algorithm updates cell's synaptic weights so that the information obtained from the cell's output is increased. According to the algorithm, model binocular cells were trained by using computer-generated stereo images as training data. As a result, cells tuned to various disparities were generated. Also, generated synaptic weight patterns of the cells were similar to Gabor-wavelets and receptive fields of simple cells in the visual cortex. Thus, they were orientation and spatial frequency selective as well as disparity selective. Gabor functions were used to fit the generated weight patterns. The fitting results indicated that the generated cells encode disparities in terms of phase disparity and/or position disparity. This result agrees with experimental findings by Anzai et al. and is consistent with ICA-based theoretical results obtained (30 refs.)

Inspec No.: 8203972

122 Rough set approach for generation of classification rules of breast cancer data

Author(s): A.E. Hassanien, J.M.H. Ali (**Affiliation:** Quantitative Methods & Inf. Syst. Dept., Kuwait Univ., Safat, Kuwait)

Journal: *Informatica (Lithuania)*, vol.15, no.1, p.23-38 (2004)

Publisher: Lithuanian Acad. Sci, Lithuania **Language:** English **ISSN:** 0868-4952

Document type: Journal article

Abstract: Extensive amounts of knowledge and data stored in medical databases require the development of specialized tools for storing, accessing, analysis, and effectiveness usage of stored knowledge and data. Intelligent methods such as neural networks, fuzzy sets, decision trees, and expert systems are, slowly but steadily, applied in the medical fields. Recently, rough set theory is a new intelligent technique was used for the discovery of data dependencies, data reduction, approximate set classification, and rule induction from databases. We present a rough set method for generating classification rules from a set of observed 360 samples of the breast cancer data. The attributes are selected, normalized and then the rough set dependency rules are generated directly from the real value attribute vector. Then the rough set reduction technique is applied to find all reducts of the data which contains the minimal subset of attributes that are associated with a class label for classification. Experimental results from applying the rough set analysis to the set of data samples are given and evaluated. In addition, the generated rules are also compared to the well-known IDS classifier algorithm. The study showed that the theory of rough sets seems to be a useful tool for inductive learning and a valuable aid for building expert systems (29 refs.)

Inspec No.: 8204839

123 Agent architecture for smart adaptive systems

Author(s): L. Lhotska, O. Stepankova (**Affiliation:** Dept. of Cybern., Czech Tech. Univ., Prague, Czech Republic)

Journal: *Trans. Inst. Meas. Control (UK)*, vol.26, no.3, p.245-60 (2004)

Publisher: Inst. Meas. Control, UK **Language:** English **ISSN:** 0142-3312, [Full text](#)

Document type: Journal article

Abstract: This paper discusses the possibility of utilizing a multiagent system as a platform for development of a smart adaptive system in medicine. A knowledge-based model of the agents' mutual awareness (social knowledge) is presented. The tri-base acquaintance (3bA) model formalizing the agent's social behaviour knowledge and co-operation neighbourhood knowledge is enhanced to support attention focussing and to ensure a holistic complex perception of the problem. The paper focuses on an analysis of the content of 3bA knowledge bases for applications in medical decision support. An architecture for a multiagent system, ADAM, for medical decision support and monitoring is proposed (18 refs.)

Inspec No.: 8205121

124 Public health situation awareness, towards a semantic approach

Author(s): P. Mirhaji, R.L. Richesson, J.P. Turley, Jiajie Zhang, J.W. Smith (**Affiliation:** Center for Biosecurity & Public Health Informatics Res., Univ. of Texas Health Sci. Center, Houston, TX, USA)

Conference: Multisensor, Multisource Information Fusion: Architectures, Algorithms, and Applications 2004, Orlando, FL, USA **Conference Date:** 14-15 April 2004 **Journal:** *Proc. SPIE - Int. Soc. Opt. Eng. (USA)*, vol.5434, no.1, p.339-50 (2004)

Publisher: SPIE-Int. Soc. Opt. Eng, USA **Language:** English

ISSN: 0277-786X, [Full text](#)

Document type: Conference paper in journal

Abstract: We propose a knowledge-based public health situation awareness system. The basis for this system is an explicit representation of public health situation awareness concepts and their interrelationships. This representation is based upon the users' (public health decision makers) cognitive model of the world, and optimized towards the efficacy of performance and relevance to the public health situation awareness processes and tasks. In our approach, explicit domain knowledge is the foundation for interpretation of public health data, as apposed to conventional systems where the statistical methods are the essence of the processes. To develop a prototype knowledge-based system for public health situation awareness and to demonstrate the utility of knowledge intensive approaches in integration of heterogeneous information, eliminating the effects of incomplete and poor quality surveillance data, uncertainty in syndrome and aberration detection and visualization of complex information structures in public health surveillance settings, particularly in the context of bioterrorism (BT) preparedness. The system employs the resource definition framework (RDF) and additional layers of more expressive languages to explicate the knowledge of domain experts into machine interpretable and computable problem-solving modules that can then guide users and computer systems in sifting through the most "relevant" data for syndrome and outbreak detection and investigation of root cause of the event. The Center for Biosecurity and Public Health Informatics Research is developing a prototype knowledge-based system around influenza, which has complex natural disease patterns, many public health implications, and is a potential agent for bioterrorism. The preliminary data from this effort may demonstrate superior performance in information integration, syndrome and aberration detection, information access through information visualization, and cross-domain investigation of the root causes of public health events (36 refs.)

Inspec No.: 8201397

125 Towards intelligent Web services for automating medical service composition

Author(s): Yugyung Lee, C. Patel (**Affiliation:** Sch. of Comput. & Eng., Missouri Univ., Kansas, MO, USA), Soon Ae Chun, J. Geller

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 384-91

Document type: Conference paper

Abstract: The vision of the Semantic Web is to reduce manual discovery and usage of Web resources (documents and services) and to allow software agents to automatically identify these Web resources, integrate them and execute them for achieving the intended goals of the user. Such a composed Web service may be represented as a workflow, called service flow. Current studies of Web services are not sufficient for automatic composition. This paper presents different types of compositional knowledge required for Web service discovery and composition: syntactic, semantic and pragmatic knowledge. As a proof of concept, we have implemented our framework in a cardiovascular domain which requires

advanced service discovery and composition across heterogeneous platforms of multiple organizations. Within this framework, we describe (1) How to represent the compositional knowledge, which plays a role in service discovery and composition, in DAML-S; (2) How heterogeneous medical services interoperate in a composed medical service flow. (3) To solve this knowledge level integration, we build on an ontology integration method called SEMIO (Semantic Interoperability) (16 refs.)

Inspec No.: 8205437

126 Promoting agent-based knowledge discovery in medical intensive care units

Author(s): A. Lourenco, O. Belo (**Affiliation:** Campus de Gualtar, Univ. do Minho, Braga, Portugal)

Journal: *WSEAS Trans. Comput. (Greece)*, vol.2, no.2, p.403-8 (April 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2750

Document type: Journal article

Abstract: Decision support has become vital in major scientific and organisational areas. Data analysis and knowledge acquisition are no longer means to an end, they have become the end itself. Knowledge discovery is a priority, constantly demanding for new, better suited efforts. Systems or tools capable of dealing with the steadily growing amount of data presented by information systems and digging through their structure, relations, contexts and ultimately, their contents, are in order. In this paper, an agent-based decision support system based in data warehousing systems contents is described, analysing its actual potential and skills in knowledge discovery as a whole and, in particular, in data preparation for data mining. Different requirements, algorithm dependent and nondependent, are pointed, analysing valid approaches. An algebraic model is used to specify the core of some of the embraced steps, as well as, to demarcate the data flow throughout the knowledge acquisition process. A case study is presented in order to demonstrate the application of the system to real world scenarios (10 refs.)

Inspec No.: 8203035

3.5 Information Science

127 Thesaurus and ontology, or formalization of the generic inheritance of conceptual features

Author(s): R. Ungvary (**Affiliation:** Orszagos Szechenyi Konyvtar, Hungary)

Journal: *Tud. Musz. Taj. (Hungary)*, vol.51, no.5, p.175-91 (2004) **Publisher:** OMIKK, Hungary **Language:** Hungarian **ISSN:** 0041-3917

Document type: Journal article

Abstract: Thesauri and classification systems are traditionally used in search and retrieval of content of documents (sources). One of their most important components is the generic chain of hierarchy of concepts (classes) in which the features of the concept of a more general meaning (a more comprehensive class) are "inherited" by the more specific concepts (classes). In the wake of aspirations toward working out the semantic Web there appeared the so-called ontologies which are made up of the generic chains of hierarchy of concepts (classes) and rules formulated according to a logic of first order linked up with them. A key task of these is to secure the aforementioned inheritance of the generic hierarchy and to enable the drawing of conclusions with it. Ontologies can indirectly be traced back, for one, to Aristotle's system of

categories and, for another, to Ranganathan's multidimensional theory of classification, a stimulus to modern classification, and by virtue of the latter, to cultures of the Far East as well. Ontologies are employed in expert systems and knowledge bases in order to provide for an information retrieval more automated in terms of semantics. Their proposal for standardization was also compiled early in 2003 (OWL). In the concluding part of the study an example of application for thesauri and classification systems is presented (27 refs.)

Inspec No.: 8205049

128 Team-based agents for proactive failure handling in dynamic composition of Web services

Author(s): Xiaocong Fan, Karthikeyan Umapathy, J. Yen, Sandeep Purao (**Affiliation:** Sch. of Inf. Sci. & Technol., Penn State Univ., University Park, PA, USA)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 782-5

Document type: Conference paper

Abstract: Currently Web services composition problems are addressed using AI planning techniques. The team-based approach, with emphases on the sharing of mental models and proactive collaboration, provides an alternative to current static approaches to Web service composition. The approach provides clear advantages for proactive handling of failures that may be encountered during execution of a complex Web service. The paper proposes a generic framework for dynamic Web-service composition, and extends the CAST architecture to realize the framework (7 refs.)

Inspec No.: 8205489

129 Towards building logical views of Web sites

Author(s): Zehua Liu, Wee Keong Ng, Ee-Peng Lim (**Affiliation:** Centre for Adv. Inf. Syst., Nanyang Technol. Univ., Singapore), Feifei Li

Journal: *Data Knowl. Eng. (Netherlands)*, vol.49, no.2, p.197-222 (May 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0169-023X, [Full text](#)

Document type: Journal article

Abstract: Information presented in a Web site is usually organized into certain logical structure that is intuitive to users. It would be useful to model Web sites with such logical structure so that extraction of Web data from these sites can be performed in a simple and efficient manner. However, the recognition and reconstruction of such logical structure by software agent is not straightforward due to the complex hyper-link structure among Web pages and the HTML formatting within each Web page. We propose the WICCAP data model, a data model that maps Web sites from their physical structure into commonly perceived logical views. To enable easy and rapid creation of such data models, we have implemented a visual tool, called the Mapping Wizard, to facilitate and automate the process of producing WICCAP data models. Using the tool, the time required to construct a logical representation for a given Web site is significantly reduced (25 refs.)

Inspec No.: 8203455

130 WICCAP: from semi-structured data to structured data

Author(s): Zhao Li (**Affiliation:** Sch. of Comput. Eng., Nanyang Technol. Univ., Singapore), Wee Keong Ng

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 86-93

Document type: Conference paper

Abstract: Web data extraction is a technique for extracting and integrating data from Web based semistructured data. Wrappers function like the kernel of Web data extraction systems providing information mediator between users and a large number of heterogeneous data sources. Typically, they process semistructured documents generated from structured databases based on rules that are usually hidden to users. Much research has been done to use various methods to represent the knowledge of hidden rules and exploit techniques such as grammar induction, inductive logic programming, etc., to discover these rules that can be used by wrappers to extract data. An important property of semistructured data is its hierarchical structure. Intuitively, we can devise a method that can use this structure information to generate wrappers. We describe a Web data extraction system - WICCAP and its internal Web Data Extraction Language (WDEL) that provides unified view of Web data resources and extracted data. We describe some rule generation features of WICCAP and provide detailed description of the internal language and its implementation. We have conducted experiments to show the ease on generating wrappers with this approach (15 refs.)

Inspec No.: 8205212

131 Ad-hoc invocation of semantic Web services

Author(s): A. Eberhart (**Affiliation:** AIFB, Karlsruhe Univ., Germany)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 116-23

Document type: Conference paper

Abstract: We present the Web service description framework (WSDF), which provides both a representation mechanism and a runtime system architecture for semantically enriched Web Services. We analyze existing languages such as BPEL4WS and OWL-S before addressing their deficiencies in our proposal. Our approach allows a client to invoke a service based solely on a shared ontology, i.e. without prior knowledge on the API, providing an important building block towards a global, flexible information infrastructure. Another main point is that WSDF can be applied to clients and services written in a conventional object oriented programming language. This is achieved by lifting data structures to an ontology level in which rich logical statements about services can be formalized. We also present a detailed system architecture that covers planning, invocation, and the automatic processing of the service results, which is

accomplished using the observer design pattern and by asserting the result in the respective model. Furthermore, the required annotations can be specified conveniently by placing comments in the source code (13 refs.)

Inspec No.: 8205406

132 Logic-based Web services composition: from service description to process model

Author(s): Jinghai Rao, Peep Kungas (**Affiliation:** Dept. of Comput. & Inf. Sci., Norwegian Univ. Sci. & Technol., Trondheim, Norway), Mihail Matskin

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 446-53

Document type: Conference paper

Abstract: This paper introduces a method for automatic composition of semantic Web services using Linear Logic (LL) theorem proving. The method uses semantic Web service language (DAML-S) for external presentation of Web services, while, internally, the services are presented by extralogical axioms and proofs in LL. We use a process calculus to present the composite service formally. The process calculus is attached to the LL inference rules in the style of type theory. Thus the process model for a composite service can be generated directly from the proof. The subtyping rules that are used for semantic reasoning are presented with LL inference figures. We propose a system architecture where the DAML-S translator, the LL theorem prover and the semantic reasoner can operate together to fulfill the task. This architecture has been implemented in Java (20 refs.)

Inspec No.: 8205444

133 Large scale, type-compatible service composition

Author(s): I. Constantinescu, B. Faltings, W. Binder (**Affiliation:** Artificial Intelligence Lab., Swiss Fed. Inst. of Technol., Lausanne, Switzerland)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 506-13

Document type: Conference paper

Abstract: Service matchmaking and composition has recently drawn increasing attention in the research community. Most existing algorithms construct chains of services based on exact matches of input/output types. However, this does not work when the available services only cover a part of the range of the input type. We present an algorithm that also allows partial matches and composes them using switches that decide on the required service at runtime based on the actual data type. We report experiments on randomly generated composition problems that show that using partial matches can decrease the failure rate of the integration algorithm using only complete matches by up to 7 times with no increase in the number of directory accesses required. This

shows that composition with partial matches is an essential and useful element of Web service composition (21 refs.)

Inspec No.: 8205451

134 Combining query and preference - an approach to fully automatize dynamic service binding

Author(s): M. Klein (**Affiliation:** Inst. for Program Structures & Data Organ., Univ. Karlsruhe, Germany), B. Konig-Ries

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 788-91

Document type: Conference paper

Abstract: Web services will only have advantages over existing technologies if the service binding can be performed dynamically. However, existing service description languages do not contain enough information for a computer agent to do the selection automatically during runtime on behalf of the user. This results from the fact that in most approaches the offer description language doubles as a re-guest language, which prevents the requestor from a precise formulation of requests and preferences. Therefore, in this paper, we emphasize the need for a distinguished service request language that allows to capture all of the requestor's preferences. We present a concrete technique to represent such preference-containing requests, which is based on fuzzy object sets (12 refs.)

Inspec No.: 8205490

135 An assistance environment for the classification of knowledge on risk project

Author(s): R. Chalal, R. Ouamane (**Affiliation:** Methodes de Conception de Systemes, Inst. Nat. d'Informatique, Alger, Algeria)

Journal: *WSEAS Trans. Comput. (Greece)*, vol.2, no.2, p.515-20 (April 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2750

Document type: Journal article

Abstract: Project risk management is making the object of several recent studies. However, it does not exist a well mastered and well known lexicon domain for risk management. In other way, the tools of assistance for terminology construction of corpus have known an important development these last years. The assistance tools for the classification of knowledge on the risk are little numerous. It is to palliate these insufficiencies that our work has been realized. It presents a assistance tool for the acquisition and the classification of knowledge on the risk. The environment is a tool, allowing the generation of concepts and relationship, from a corpus of text, in order to lead to an ontology of the domain (16 refs.)

Inspec No.: 8203040

136 Interval set clustering of Web users with rough K-means

Author(s): P. Lingras, C. West (**Affiliation:** Saint Mary's Univ., Halifax, NS, Canada)

Journal: *J. Intell. Inf. Syst., Integr. Artif. Intell. Database Technol. (Netherlands)*, vol.23, no.1, p.5-16 (July 2004)

Publisher: Kluwer Academic Publishers, Netherlands

Language: English **ISSN:** 0925-9902, [Full text](#)

Document type: Journal article

Abstract: Data collection and analysis in Web mining faces certain unique challenges. Due to a variety of reasons inherent in Web browsing and Web logging, the likelihood of bad or incomplete data is higher than conventional applications. The analytical techniques in Web mining need to accommodate such data. Fuzzy and rough sets provide the ability to deal with incomplete and approximate information. Fuzzy set theory has been shown to be useful in three important aspects of Web and data mining, namely clustering, association, and sequential analysis. There is increasing interest in research on clustering based on rough set theory. Clustering is an important part of Web mining that involves finding natural groupings of Web resources or Web users. Researchers have pointed out some important differences between clustering in conventional applications and clustering in Web mining. For example, the clusters and associations in Web mining do not necessarily have crisp boundaries. As a result, researchers have studied the possibility of using fuzzy sets in Web mining clustering applications. Recent attempts have used genetic algorithms based on rough set theory for clustering. However, the genetic algorithms based clustering may not be able to handle the large amount of data typical in a Web mining application. This paper proposes a variation of the K-means clustering algorithm based on properties of rough sets. The proposed algorithm represents clusters as interval or rough sets. The paper also describes the design of an experiment including data collection and the clustering process. The experiment is used to create interval set representations of clusters of Web visitors (23 refs.)

Inspec No.: 8201200

137 Describing and reasoning on Web services using process algebra

Author(s): G. Salaun, L. Bordeaux, M. Schaerf (**Affiliation:** DIS, Univ. di Roma "La Sapienza", Italy)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 43-50

Document type: Conference paper

Abstract: We argue that essential facets of Web services, and especially those useful to understand their interaction, can be described using process-algebraic notations. Web service description and execution languages such as BPEL are essentially process description languages; they are based on primitives for behaviour description and message exchange which can also be found in more abstract process algebras. One legitimate question is therefore whether the formal approach and the sophisticated tools introduced for process algebra can be used to improve the effectiveness and the reliability of Web service development. Our investigations suggest a positive answer, and we claim that process algebras provide a very complete and satisfactory assistance to the whole process of Web service development. We show on a case study that readily available tools based on process algebra are effective at verifying that Web services conform to their requirements and respect properties. We advocate their use both at the design stage and for reverse engineering

issues. More prospectively, we discuss how they can be helpful to tackle choreography issues (26 refs.)

Inspec No.: 8205398

138 Reputation = f(user ranking, compliance, verity)

Author(s): S. Kalepu, S. Krishnaswamy, Seng Wai Loke
(**Affiliation:** Sch. of Comput. Sci. & Software Eng., Monash Univ., Clayton, Vic., Australia)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 200-7

Document type: Conference paper

Abstract: The selection of Web services is typically based on both functional and nonfunctional attributes of the service, such as the quality of service (QoS) levels. Reputation, a widely acknowledged nonfunctional QoS attribute is currently expressed as the average of user ratings given to the service. However, this expression confines reputation to the subjective perception of the end user and is limited by the lack of an objective representation of performance history. In this paper, we address the need for a reputation mechanism that couples the subjective perception of the end user with the objective view of performance history. To represent performance history, we propose a novel QoS metric termed verity. Verity measures the degree of consistency exhibited by the service provider in delivering the quality levels laid out in the service contract, over a range of previous transactions. We express reputation as a composition of user rating, the compliance levels exhibited by the provider and the verity value. We contend that this reputation expression is a more viable attribute of quality than user rating alone (20 refs.)

Inspec No.: 8205416

139 Constraint specification and processing in Web services publication and discovery

Author(s): S. Degwekar, S.Y.W. Su, H. Lam (**Affiliation:** Dept. of Comput. & Inf. Sci. & Eng., Florida Univ., Gainesville, FL, USA)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 210-17

Document type: Conference paper

Abstract: Much effort is being made by the IT industry towards the establishment of a Web services infrastructure and the refinement of its component technologies to enable the sharing of heterogeneous application resources. Traditional roles of the service provider, service requestor and service broker and their interactions are now being improved upon to enable more effective services. The implementation of the Web service broker is currently limited to being an interface to the service repository for service registration, browsing and/or programmatic access. In this work, we have extended the functionality of the Web services broker to include constraint specification and processing, which enables the broker to find a good match between a service provider's capabilities and a service requestor's requirements. This paper presents the

extension made to the Web Services Description Language to include constraint specifications in service descriptions and requests, the architecture of a constraint-based broker, the constraint matching technique, some implementation details, and preliminary evaluation results (19 refs.)

Inspec No.: 8205417

140 Adapting the communication of Web services to the language of their user community

Author(s): K. Englmeier (**Affiliation:** Fachbereich Informatik, Hamburg Univ., Germany), J. Mothe, F. Murtagh, M. Stempinhuber

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 252-9

Document type: Conference paper

Abstract: This paper presents the WS-Talk interface layer, a concept for a structured natural language interface which virtualizes the composition of Web services at the end-user's side. As the number of available Web services grows, the representation of the context in which they are used becomes more and more an issue for both service providers and service consumers. While providers concentrate more on the technical levels of activation and communication within a service network, the users, i.e. the service consumers, want to form ad-hoc collaborations between Web services. To shield users of Web services from the technical details of finding and combining them, a semantic level has to be found that suits their specific needs and expertise. Through a semantic layer, WS Talk tries to create a level of abstraction that enables views on services expressed in natural language. To reduce the complexity of natural language, domain-specific languages are defined and their semantics are mapped onto Web services and their attributes. The domain of economics is currently used for testing architecture and usage aspects of the WS-Talk interface layer (16 refs.)

Inspec No.: 8205422

141 Discovery of Web services in a federated registry environment

Author(s): Kaarthik Sivashanmugam, Kunal Verma, Amit Sheth (**Affiliation:** Dept. of Comput. Sci., Georgia Univ., Athens, GA, USA)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 270-8

Document type: Conference paper

Abstract: The potential of a large scale growth of private and semi-private registries is creating the need for an infrastructure which can support discovery and publication over a group of autonomous registries. Recent versions of UDDI have made changes to accommodate interactions between distributed registries. In this paper, we discuss METEOR-S Web service Discovery Infrastructure, which provides an ontology-based infrastructure to access a group of registries that are divided based on business domains and

grouped into federations. We also discuss how Web service discovery is carried out within a federation (20 refs.)

Inspec No.: 8205424

142 Web services for information extraction from the Web

Author(s): B. Habegger (**Affiliation:** Lab. d'Informatique de Nantes Atlantique, Nantes Univ., France), M. Quafafou

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 279-86

Document type: Conference paper

Abstract: Extracting information from the Web is a complex task with different components which can either be generic or specific to the task, going from downloading a given page, following links, querying a Web-based applications via an HTML form and the HTTP protocol, querying a Web service via the SOAP protocol, etc. Therefore building Web services which proceed to executing an information tasks can not be simply hard coded (i.e. written and compiled once and for all in a given programming language). In order to be able to build flexible information extraction Web Services we need to be able to compose different sub tasks together. We propose a, XML-based language to describe information extraction Web services as the compositions of existing Web services and specific functions. The usefulness the proposed framework is demonstrated by three real world applications. (1) Search engines: we show how to describe a task which queries Google's Web service, retrieves more information on the results by querying their respective HTTP servers, and filters them according to this information. (2) E-commerce sites : an information extraction Web service giving access to an existing HTML-based e-commerce online application such as Amazon is built. (3) Patent extraction: a last example shows how to describe an information extraction Web service which allows to query a Web-based application, extract the set of result links, follow them, and extract the needed information on the result pages. In all three applications the generated description can be easily modified and completed to further respond the user's needs and create value-added Web services (8 refs.)

Inspec No.: 8205425

143 DAML-QoS ontology for Web services

Author(s): Chen Zhou, Liang-Tien Chia, Bu-Sung Lee (**Affiliation:** Sch. of Comput. Eng., Nanyang Technol. Univ., China)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 472-9

Document type: Conference paper

Abstract: As more and more Web services are deployed, Web service's discovery mechanisms become essential. Similar services can have quite different QoS levels. For service selection and management purpose, it is necessary to explicitly, precisely, and unambiguously specify various

constraints and QoS metrics for Web services descriptions. This paper provides a novel DAML-QoS ontology as a complement for DAML-S ontology to provide a better QoS metrics model. Three layers are defined together with clear role descriptions for developments. Cardinality constraints are utilized to describe the QoS property constraints. Basic profile is presented for general Web service's description and the speed startup of ontology definition. Matchmaking algorithm for QoS property constraints is presented and different matching degrees are described. When incorporated with DAML-S, multiple service levels can be described through attaching multiple QoS profiles to one service profile. Well-defined Metrics can be further utilized by measurement organizations to guarantee the promised service level (15 refs.)

Inspec No.: 8205447

144 Provisioning resilient, adaptive Web services-based workflow: a semantic modeling approach

Author(s): C. Patel, K. Supekar, Yugyung Lee (**Affiliation:** Sch. of Comput. & Eng., Missouri Univ., Kansas, TX, USA)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 480-7

Document type: Conference paper

Abstract: Web Services are emerging technologies that enable application-to-application communication and reuse of autonomous services over the Web. Recent efforts, OWLS, model the semantics of Web Services that includes the capabilities of the service, the service interaction protocol, and the actual messages for service exchanges. However, there is a need to automate discovery, selection and execution of OWL-S services. Further, a framework that meets the quality of service (QoS) requirements for ad hoc Internet based services is rarely provided. In this paper, we have proposed a rule-based framework, called SetnWebQ, which manages workflows composed of Semantic Web Services. SemWebQ is capable of conducting QoS-based adaptive selection as well as dynamic binding and execution of Web Services according to the semantics of workflow, thereby rendering a resilient and adaptive Web based service flow. A series of experiments performed on the SemWebQ with real Web Services have confirmed the effectiveness of proposed framework with respect to adaptive selection and execution of the Web Services in Web based workflows (19 refs.)

Inspec No.: 8205448

145 Dynamic workflow composition using Markov decision processes

Author(s): P. Doshi (**Affiliation:** Dept. of Comput. Sci., Illinois Univ., Chicago, IL, USA), R. Goodwin, R. Akkiraju, K. Verma

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 576-82

Document type: Conference paper

Abstract: The advent of Web services has made automated workflow composition relevant to Web based applications. One technique, that has received some attention, for automatically composing workflows is AI-based classical planning. However, classical planning suffers from the paradox of first assuming deterministic behavior of Web services, then requiring the additional overhead of execution monitoring to recover from unexpected behavior of services. To address these concerns, we propose using Markov decision processes (MDPs), to model workflow composition. Our method models both, the inherent stochastic nature of Web services, and the dynamic nature of the environment. The resulting workflows are robust to nondeterministic behaviors of Web services and adaptive to a changing environment. Using an example scenario, we demonstrate our method and provide empirical results in its support (12 refs.)

Inspec No.: 8205459

146 A dynamic services discovery framework for traversing Web Services representation chain

Author(s): Liang-Jie Zhang (**Affiliation:** T.J. Watson Res. Center, IBM, Yorktown Heights, NY, USA), Qun Zhou, Tian Chao

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 632-9

Document type: Conference paper

Abstract: In this paper, we propose a dynamic service discovery framework from Web services representation chain. The major components include Web service chains precategorizing module, service exploration engine, services container, chain change detection module with associated control parameters. From the working research prototype, our proposed mechanism enables businesses to easily retrieve up-to-date Web services linked and nested multi-level deep in the service description documents. In addition, a federated Web services discovery portal is illustrated to show how the research results are aggregated from hyperlinked WSIL documents as well as UDDI registries for achieving more accurate services exploration result (13 refs.)

Inspec No.: 8205466

147 Collecting, organizing, and managing non-contextualised data by using MVML to develop a human-computer interface

Author(s): M. Verhaart, J. Jamieson (**Affiliation:** Eastern Inst. of Technol., Hawke's Bay, New Zealand), Kinshuk

Editor(s): M.Masoodian, S.Jones, B.Rogers

Conference: Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings, Rotorua, New Zealand **Conference Date:** 29 June-2 July 2004 **Sponsor(s):** Dept. of Comput. Sci., Univ. of Waikato; Univ. of Waikato Usability Lab.; IFIP Tech. Committee No. 13 on Human-Comput. Interaction (IFIP TC.13); AMC SIGCHI, New Zealand Chapter (SIGCHI-NZ); COMPUTERWORLD; Human Interface Technol. Lab. New Zealand (HITLabNZ); Orbit Corporate Travel **Publication:** Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3101) **Publisher:** Springer-Verlag,

Germany, 2004 **Language:** English **ISBN:** 3 540 22312 6 **Page:** 511-20

Document type: Conference paper

Abstract: One aspect of information technology increasingly being researched is organizing and managing the huge repository of data and information available. This is particularly relevant in the context of the explosion in Internet use. As knowledge is gathered from the Internet the ability to retain and recall the knowledge is becoming more and more difficult and complex. Computer based techniques are often used to store and manage this data. Managing the data typically takes two forms; first cross-referencing to the original source as in a bibliography or reference list, or secondly collected by an individual, such as purchasing a book, creating a hard copy of a Web page, and so forth. With the Internet, Web based tools and techniques are frequently employed to manage this information. This may be by maintaining a list of links as in a portal to actual Web content, by using the available Web search engines, or saving the content into a personal electronic knowledge space. This paper looks at the ways this knowledge could be collected and the smallest unit required to organize small pieces of data. The ability to map this into an electronic medium is then explored, and a prototype metadata schema is then discussed. Finally, the human-computer interface is discussed that could enable the data to be organized in a variety of ways based on the metadata scheme (13 refs.)

Inspec No.: 8205715

148 Word proximity constraints: information retrieval meets temporal reasoning

Author(s): M. Koubarakis (**Affiliation:** Dept. of Electron. & Comput. Eng., Crete Tech. Univ., Greece)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004 **Sponsor(s):** IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 5-6

Document type: Conference paper

Abstract: We study the data models WP and AWP that have been widely used for many years in the area of information retrieval. WP and AWP can be used to represent and query textual information under the Boolean model using the concepts of attributes with values of type text, and word proximity constraints. Variations of WP and AWP are in use in most deployed digital libraries using the Boolean model, text extenders for relational database systems (e.g., Oracle text) and the search engine Altavista. We present the syntax, semantics and model theory of WP and AWP and analyze the complexity of query satisfiability and entailment. Since word proximity constraints are very similar to temporal constraints, the techniques we use in our analysis are similar to the ones developed in previous work on first-order theories of temporal constraints and temporal constraint databases (16 refs.)

Inspec No.: 8205894

149 The e-salesman system

Author(s): M.P. Ting, V. Seth, J. Gao (**Affiliation:** San Jose State Univ., CA, USA)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 353-7

Document type: Conference paper

Abstract: Current e-commerce Websites employ designs that are elaborate displays of product catalogs. The drawback is the presence of complex navigation and keyword searches to find products. Customers rely solely on their own decision to purchase a product. In addition, it's difficult to distinguish online retailers from one another based on the Web-front; the identity of the Web business and the portrayal of genuineness are lost. The human and personal element seen in traditional shopping is most often not present when shopping online. This paper discusses and proposes a solution to these problems. The e-salesman system (eSS) aims to simulate the human element of traditional shopping for online sales (8 refs.)

Inspec No.: 8206189

150 Informalising formal mathematics: searching the Mizar library with latent semantics

Author(s): P. Cairns (**Affiliation:** UCL Interaction Centre, Univ. Coll. London, UK)

Editor(s): A.Aspteri, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 58-72

Document type: Conference paper

Abstract: Finding required information in a library of mathematics can be problematic, just as in any other library. However, so far, there are no strong search methods based on the semantics of formal mathematics. This paper describes a new approach based on latent semantic indexing (LSI). Using this, the semantics of terms need not be explicitly defined but is indirectly inferred from a body of documents in which the terms occur. The Mizar library is used as it is a substantial resource of formal mathematics. The system described in the paper adapts Mizar articles to produce an appropriate body of documents that can be used by LSI. Preliminary tests suggest that this approach is able to provide a useful mechanism for the search and retrieval of formal mathematics (24 refs.)

Inspec No.: 8206322

151 The role of cooperation in creating a library online tutorial

Author(s): E.M. Diel, T.K. Flett (**Affiliation:** St. Charles Community Coll., St. Peters, MO, USA)

Journal: *Ref. Libr. (USA)*, no.83-84, p.175-82 (2003) **Publisher:** Haworth Press, USA **Language:** English **ISSN:** 0276-3877, [Full text](#)

Document type: Journal article

Abstract: Reference librarians at St. Charles Community College discovered the benefits of cooperation when they

began the process of creating an online library tutorial. In the fall of 1999, librarians realized that their walk-in library sessions were becoming ineffective due to poor attendance by students. The tours didn't fit into on-campus students' busy schedules, and failed to serve distance students at all. Two reference librarians decided to work together to create a Web-based tutorial introducing students to library online resources. It was an informal process that started with information gathering on what tutorials exist, how they were developed and what type of software was used. After the librarians decided on the format and appropriate software, they brought together their creative and technical strengths to design an appealing and functional tutorial. To create a "virtual tour" of the library's physical layout, the librarians also collaborated with the Instructional Support Center, a group of educational technology specialists who are part of SCC's community college consortium. The tutorial was completed in only a year, partly because working in a small library allows for constant contact between the librarians, but also due to the efficiency of using cooperation. This successful collaborative project eventually won the Missouri Community College Association's 2001 Technology Innovation Award (7 refs.)

Inspec No.: 8201826

152 Librarians as researchers and the digital research library

Author(s): R. Wallach (**Affiliation:** Archit. & Fine Arts Libr., Southern California Univ., USA)

Journal: *New Rev. Inf. Libr. Res. (UK)*, vol.9, p.135-43 (2003)

Publisher: Taylor Graham Publishing, UK **Language:** English **ISSN:** 1361-455X, [Full text](#)

Document type: Journal article

Abstract: This paper discusses the changing relationship between academic institutions and their libraries, particularly in areas of knowledge management. It also discusses the role of the librarian/researcher, and posits the possibility that this role and the librarian's relationship to the larger institution may be changing. The paper uses as case study the Public Art in Los Angeles Web site, developed by the author. The project, which is ongoing, started in 1996 out of curiosity was there public art in Los Angeles, one of the world's largest cities, which grew exponentially in the 20th century, and was heavily car-oriented? This on-going project is driven by the interests of an individual librarian, with some support from the institution. Issues raised in the paper are: (1) Who owns the digital and the print information? (2) Who controls and provides access? (3) What is the usage of the collection? (7 refs.)

Inspec No.: 8202629

153 Improving ranking of the PLIR system by local and global approaches

Author(s): E. Ashoori, F. Orumchian, B.N. Araabi (**Affiliation:** Dept. of Electr. & Comput. Eng., Tehran Univ., Iran)

Journal: *WSEAS Trans. Syst. (Greece)*, vol.2, no.3, p.724-33 (July 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2777

Document type: Journal article

Abstract: The PLIR system is an experimental high precision retrieval system based on the theory of the plausible reasoning of Collins and Michalsky. The PLIR system retrieves documents through plausible inferences. These inferences could be considered as sources of evidence of relevance of a document to a query or query term. A series of experiments were conducted to improve the quality of ranking

of the PLIR system. For these experiments the application of Dempster-Shafer theory (DS) of evidence was considered for combining the evidence gathered through plausible inferences. The experiments were conducted with different assumptions and settings using the titles and abstracts of the CACM text collection. Two approaches in combining evidences of plausible inferences were discussed and it was observed that the application of DS theory and weighting inferences based on their overall usefulness in retrieval has improved the quality of ranking of the system (11 refs.)

Inspec No.: 8203015

3.6 Natural Sciences

154 Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119)

Editor(s): A.Aspteri, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004

Sponsor(s): Centrum Informatyki ZETO s.a., Bialystok, Poland

Publisher: Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7

Document type: Conference proceedings

Abstract: The following topics are dealt with: mathematical knowledge management; mathematical statement retrieval; set theory; formal mathematics; mathematical service matching; differential equations; World Wide Web; Internet; mathematical knowledge repository; mathematical knowledge representation; predicate logic; digital library; theorem proving; proof verification; mathematics learning; matrices; proof assistant environments; mathematical formula parsing; and mathematical Web services

Inspec No.: 8206320

155 Mathematical service matching using description logic and OWL

Author(s): O. Caprotti (**Affiliation:** RISC-Linz, Johannes Kepler Univ., Linz, Austria), M. Dewar, D. Turi

Editor(s): A.Aspteri, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004

Sponsor(s): Centrum Informatyki ZETO s.a., Bialystok, Poland

Publication: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 73-87

Document type: Conference paper

Abstract: Web service technology is increasingly being used to develop distributed applications, however the convention is to describe individual services in terms of the interfaces that they expose, rather in terms of the function that they perform. In this paper we describe a mechanism for encoding information about mathematical Web services which is rich enough to allow a potential client to identify automatically all those services which may be capable of performing a particular task. This mechanism makes use of the Web Ontology Language (OWL) and a novel approach to description logic reasoning exploiting enterprise database technologies (24 refs.)

Inspec No.: 8206323

156 Effect of molecular descriptor feature selection in support vector machine classification of pharmacokinetic and toxicological properties of chemical agents

Author(s): Y. Xue (**Affiliation:** Dept. of Comput. Sci., Singapore Nat. Univ., Singapore), Z.R. Li, C.W. Yap, L.Z. Sun, X. Chen, Y.Z. Chen

Journal: *J. Chem. Inf. Comput. Sci. (USA)*, vol.44, no.5, p.1630-8 (Sept.-Oct. 2004) **Publisher:** ACS, USA **Language:** English **ISSN:** 0095-2338, [Full text](#)

Document type: Journal article

Abstract: Statistical-learning methods have been developed for facilitating the prediction of pharmacokinetic and toxicological properties of chemical agents. These methods employ a variety of molecular descriptors to characterize structural and physicochemical properties of molecules. Some of these descriptors are specifically designed for the study of a particular type of properties or agents, and their use for other properties or agents might generate noise and affect the prediction accuracy of a statistical learning system. This work examines to what extent the reduction of this noise can improve the prediction accuracy of a statistical learning system. A feature selection method, recursive feature elimination (RFE), is used to automatically select molecular descriptors for support vector machines (SVM) prediction of P-glycoprotein substrates (P-gp), human intestinal absorption of molecules (HIA), and agents that cause torsades de pointes (TdP), a rare but serious side effect. RFE significantly reduces the number of descriptors for each of these properties thereby increasing the computational speed for their classification. The SVM prediction accuracies of P-gp and HIA are substantially increased and that of TdP remains unchanged by RFE. These prediction accuracies are comparable to those of earlier studies derived from a selective set of descriptors. Our study suggests that molecular feature selection is useful for improving the speed and, in some cases, the accuracy of statistical learning methods for the prediction of pharmacokinetic and toxicological properties of chemical agents (78 refs.)

Inspec No.: 8203421

157 Nonlinear prediction of quantitative structure-activity relationships

Author(s): P. Tino (**Affiliation:** Sch. of Comput. Sci., Birmingham Univ., UK), I.T. Nabney, B.S. Williams, J. Losel, Yi Sun

Journal: *J. Chem. Inf. Comput. Sci. (USA)*, vol.44, no.5, p.1647-53 (Sept.-Oct. 2004) **Publisher:** ACS, USA **Language:** English **ISSN:** 0095-2338, [Full text](#)

Document type: Journal article

Abstract: Predicting the log of the partition coefficient P is a long-standing benchmark problem in quantitative structure-activity relationships (QSAR). In this paper we show that a relatively simple molecular representation (using 14 variables) can be combined with leading edge machine learning algorithms to predict logP on new compounds more accurately than existing benchmark algorithms which use complex molecular representations (23 refs.)

Inspec No.: 8203423

158 A comparative study on feature selection methods for drug discovery

Author(s): Ying Liu (**Affiliation:** Coll. of Comput., Georgia Inst. of Technol., Atlanta, GA, USA)

Journal: *J. Chem. Inf. Comput. Sci. (USA)*, vol.44, no.5, p.1823-8 (Sept.-Oct. 2004) **Publisher:** ACS, USA **Language:** English **ISSN:** 0095-2338, [Full text](#)
Document type: Journal article

Abstract: Feature selection is frequently used as a preprocessing step to machine learning. The removal of irrelevant and redundant information often improves the performance of learning algorithms. This paper is a comparative study of feature selection in drug discovery. The focus is on aggressive dimensionality reduction. Five methods were evaluated, including information gain, mutual information, a χ^2 -test, odds ratio, and GSS coefficient. Two well-known classification algorithms, naive Bayesian and support vector machine (SVM), were used to classify the chemical compounds. The results showed that naive Bayesian benefited significantly from the feature selection, while SVM performed better when all features were used. In this experiment, information gain and χ^2 -test were most effective feature selection methods. Using information gain with a naive Bayesian classifier, removal of up to 96% of the features yielded an improved classification accuracy measured by sensitivity. When information gain was used to select the features, SVM was much less sensitive to the reduction of feature space. The feature set size was reduced by 99%, while losing only a few percent in terms of sensitivity (from 58.7% to 52.5%) and specificity (from 98.4% to 97.2%). In contrast to information gain and χ^2 -test, mutual information had relatively poor performance due to its bias toward favoring rare features and its sensitivity to probability estimation errors (33 refs.)

Inspec No.: 8203437

159 Agent architecture for smart adaptive systems

Author(s): L. Lhotska, O. Stepankova (**Affiliation:** Dept. of Cybern., Czech Tech. Univ., Prague, Czech Republic)
Journal: *Trans. Inst. Meas. Control (UK)*, vol.26, no.3, p.245-60 (2004) **Publisher:** Inst. Meas. Control, UK **Language:** English **ISSN:** 0142-3312, [Full text](#)
Document type: Journal article

Abstract: This paper discusses the possibility of utilizing a multiagent system as a platform for development of a smart adaptive system in medicine. A knowledge-based model of the agents' mutual awareness (social knowledge) is presented. The tri-base acquaintance (3bA) model formalizing the agent's social behaviour knowledge and co-operation neighbourhood knowledge is enhanced to support attention focussing and to ensure a holistic complex perception of the problem. The paper focuses on an analysis of the content of 3bA knowledge bases for applications in medical decision support. An architecture for a multiagent system, ADAM, for medical decision support and monitoring is proposed (18 refs.)

Inspec No.: 8205121

160 On diagrammatic representation of mathematical knowledge

Author(s): Z. Kulpa (**Affiliation:** Inst. of Fundamental Technol. Res., Polish Acad. of Sci., Warsaw, Poland)
Editor(s): A.Asperiti, G.Bancerek, A.Trybulec
Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004
Sponsor(s): Centrum Informatyki ZETO s.a., Bialystok, Poland
Publication: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture

Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 190-204

Document type: Conference paper

Abstract: The more extensive use of diagrammatic representations as a tool for managing complexity and communication problems of mathematical knowledge is advocated in the paper. The specifics of this representation tool are introduced, including the problems with using diagrams in mathematics, issues of proper design of diagrams, specification of main usage types of mathematical diagrams and ways of their implementation. The discussion is illustrated by a number of diagrams, mostly taken from the diagrammatic notation for interval algebra recently developed by the author. These and other issues of diagrammatic representation and reasoning are investigated by the recently emerging discipline of diagrammatics (27 refs.)

Inspec No.: 8206325

161 Efficient ambiguous parsing of mathematical formulae

Author(s): C.S. Coen, S. Zacchiroli (**Affiliation:** Dept. of Comput. Sci., Bologna Univ., Italy)
Editor(s): A.Asperiti, G.Bancerek, A.Trybulec
Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004
Sponsor(s): Centrum Informatyki ZETO s.a., Bialystok, Poland
Publication: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 347-62

Document type: Conference paper

Abstract: Mathematical notation has the characteristic of being ambiguous: operators can be overloaded and information that can be deduced is often omitted. Mathematicians are used to this ambiguity and can easily disambiguate a formula making use of the context and of their ability to find the right interpretation. Software applications that have to deal with formulae usually avoid these issues by fixing an unambiguous input notation. This solution is annoying for mathematicians because of the resulting tricky syntaxes and becomes a show stopper to the simultaneous adoption of tools characterized by different input languages. In this paper we present an efficient algorithm suitable for ambiguous parsing of mathematical formulae. The only requirement of the algorithm is the existence of a "validity" predicate over abstract syntax trees of incomplete formulae with placeholders. This requirement can be easily fulfilled in the applicative area of interactive proof assistants, and in several other areas of mathematical knowledge management (13 refs.)

Inspec No.: 8206328

162 Gene prediction using multinomial probit regression with Bayesian gene selection

Author(s): Xiaobo Zhou (**Affiliation:** Dept. of Electr. Eng., Texas A&M Univ., College Station, TX, USA), Xiadong Wang, E.R. Dougherty
Journal: *EURASIP J. Appl. Signal Process. (USA)*, vol.2004, no.1, p.115-24 (1 Jan. 2004) **Publisher:** Hindawi, USA
Language: English **ISSN:** 1110-8657, [Full text](#)
Document type: Journal article

Abstract: This paper presents some maximum number of predictors to be used, a full search of all possible predictor sets is combinatorially prohibitive except for small predictors sets, and even then, may require supercomputing. Hence, suboptimal approaches to finding predictor sets and network topologies are desirable. Also considers Bayesian variable selection for prediction using a multinomial probit regression model with data augmentation to turn the multinomial problem into a sequence of smoothing problems. There are multiple regression equations and strongest genes are selected for all regression equations to constitute a target predictor set or, in the context of a genetic network, the dependency set for the target. The probit regressor is approximated as a linear combination of the genes and a Gibbs sampler is employed to find the strongest genes. Numerical techniques to speed up the computation are discussed. After finding the strongest genes, the target genes are predicted based on the strongest genes, with the coefficient of determination being used to measure predictor accuracy. Using malignant melanoma microarray data, comparison of two predictor models are made, the estimated probit regressors themselves and the optimal full-logic predictor based on the selected strongest genes, and comparison of these to optimal prediction without feature selection (30 refs.)

Inspec No.: 8202441

163 A genetic programming method for the identification of signal peptides and prediction of their cleavage sites

Author(s): D. Lennartsson, P. Nordin (**Affiliation:** Saida Med. AB, Goteborg, Sweden)

Journal: *EURASIP J. Appl. Signal Process. (USA)*, vol.2004, no.1, p.138-45 (1 Jan. 2004) **Publisher:** Hindawi, USA

Language: English **ISSN:** 1110-8657, [Full text](#)

Document type: Journal article

Abstract: A novel approach to signal peptide identification is presented. We use an evolutionary algorithm for automatic evolution of classification programs, so-called programmatic motifs. The variant of evolutionary algorithm used is called genetic programming where a population of solution candidates in the form of full computer programs is evolved, based on training examples consisting of signal peptide sequences. The method is compared with a previous work using artificial neural network (ANN) approaches. Some advantages compared to ANNs are noted. The programmatic motif can perform computational tasks beyond that of feedforward neural networks and has also other advantages such as readability. The best motif evolved was analyzed and shown to detect the h-region of the signal peptide. A powerful parallel computer cluster was used for the experiment (13 refs.)

Inspec No.: 8202444

164 Beyond the Floquet theorem: generalized Floquet formalisms and quasienergy methods for atomic and molecular multiphoton processes in intense laser fields

Author(s): S.-I. Chu (**Affiliation:** Dept. of Chem., Kansas Univ., Lawrence, KS, USA), D.A. Telnov

Journal: *Phys. Rep. (Netherlands)*, vol.390, no.1-2, p.1-131 (Feb. 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0370-1573, [Full text](#)

Document type: Journal article

Abstract: The advancement of high-power and short-pulse laser technology in the past two decades has generated considerable interest in the study of multiphoton and very high-order nonlinear optical processes of atomic and molecular systems in intense and superintense laser fields, leading to the discovery of a host of novel strong-field phenomena which cannot be understood by the conventional perturbation theory. The Floquet theorem and the time-independent Floquet Hamiltonian method are powerful theoretical framework for the study of bound-bound multiphoton transitions driven by periodically time-dependent fields. However, there are a number of significant strong-field processes which cannot be directly treated by the conventional Floquet methods. In this review article, we discuss several recent developments of generalized Floquet theorems, formalisms, and quasienergy methods, beyond the conventional Floquet theorem, for accurate nonperturbative treatment of a broad range of strong-field atomic and molecular processes and phenomena of current interests. Topics covered include (a) artificial intelligence (AI)-most-probable-path approach (MPPA) for effective treatment of ultralarge Floquet matrix problem; (b) nonHermitian Floquet formalisms and complex quasienergy methods for nonperturbative treatment of bound-free and free-free processes such as multiphoton ionization (MPI) and above-threshold ionization (ATI) of atoms and molecules, multiphoton dissociation (MPD) and above-threshold dissociation (ATD) of molecules, chemical bond softening and hardening, charge-resonance enhanced ionization (CREI) of molecular ions, and multiple high-order harmonic generation (HHG), etc.; (c) many-mode Floquet theorem (MMFT) for exact treatment of multiphoton processes in multicolor laser fields with nonperiodic time-dependent Hamiltonian; (d) Floquet-Liouville supermatrix (FLSM) formalism for exact nonperturbative treatment of time-dependent Liouville equation (allowing for relaxations and dephasing mechanisms) and high-order nonlinear optical processes (such as intensity-dependent nonlinear optical susceptibilities and multiphoton resonance fluorescence, etc.); (e) generalized Floquet approaches for the treatment of nonadiabatic and complex geometric phases involving multiphoton transitions; (f) generalized Floquet techniques for the treatment of multiphoton processes in intense laser pulse fields with nonperiodic time-dependent Hamiltonians; (g) Floquet formulations of time-dependent density functional theory (DFT) and time-dependent current DFT for nonperturbative treatment of multiphoton processes of many-electron quantum systems in periodic or polychromatic (quasiperiodic) laser fields. For each generalized Floquet approach, we present also the corresponding development of new computational techniques for facilitating the study of strong-field processes and phenomena. The advancement of these generalized Floquet formalisms and quasienergy methods provides powerful new theoretical frameworks and accurate computational methods for nonperturbative and ab initio treatment of a wide range of interesting and challenging laser-induced chemical and physical processes and insightful exploration of strong-field atomic and molecular physics (273 refs.)

Inspec No.: 8204837

165 Towards intelligent Web services for automating medical service composition

Author(s): Yugyung Lee, C. Patel (**Affiliation:** Sch. of Comput. & Eng., Missouri Univ., Kansas, MO, USA), Soon Ae Chun, J. Geller

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 384-91

Document type: Conference paper

Abstract: The vision of the Semantic Web is to reduce manual discovery and usage of Web resources (documents and services) and to allow software agents to automatically identify these Web resources, integrate them and execute them for achieving the intended goals of the user. Such a composed Web service may be represented as a workflow, called service flow. Current studies of Web services are not sufficient for automatic composition. This paper presents different types of compositional knowledge required for Web service discovery and composition: syntactic, semantic and pragmatic knowledge. As a proof of concept, we have implemented our framework in a cardiovascular domain which requires advanced service discovery and composition across heterogeneous platforms of multiple organizations. Within this framework, we describe (1) How to represent the compositional knowledge, which plays a role in service discovery and composition, in DAML-S; (2) How heterogeneous medical services interoperate in a composed medical service flow. (3) To solve this knowledge level integration, we build on an ontology integration method called SEMIO (Semantic Interoperability) (16 refs.)

Inspec No.: 8205437

166 IRIS: a framework for mediator-based composition of service-oriented software

Author(s): U. Radetzki, A.B. Cremers (**Affiliation:** Dept. of Comput. Sci., Bonn Univ., Germany)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 752-6

Document type: Conference paper

Abstract: Due to a rapidly increasing volume of information and autonomous services, domain experts spend considerable time and effort in correlating and integrating these resources into meaningful business processes. In this paper we present IRIS, a framework that effectively supports users in creating service compositions with the special focus on interoperability. We illustrate our work by means of a scientific business process from the area of life science informatics. For the sake of adaptability and reusability our approach relies on a component-based mediator model. The problem of mediator discovery and composition is addressed by an ontology-based registry approach (6 refs.)

Inspec No.: 8205482

167 SWARM simulation of multi-agent fault mitigation in large-scale, real-time embedded systems

Author(s): D. Messie, J.C. Oh (**Affiliation:** Dept. of Electr. Eng. & Comput. Sci., Syracuse Univ., NY, USA)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004,

Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 71-6

Document type: Conference paper

Abstract: This paper presents a SWARM multi-agent simulation of fault mitigation within BTeV, a large-scale, real-time embedded system. The real-time embedded systems (RTES) group collaborates on designing real-time embedded intelligent software to ensure data integrity and fault tolerance within BTeV, a triggering and data acquisition system for particle-accelerator-based high energy Physics experiments at Fermi National Laboratory. The hardware layout spans 2,500 digital signal processors and approximately 2,500 Linux computers. Adaptive and small in footprint, very lightweight agents were designed to apply both reactive and proactive rules to accomplish fault tolerance within the system. The scale and real-time requirements of this system make it ineffective to design a traditional expert system that relies on centralized processing to determine appropriate rule actions for every possible system state. Instead, a decentralized approach based on Rodney Brooks' subsumption architecture is presented. A SWARM simulation is used to investigate emergent behavior of the multi-layered, distributed approach to fault mitigation within the RTES/BTeV environment (9 refs.)

Inspec No.: 8205732

168 Parallelization of Bayesian network based SNPs pattern analysis and performance characterization on SMP/HT

Author(s): J. Song, E. Li, Wei Hu, S. Ge, Chunrong Lai, Yimin Zhang (**Affiliation:** Corporate Technol. Group, Intel Corp., Santa Clara, CA, USA), Xuegong Zhang, Wenguang Chen, Weimin Zheng

Editor(s): N.-F.Tzeng

Conference: Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004), Newport Beach, CA, USA **Conference Date:** 7-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Parallel Processing; IEEE Comput. Soc. Tech. Committee on Distributed Processing **Publication:** Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004) **Publisher:** IEEE Comput. Soc., USA, 2004 **Language:** English **ISBN:** 0 7695 2152 5 **Page:** 315-22

Document type: Conference paper

Abstract: Single nucleotide polymorphisms (SNPs) is subtle variation in a genomic DNA sequence of individuals of the same species. It plays a key role in the pharmaceutical industry to understand variations in drug treatment responses between individuals at the molecular level. Discovering patterns around SNPs loci is very important for better understanding the possible origin of SNPs in evolution. Bayesian network has been applied to this problem and got promising results. Since Bayesian network based SNPs pattern analysis demonstrates high computational complexity, we parallelized this workload on Intel Xeon SMP systems. SNPs' task level parallelism is exploited. Experiment results show that memory is bottleneck: on 8-way Xeon SMP hyper-threading enabled system, system memory bandwidth is fully saturated and memory load access latency is roughly 50% longer than on single processor system. Another interesting result is that Intel's hyper-threading technology helps improve the multithreaded workload's performance by 1.6X speedup. Workload profiling shows that parallel SNPs' data sharing

nature matches hyper-threading's cache sharing mechanism, and thus greatly reduces cache coherency protocol traffic on shared front side bus. Scalability analysis shows that imbalance and locks are two major factors that may limit the parallel workload speedup on more processor platforms (11 refs.)

Inspec No.: 8206085

169 Formalizing set theory as it is actually used

Author(s): A. Avron (**Affiliation:** Sch. of Comput. Sci., Tel Aviv Univ., Israel)

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 32-43

Document type: Conference paper

Abstract: We present a formalization of the axiomatic set theory ZF which reflects real mathematical practice, and is easy for mechanical manipulation and interactive theorem proving. Unlike the standard first-order formalizations, our version provides a rich class of abstraction terms denoting sets on the one hand, and is based on purely syntactical (rather than semantic) considerations on the other hand (9 refs.)

Inspec No.: 8206321

170 Informalising formal mathematics: searching the Mizar library with latent semantics

Author(s): P. Cairns (**Affiliation:** UCL Interaction Centre, Univ. Coll. London, UK)

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 58-72

Document type: Conference paper

Abstract: Finding required information in a library of mathematics can be problematic, just as in any other library. However, so far, there are no strong search methods based on the semantics of formal mathematics. This paper describes a new approach based on latent semantic indexing (LSI). Using this, the semantics of terms need not be explicitly defined but is indirectly inferred from a body of documents in which the terms occur. The Mizar library is used as it is a substantial resource of formal mathematics. The system described in the paper adapts Mizar articles to produce an appropriate body of documents that can be used by LSI. Preliminary tests suggest that this approach is able to provide a useful mechanism for the search and retrieval of formal mathematics (24 refs.)

Inspec No.: 8206322

171 C-CoRN, the Constructive Coq Repository at Nijmegen

Author(s): L. Cruz-Filipe, H. Geuvers, F. Wiedijk (**Affiliation:** NIII, Radboud Univ., Nijmegen, Netherlands)

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 88-103

Document type: Conference paper

Abstract: We present C-CoRN, the Constructive Coq Repository at Nijmegen. It consists of a mathematical library of constructive algebra and analysis formalized in the theorem prover Coq. We explain the structure and the contents of the library and we discuss the motivation and some (possible) applications of such a library. The development of C-CoRN is part of a larger goal to design a computer system where 'a mathematician can do mathematics', which covers the activities of defining, computing and proving. An important proviso for such a system to be useful and attractive is the availability of a large structured library of mathematical results that people can consult and build on. C-CoRN wants to provide such a library, but it can also be seen as a case study in developing such a library of formalized mathematics and deriving its requirements. As the actual development of a library is very much a technical activity, the work on C-CoRN is tightly bound to the proof assistant Coq (33 refs.)

Inspec No.: 8206324

172 Adaptive access to a proof planner

Author(s): E. Melis, A. Meier (**Affiliation:** German Res. Inst. for Artificial Intelligence, Saarbrücken, Germany), M. Pollet

Editor(s): A.Asperti, G.Bancerek, A.Trybulec

Conference: Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings, Bialowieza, Poland **Conference Date:** 19-21 Sept. 2004 **Sponsor(s):** Centrum Informatyki ZETO s.a., Bialystok, Poland **Publication:** Mathematical Knowledge Management. Third International Conference, MKM 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3119) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 23029 7 **Page:** 251-64

Document type: Conference paper

Abstract: Mathematical tools such as computer algebra systems and interactive and automated theorem provers are complex systems and can perform difficult computations. Typically, such tools are used by a (small) group of particularly trained and skilled users to assist in mathematical problem solving. They can also be used as back-engines for interactive exercises in learning environments. This, however, suggests the adaptation of the choice of functionalities of the tool to the learner. This paper addresses the adaptive usage of the proof planner Multi for the learning environment ActiveMath. The proof planner is a back-engine for interactive proof exercises. We identify different dimensions in which the usage of such a service system can be adapted and investigate the architecture realizing the adaptive access to Multi (13 refs.)

Inspec No.: 8206326

173 Promoting agent-based knowledge discovery in medical intensive care units

Author(s): A. Lourenco, O. Belo (**Affiliation:** Campus de Gualtar, Univ. do Minho, Braga, Portugal)

Journal: *WSEAS Trans. Comput. (Greece)*, vol.2, no.2, p.403-8 (April 2003) **Publisher:** WSEAS, Greece **Language:** English **ISSN:** 1109-2750

Document type: Journal article

Abstract: Decision support has become vital in major scientific and organisational areas. Data analysis and knowledge acquisition are no longer means to an end, they have become the end itself. Knowledge discovery is a priority, constantly demanding for new, better suited efforts. Systems or tools capable of dealing with the steadily growing amount of data presented by information systems and digging through their structure, relations, contexts and ultimately, their contents, are in order. In this paper, an agent-based decision support system based in data warehousing systems contents is described, analysing its actual potential and skills in knowledge discovery as a whole and, in particular, in data preparation for data mining. Different requirements, algorithm dependent and nondependent, are pointed, analysing valid approaches. An algebraic model is used to specify the core of some of the embraced steps, as well as, to demarcate the data flow throughout the knowledge acquisition process. A case study is presented in order to demonstrate the application of the system to real world scenarios (10 refs.)

Inspec No.: 8203035

3.7 Electrical and Electronic Engineering

174 Expert system for hydropower plants real-time monitoring

Author(s): Shi Sai-jun (**Affiliation:** Sch. of Electr. Eng., Wuhan Univ., China), Mao Jing-ming

Journal: *Proc. CSU-EPSCA (China)*, vol.16, no.3, p.9-28 (2004) **Publisher:** Editorial Board of the Proceedings of the CSU-EPSCA, China **Language:** Chinese **ISSN:** 1003-8930

Document type: Journal article

Abstract: A supervisory control and data acquisition expert system for water power stations is discussed in this paper. The system can gather the real time data in the power stations and substations, and make tables and graphs from the data. All reports and curve sketches are in Chinese format and can be printed by parallel printers. The system can record the important information in the accident moment. It achieves high level automation in power stations monitoring (4 refs.)

Inspec No.: 8202988

175 Intelligent gas sensor array testing system

Author(s): Jia Jian, Jian Ping, Gao Xiao-guang, He Xiu-li (**Affiliation:** Inst. of Electron., Chinese Acad. of Sci., Beijing, China)

Journal: *Instrum. Technique Sens. (China)*, no.3, p.20-3 (2004) **Publisher:** Editorial Board of the Instrument Technique and Sensor J, China **Language:** Chinese **ISSN:** 1002-1841

Document type: Journal article

Abstract: This paper describes an intelligent gas sensor array testing system with automated flow-injection apparatus for dynamic testing. Sample injection and data acquisition are realized automatically through PC real time control thereby it

assures the reproducibility and the reliability of the sensor testing (7 refs.)

Inspec No.: 8203677

176 Strengthen the protection relay and urgency control systems to improve the capability of security in the interconnected power network

Author(s): Zhang Bao-hui (**Affiliation:** Sch. of Electr. Eng., Xi'an Jiaotong Univ., China)

Journal: *Proc. CSEE (China)*, vol.24, no.7, p.1-6 (July 2004) **Publisher:** Chinese Soc. Electr. Eng., China **Language:** Chinese **ISSN:** 0258-8013

Document type: Journal article

Abstract: This paper presents a fourfold line of defence that should be set up instead of a threefold for a security and stability control system, and points out the basic principles and the key techniques which is required to further research for developing the fourfold line of defence. These principles and techniques are introduced as follows. The first is to develop a new type of protection relay system with rapidity of action and better transient performance, including the research on the new type of protection relay useful to the transmission line inclusive of electronic apparatus, the new type of protection relay of 750 kV system and the new theories of protection using transient component of the fault. The second is to research into power system transient stability prediction and urgency control systems independent of the system models and parameters based on the information network of synchronous phasor to improve transmission capacity and to prevent synchronous instability, including the study on generator swing trajectory prediction based on the information of synchronous phasor measurement units, the expression of transient instability boundary and the principle of instability prediction, and realization techniques of transient stability real-time control system of the whole power system. Based on information network of wide area, the third is to construct a network protective system, which should coordinate the performance of back-up protection relays and other auto-mechanisms in the whole system network, to maintain the stability of the truss of network and prevent a unorderly trips, including the optimal cooperation of the protection relay of transmission line and auto-reclosure, the research on the protective relaying system with "transmission section", and the network protection based on multi-agent system. The fourth is to construct a self-adaptive network disconnection system to prevent the collapse of interconnection power system and shorten the duration of power cut, including the principle of fast identifying the mode of out-of-step generators, the automatic catching of oscillation center and the self-adaptive decision of network disconnection section. The fifth is to arouse the enthusiasm of participators in the power market for ensuring the security of power system, through studying the principle of taking on economic responsibility of the protection and urgency control system in a power market (12 refs.)

Inspec No.: 8203826

177 Diagnostic model of insulation faults in power equipment based on rough set theory

Author(s): Mo Juan, Wang Xue, Dong Ming, Yan Zhang (**Affiliation:** Sch. of Electr. Eng., Xi'an Jiaotong Univ., China)

Journal: *Proc. CSEE (China)*, vol.24, no.7, p.162-7 (July 2004) **Publisher:** Chinese Soc. Electr. Eng., China **Language:** Chinese **ISSN:** 0258-8013

Document type: Journal article

Abstract: Due to the incompleteness and complexity of fault diagnosis for power transformer, a specific fault diagnostic model with self-improvement method based on the rough-set theory is given in this paper. After the statistic analysis on the collected fault examples of oil-immersed transformer, the results were reduced according to rough set theory and then the diagnostic rules were gotten also. In this paper, how to use this model for fault diagnosis under various conditions was described. Especially lacking key information, the reduced decision table may be gotten by synthetically matching the information came from artificial intelligences, i.e. Euclid distance, artificial neural network and fuzzy mathematics, and then the diagnosis might be completed by the reduced decision table and the accordingly rule set. Meanwhile the effectiveness of this model can be enhanced by the self-improvement method, which is achieved by modifying the decision table through richening training sample. The case studies show it is effective and useful (14 refs.)

Inspec No.: 8203851

178 An e-learning system based on the top-down method and the cellular models

Author(s): N. Fujii, S. Yukita, N. Koike (**Affiliation:** Hosei Univ., Tokyo, Japan), T.L. Kunii

Journal: *Int. J. Distance Educ. Technol. (USA)*, vol.2, no.3, p.77-93 (Oct.-Dec. 2004) **Publisher:** Idea Group Publishing, USA **Language:** English **ISSN:** 1539-3100

Document type: Journal article

Abstract: As the broadband connectivity to the Internet becomes common, Web based e-learning and distance learning have come to play the central roles for self-learning, where learners are given much flexibility in choosing place and time to study. However, the learners still have to spend a lot of time before reaching the learning goal. This discourages the learners from continuing their studies and diminishes their motivation. To overcome this problem and to let the learners keep focusing on their primary interests, we propose a top-down e-learning system called TDeLS. The TDeLS can offer learners the learning materials based on the top-down (i.e., goal-oriented) method, according to the learners' demands and purposes. Moreover, the TDeLS can distribute them to the learners through the Internet, and manage the database for learning materials. In order to share learning materials among learners through the Web, these learning materials are wrapped in XML with a specially designed vocabulary for TDeLS. We employed the cellular models that ensure the consistency among design modules and support a top-down design methodology. In this paper, we present the TDeLS for hardware logic design courses based on the cellular models. The primary goal is to design complex logic circuits in Verilog HDL, which is an industrial-standard hardware description language. This paper also presents the basic XML vocabulary designed to describe hardware modules efficiently, and a brief introduction to the structure and functions of the proposed system that implements the TDeLS (15 refs.)

Inspec No.: 8204454

179 Simulation of the electric power market considering load fluctuation and form of transaction

Author(s): T. Shinmichi, T. Oyama (**Affiliation:** Yokohama Nat. Univ., Japan)

Journal: *Trans. Inst. Electr. Eng. Jpn. B (Japan)*, vol.124-B, no.5, p.683-9 (May 2004) **Publisher:** Inst. Electr. Eng. Japan, Japan **Language:** English **ISSN:** 0385-4213, [Full text](#)

Document type: Journal article

Abstract: As the deregulation in the electric industry progress, the electric power market is becoming more competitive. In this study, the simulation of the electric power market using the multi agent system was performed. Modeling and the simulation of pool market were performed first, additionally in the case of monopoly. The effect of the bilateral contract on the power market is also considered. The influence of difference of power market on bid behavior is studied (5 refs.)

Inspec No.: 8205115

180 Reliability and power quality evaluation of high-voltage supplied customer - estimation of outage cost by fuzzy reasoning

Author(s): J. Yoshino, H. Kita, E. Tanaka, J. Hasegawa (**Affiliation:** Div. of Syst. & Inf. Eng., Hokkaido Univ., Japan), H. Kubo, S. Yonaga

Journal: *Trans. Inst. Electr. Eng. Jpn. B (Japan)*, vol.124-B, no.5, p.723-32 (May 2004) **Publisher:** Inst. Electr. Eng. Japan, Japan **Language:** Japanese **ISSN:** 0385-4213, [Full text](#)

Document type: Journal article

Abstract: Recently, a number of electric consumers have concerned about the reliability of electricity to be served. For example, some consumers need the electricity with a higher reliability by the automation of manufacturing processes. On the other hand, some consumers need the electricity of a cheaper price even if the reliability becomes a little worse. Under such circumstances, it is necessary that power suppliers evaluate the needs of every consumer precisely and propose the most desirable measures for meeting their requirements. This paper develops a tool to analyze the reliability for high-voltage supplied consumers quantitatively. Further, this paper presents a method for evaluating the outage cost of consumers to help them choose the most appropriate measures for maintaining the reliability. The proposed method applies the fuzzy reasoning approach. The validity of the proposed method is ascertained through some numerical simulations (8 refs.)

Inspec No.: 8205118

181 Routine high-return human-competitive evolvable hardware

Author(s): J.R. Koza (**Affiliation:** Stanford Univ., CA, USA), M. A. Keane, M.J. Streeter

Conference: 2004 NASA/DoD Conference on Evolvable Hardware, Seattle, WA, USA **Conference Date:** 24-26 June 2004 **Sponsor(s):** Nat. Aeronautics and Space Administration (NASA); Dept. of Defense (DoD) **Publication:** 2004 NASA/DoD Conference on Evolvable Hardware **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2145 2 **Page:** 3-17

Document type: Conference paper

Abstract: This paper reviews the use of genetic programming as an automated invention machine for the synthesis of both the topology and sizing of analog electrical circuits. The paper focuses on the importance of the developmental representation in this process. The paper makes the point that genetic programming now routinely delivers high-return human-competitive machine intelligence. It also makes the point that genetic programming has delivered a progression of qualitatively more substantial results in synchrony with five approximately order-of-magnitude increases in the expenditure of computer time. The paper shows six examples where genetic programming has synthesized a circuit that duplicates the functionality or infringes a 21st-century

patented electrical circuit. Finally, the paper discusses how genetic programming can be enhanced in order to potentially enable it to deliver more complex industrial-strength results (31 refs.)

Inspec No.: 8205826

182 Team enabling architectures for very large sensor networks

Author(s): R. Newman, E. Gaura (**Affiliation:** Sch. of Math. & Inf. Sci., Coventry Univ., UK)

Conference: Proceedings. 2004 International Conference on Intelligent Mechatronics and Automation, Chengdu, China
Conference Date: 26-31 Aug. 2004 **Publication:** Proceedings. 2004 International Conference on Intelligent Mechatronics and Automation (IEEE Cat. No.04EX952) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8748 1 **Page:** 927-32

Document type: Conference paper

Abstract: This paper considers the problems involved in the design of 'dream applications' of networked intelligent sensors. These applications propose global scale networks of millions of sensors, requiring new paradigms of system design. One requirement of any suitable design methodology is that it supports the complete, multidisciplinary team that is needed for this scale and type of application. The problems preventing effective working of such teams are discussed. It is argued that the adoption of suitable design principles and architectures can provide the well defined functions and interfaces which help a design team to define their several goals and responsibilities and therefore allow more effective development. Some established distributed design principles are put forward as being a suitable starting point, and the outline of an architecture is proposed (7 refs.)

Inspec No.: 8206419

183 Novel admission control algorithm for GPRS/EGPRS based on fuzzy logic

Author(s): D. Todinca (**Affiliation:** Dept. of Comput., Univ. Politehnica Timisoara, Romania), H. Graja, P. Perry, J. Murphy

Conference: Fifth IEE International Conference on 3G Mobile Communication Technologies (3G 2004) The Premier Technical Conference for 3G and Beyond, London, UK
Conference Date: 18-20 Oct. 2004 **Publication:** Fifth IEE International Conference on 3G Mobile Communication Technologies (3G 2004) The Premier Technical Conference for 3G and Beyond **Publisher:** IEE, UK, 2004 **Language:** English **ISBN:** 0 86341 388 9 **Page:** 342-6

Document type: Conference paper

Abstract: As users in a GPRS/EGPRS network have different quality of service (QoS) demands, the network aims to satisfy their demands while maximizing the utilization of the existing resources. A crucial aspect of the resource allocation problem in cellular data networks is the admission control (AC). We propose a new session admission control algorithm that overcomes most of the difficulties encountered by the existing AC algorithms. The novelty of our approach is the use of fuzzy logic (fuzzy inference) for the AC. We demonstrate the efficiency of our AC algorithm by simulations (15 refs.)

Inspec No.: 8206600

3.8 Other Engineering

184 Modeling of real-time monitoring and simulation for plastic injection molding process

Author(s): Xiaoping Liao, Jianxin Lao, Meilian Jiang (**Affiliation:** Sch. of Mech. Eng., Guangxi Univ., Nanning, China)

Conference: Proceedings. 2004 International Conference on Intelligent Mechatronics and Automation, Chengdu, China
Conference Date: 26-31 Aug. 2004 **Publication:** Proceedings. 2004 International Conference on Intelligent Mechatronics and Automation (IEEE Cat. No.04EX952) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8748 1 **Page:** 60-5

Document type: Conference paper

Abstract: This paper proposes a component-based modeling technique of real-time monitoring and simulation system developed for plastic injection molding process, which consists of three functional modules: graphic simulation module, communication interface module and the module of process establishment and trouble-shooting analysis. Using component-based and object-oriented technology, the graphic simulation module is built; The bit structure is adopted to compress data and improve the response speed of the interface communication; The parameters of molding process are initialized by the case-based reasoning technology. Through monitoring and analyzing the molding process parameters, the equipment trouble and molding products quality on the basis of rule constraints, the developed system may find out the trouble-shooting and present the suggestion of revising the molding parameter. This model serves to enhance the progressive introduction of machine intelligence to the whole system, and therefore provides a platform for further research in terms of intelligent control of molding processes (5 refs.)

Inspec No.: 8206348

185 Research on a fault diagnosis expert system based on the fault tree analysis method for pressure vessels

Author(s): Hu Bo (**Affiliation:** Sch. of Mater. Sci. & Eng., WUT, China), Zhang Mingyou

Journal: *J. Wuhan Univ. Technol. Inf. Manag. Eng. (China)*, vol.26, no.2, p.71-4 (April 2004) **Publisher:** Editorial Board J. Wuhan Univ. Technol, China **Language:** Chinese **ISSN:** 1007-144X

Document type: Journal article

Abstract: Research on a fault diagnosis expert system based on the fault tree analysis (FTA) method for pressure vessel is made. FTA is one of the effective methods for analyzing the failure of a complicated system. Information and rules of the FTA method for pressure vessels are adopted to build the knowledge bank. The expert system is made up of an obverse reasoning mechanism, and the fault diagnosis for pressure vessels is realized by the searching and matching through a man-machine interface. Hence, the importance of FTA is stressed in building a large complicated fault diagnosis expert system (9 refs.)

Inspec No.: 8203734

186 Automatic entity extraction and activity detection for aerospace communications

Author(s): A.D. Lawson (**Affiliation:** Res. Associates for Defense Conversion, Rome, NY, USA)

Conference: 2004 IEEE Aerospace Conference Proceedings, Big Sky, MT, USA **Conference Date:** 6-13 March 2004

Sponsor(s): Aerosp. and Electron. Syst. Soc **Publication:** 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No.04TH8720) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8155 6 **Page:** 3098-109 Vol.5

Document type: Conference paper

Abstract: This paper presents a technical approach to training and execution of an information extraction system for aerospace and other applications. The bulk of this project involved developing algorithms for the automation of training for document classification and entity extraction, using medical data and newswire as testing material. A self-organizing clusters approach to unsupervised document classification was developed as was a 'feed and seed' approach to entity extraction training. Semi-automatic approaches to training were developed to create an analyst toolkit, which would facilitate porting the system to new domains with some human intervention. Executing the system is accomplished via the metonymic bracketing algorithm and through databases, regular expressions and rules sets. This project's primary applications goal was to build advanced entity extraction capabilities on top of small-vocabulary speech recognition technology to: 1) improve identification of essential information, such as call signs, headings, tower directives; 2) to provide an automated alerts systems that notifies air traffic of probable errors and corrections in communications; 3) to categorize and route messages based on their content and probable destination; and 4) to automatically complete and clarify partial messages based on long-distance inferencing. These goals are accomplished through a combination of artificial intelligence techniques, beginning with a custom-trained speech recognition system geared towards air traffic under noisy conditions. Using the recognized output, basic information is identified in the communication using easily extended XML-formatted content files, which list information about the essential components of air traffic communication, e.g. plane names, digits, alphabets. These components are compiled into meaningful entities via general context free grammar (CFG) rulesets that define forms of entities like headings, speeds, call signs, etc. The final stage verifies the completeness of the extracted meaningful entity set and automatically: 1) completes garbled or truncated entities; 2) warns participants of errors and corrections; 3) categorizes the event by activity type, e.g. landing directive, acknowledgement, etc.; and 4) isolates tower directives. The resulting information is presented to air traffic with the goal of reducing communication errors, facilitating accurate transfer of information and providing a text record of communications. The extracted output can be further processed to provide additional automated alerts or extended to other flight scenarios (4 refs.)

Inspec No.: 8206470

187 Advanced diagnostic/prognostic reasoning and evidence transformation techniques for improved avionics maintenance

Author(s): C.S. Byington, P.W. Kalgren, B.K. Dunkin, B.P. Donovan (**Affiliation:** Impact Technol., LLC, State College, PA, USA)

Conference: 2004 IEEE Aerospace Conference Proceedings, Big Sky, MT, USA **Conference Date:** 6-13 March 2004 **Sponsor(s):** Aerosp. and Electron. Syst. Soc **Publication:** 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No.04TH8720) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8155 6 **Page:** 3424-34 Vol.5

Document type: Conference paper

Abstract: Techniques used in the design and implementation of modern avionics suggest an opportunity to re-examine the maintenance and repair process for current and future systems. The authors have developed a framework with automated evidence collection, data representation and storage, and, advanced automated reasoning techniques to implement within an avionics health management system. This paradigm shift approach utilizes advanced capture and representation of environmental, operational, and component inter-relationship evidence at a systems level to reduce diagnostic ambiguity, guide at-wing testing, and provide prognostics. Modular and reusable software and data elements that combine built-in-test (BIT) with contextual information, component usage models, and evidentiary reasoning techniques are described. A use case scenario is presented that illustrates evidence collection and XML transformation, automated database storage and retrieval, and automated advanced reasoning for diagnostics and prognostics, and finally, an example of at wing ambiguity reduction using an advanced evidence-based reasoner is presented (19 refs.)

Inspec No.: 8206480

188 NASA's swarm missions: the challenge of building autonomous software

Author(s): W. Truszkowski (**Affiliation:** Adv. Archit. & Autom., NASA Goddard Space Flight Center, Greenbelt, MD, USA), M. Hinchey, J. Rash, C. Rouff

Journal: *IT Prof. (USA)*, vol.6, no.5, p.47-52 (Sept.-Oct. 2004)

Publisher: IEEE, USA **Language:** English **ISSN:** 1520-9202,

[Full text](#)

Document type: Journal article

Abstract: The National Aeronautics and Space Administration (NASA) introduces its new millenium mission class. Motivated by the need to gather more data than is possible with a single spacecraft, scientists have developed a new class of missions based on the efficiency and cooperative nature of a hive culture. The missions, aptly dubbed nanoswarm is a little more than mechanized colonies cooperating in their exploration of the solar system. Each swarm mission can have hundreds or even thousands of cooperating intelligent spacecraft that work in teams. One swarm mission under concept development for 2020 to 2030 is the autonomous nano technology swarm (ANTS). For software and systems development, this is uncharted territory that calls for revolutionary techniques

Inspec No.: 8202596

189 Integration of dependability in a task allocation problem

Author(s): F. Bicking (**Affiliation:** Nancy Res. Centre in Autom. Control, France), B. Conrard, J.-M. Thiriet

Journal: *IEEE Trans. Instrum. Meas. (USA)*, vol.53, no.6, p.1455-63 (Dec. 2004) **Publisher:** IEEE, USA **Language:** English **ISSN:** 0018-9456, [Full text](#)

Document type: Journal article

Abstract: The task allocation on various components or processing units composing an automation system needs a design-aided methodology in order to obtain both relevant architecture of this automation system and a sufficiently efficient task allocation. The dependability, mainly illustrated in our approach by the availability of the process and by the reliability that characterizes the risk of incidents or accident occurrences, can be taken into account in the earliest stage of the automation system design. The main interest is that it

allows for the definition of reliable operational and functional architectures. To achieve the task allocation, the search method used must integrate a notion of cost that includes both technological and functional aspects and those linked to the dependability. A genetic approach is used for the task allocation, giving a very satisfactory result in a reasonable computational time (14 refs.)

Inspec No.: 8202859

190 Asteroid exploration with autonomic systems

Author(s): W. Truskowski, J. Rash (**Affiliation:** NASA Goddard Space Flight Center, New York, NY, USA), C. Rouff, M. Hinchey

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 484-9

Document type: Conference paper

Abstract: NASA is studying advanced technologies for a future robotic exploration mission to the asteroid belt. The prospective ANTS (Autonomous Nano Technology Swarm) mission comprises autonomous agents including "worker" agents (small spacecraft) designed to cooperate in asteroid exploration under the overall authority of at least one "ruler" agent (a larger spacecraft) whose goal is to cause science data to be returned to Earth. The ANTS team (ruler plus workers and messenger agents), but not necessarily any individual on the team, exhibit behaviors that qualify it as an autonomic system, where an autonomic system is defined as a system that self-reconfigures, self-optimizes, self-heals, and self-protects. Autonomic system concepts lead naturally to realistic, scalable architectures rich in capabilities and behaviors. In-depth consideration of a major mission like ANTS in terms of autonomic systems brings new insights into alternative definitions of autonomic behavior. This paper gives an overview of the ANTS mission and discusses the autonomic properties of the mission (11 refs.)

Inspec No.: 8205263

191 Some autonomic properties of two legacy multi-agent systems - LOGOS and ACT

Author(s): W. Truskowski, J. Rash (**Affiliation:** NASA Goddard Space Flight Center, New York, NY, USA), C. Rouff, M. Hinchey

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 490-8

Document type: Conference paper

Abstract: To reduce the cost of future space flight missions and to perform new science, NASA has been investigating autonomous ground and space flight systems. These goals of cost reduction have been further complicated by NASA's plans to use constellations and swarms of nanosatellites for future

science data-gathering which may entail large communications delays and loss of contact with ground control for extended periods of time. This paper describes two prototype agent-based systems, the Lights-out Ground Operations System (LOGOS) and the Agent Concept Testbed (ACT), and their autonomic properties that were developed at NASA Goddard Space Flight Center (GSFC) to demonstrate autonomous operations of future space flight missions. The paper discusses the architecture of the two agent-based systems, operational scenarios of both, and the two systems' autonomic properties (9 refs.)

Inspec No.: 8205264

192 Quantitative analysis of human behavior and implied user interface in 3D sketching

Author(s): S. Tano, T. Matsumoto, M. Iwata (**Affiliation:** Electro-Commun. Univ., Tokyo, Japan)

Editor(s): M.Masoodian, S.Jones, B.Rogers

Conference: Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings, Rotorua, New Zealand **Conference Date:** 29 June-2 July 2004 **Sponsor(s):** Dept. of Comput. Sci., Univ. of Waikato; Univ. of Waikato Usability Lab.; IFIP Tech. Committee No. 13 on Human-Comput. Interaction (IFIP TC.13); AMC SIGCHI, New Zealand Chapter (SIGCHI-NZ); COMPUTERWORLD; Human Interface Technol. Lab. New Zealand (HITLabNZ); Orbit Corporate Travel **Publication:** Computer Human Interaction. 6th Asia Pacific Conference, APCHI 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3101) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 22312 6 **Page:** 481-90

Document type: Conference paper

Abstract: Designers normally create three-dimensional images on paper. As a natural extension of this, we are developing a design support system that enables people to draw 3D lines in 3D space directly. However, we first need to better understand how people behave when sketching in three-dimensional space and then design a user interface that supports natural 3D sketching. This paper describes the experiment we conducted to quantitatively analyze the diverse aspects of this behavior. The experiment clarified the characteristics of 3D drawing, particularly the effect of the drawing location, the perception of depth, the sense of balance, the direct/indirect input method, and the preferred drawing direction. We propose improving the user interface by using the metaphor of "shadow" and the metaphor of "hand mirror" to promote the user's awareness of errors and/or by using "adaptive rotation" to improve the user's comfort when sketching (10 refs.)

Inspec No.: 8205714

193 Automated envisioning for spacecraft contingencies

Author(s): G. Alsac (**Affiliation:** United Space Alliance, Houston, TX, USA), M.R. Barry

Conference: 2004 IEEE Aerospace Conference Proceedings, Big Sky, MT, USA **Conference Date:** 6-13 March 2004 **Sponsor(s):** Aerosp. and Electron. Syst. Soc **Publication:** 2004 IEEE Aerospace Conference Proceedings (IEEE Cat. No.04TH8720) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7803 8155 6 **Page:** 2885-91 Vol.5

Document type: Conference paper

Abstract: Space shuttle and space station flight controllers frequently perform a kind of reasoning we call envisioning. They imagine scenarios in which certain faults have occurred

to see how it affects the mission. This allows them not only to respond more quickly should the imagined contingency arise but also to configure the spacecraft in advance to lessen the impact of the contingency should it occur. We present here the advisor system that automates envisioning. Given a planner written for any domain, it graphically displays what mission-critical contingencies it has found so far, what contingencies it is currently processing, and how much longer it will be until it has completely imagined all possible contingencies - i.e., finished envisioning. We also present our results with using it to envision in the Space Shuttle RCS domain (8 refs.)

Inspec No.: 8206465

3.9 Applications to Software

194 Desing of reasoning network and nonstationary model system for detecting network traffic anomalies

Author(s): Ying Ming, Li Jian-hua, Tie Ling (**Affiliation:** Dept. of Electron. Eng., Shanghai Jiaotong Univ., China)

Journal: *J. Shanghai Jiaotong Univ. (China)*, vol.38, no.5, p.791-3 (May 2004) **Publisher:** Shanghai Jiaotong University Press, China **Language:** Chinese **ISSN:** 1006-2467

Document type: Journal article

Abstract: In order to solve the problem of high rate of false positive that lies in the network traffic anomaly detection method of network intrusion detection, this paper presented a system of reasoning network and nonstationary model. The system adopts the idea of evaluating the degree of anomaly based on the pre-possibility of the occurrence of new events and the trend. It builds nonstationary models with the approach of machine learning, which leads to the set of basic anomaly events. From that, it goes further to mine the information of the date by reasoning network. Finally, the design of the prototype of the synthesis system was created (5 refs.)

Inspec No.: 8203788

195 Advanced Bayesian optimization algorithms applied in decomposition problems

Author(s): J. Schwarz (**Affiliation:** Dept. of Comput. Syst., Brno Univ. of Technol., Czech Republic), J. Ocenasek, J. Jaros

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 102-11

Document type: Conference paper

Abstract: We deal with the usage of Bayesian optimization algorithm (BOA) and its advanced variants for solving complex NP-complete combinatorial optimization problems. We focus on the hypergraph-partitioning problem and multiprocessor scheduling problem, which belong to the class of frequently solved decomposition tasks. One of the goals is to use these problems to experimentally compare the performance of the recently proposed Mixed Bayesian optimization algorithm (MBOA) with the performance of several other evolutionary algorithms based on the estimation and sampling of probabilistic model. We also propose the utilization of prior knowledge about the structure of hypergraphs and task graphs to increase the convergence speed and the quality of

solutions. The performance of knowledge based MBOA (KMBOA) algorithms on the multiprocessor scheduling problem is empirically investigated and confirmed (30 refs.)

Inspec No.: 8205214

196 A new interaction model for agent based simulation

Author(s): F. Balbo (**Affiliation:** Univ. Paris-Dauphine, Paris, Greece)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004, Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 357-63

Document type: Conference paper

Abstract: In this paper, we propose an interaction model which allows more realistic interactions for simulations of human societies and limits the communication cost. Interaction is a central concept when designing a multi-agent system. Classically, interaction between two agents contains two elements: communication and the action which is the result of this information exchange. This view of interaction has underpinned most research in the multi-agent domain. However, if we draw an analogy with human behavior, the notion of interaction is more complex and we show that the classical ways of interaction managing are not adapted. To reach this objective, the environment could be used to mediate interaction between agents. A toy problem shows that our proposition helps the agents to adapt their perception of communications in the light of their interest and limits the communication cost in case of complex interactions between agents. A real application stemming from the transportation domain illustrates the use of our proposition to help agents to adapt their behavior to the context (12 refs.)

Inspec No.: 8205745

197 Mapping calendar expressions into periodical granularities

Author(s): C. Bettini, S. Mascetti (**Affiliation:** DI Co, Milan Univ., Italy), X. Sean Wang

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004 **Sponsor(s):** IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 96-102

Document type: Conference paper

Abstract: An effort has been devoted in the recent years to study and formalize the concept of time granularity and to design applications and services using the formalization. Among other proposals, a calendar algebra has been defined to facilitate the specification of new granularities and to perform conversions among them. This paper shows how granularities defined as algebraic calendar expressions can be represented as periodical sets of instants. More precisely the

paper shows how each algebraic operator changes the periodical structure of the granularities given as operands. These results have an immediate application enabling users to easily specify new granularities and using them in the only constraint solver supporting time granularities that is currently available (8 refs.)

Inspec No.: 8205908

198 Using temporal logics of knowledge in the formal verification of security protocols

Author(s): C. Dixon, M.-C.F. Gago, M. Fisher, W. van der Hoek (**Affiliation:** Dept. of Comput. Sci., Liverpool Univ., UK)

Editor(s): C. Combi, G. Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatihou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 148-51

Document type: Conference paper

Abstract: Temporal logics of knowledge are useful for reasoning about situations where the knowledge of an agent or component is important, and where change in this knowledge may occur over time. Here we use temporal logics of knowledge to reason about security protocols. We show how to specify part of the Needham-Schroeder protocol using temporal logics of knowledge and prove various properties using a clausal resolution calculus for this logic (14 refs.)

Inspec No.: 8205915

199 A Markov-decision-based price comparison problem for mobile agent-based Internet commerce system (MAGICS)

Author(s): P.P.Y. Lam, H.C.B. Chan (**Affiliation:** Dept. of Comput., Hong Kong Polytech. Univ., China)

Editor(s): M. Bichler, J.-Y. Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 34-41

Document type: Conference paper

Abstract: Mobile agents are well suited for facilitating the consumer buying process on the Internet, particularly for comparing prices. In this paper, we present a mobile agent-based e-commerce system, called MAGICS, that addresses some issues concerning price comparisons. After providing an overview of the MAGICS architecture, we use the Markov decision theory to formulate a price comparison problem. The objective of the problem is to obtain an optimal decision policy in order to buy a product at the lowest expected cost, including the traveling costs. The problem was solved using the backward induction algorithm. Some analytical results are presented and discussed (17 refs.)

Inspec No.: 8206147

200 On composite Web services provisioning in an environment of fixed and mobile computing resources

Author(s): Z. Maamar (**Affiliation:** Coll. of Inf. Syst., Zayed Univ., Dubai, United Arab Emirates), Q.Z. Sheng, B. Benatallah

Journal: *Inf. Technol. Manag. (Netherlands)*, vol.5, no.3-4, p.251-70 (July-Oct. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1385-951X, [Full text](#)

Document type: Journal article

Abstract: We present a framework for Web services provisioning in a hybrid environment of fixed and mobile computing resources. Several obstacles still hinder the seamless provisioning of Web services in mobile environments. Examples of such obstacles are: throughput and connectivity of wireless networks, limited computing resources of mobile devices, and risks of communication channel disconnections. In the proposed framework, software agents represent users, providers of services, and providers of resources. The business logic of composite services is expressed as a process model using statecharts formalism. Among other things, the use of agents provides an infrastructure that has the ability to handle disconnections during service preparation for execution. The framework also integrates a service execution planning approach to optimally select computing resources (fixed or mobile) on top of which services executed (29 refs.)

Inspec No.: 8200715

201 Configurable immunity for evolving human-computer systems

Author(s): M. Burgess (**Affiliation:** Oslo Univ. Coll., Norway)

Journal: *Sci. Comput. Program. (Netherlands)*, vol.51, no.3, p.197-213 (June 2004) **Publisher:** Elsevier, Netherlands **Language:** English **ISSN:** 0167-6423, [Full text](#)

Document type: Journal article

Abstract: The immunity model, as used in the GNU cfengine project, is a distributed framework for performing policy conformant system administration, used on hundreds of thousands of Unix-like and Windows systems. This paper describes the idealized approach to policy-guided maintenance, that is approximated by cfengine, building on the notion of 'convergent' operations, i.e. those that reach stable equilibrium. Agents gravitate towards a policy-determined configurations, through the repeated application of unintelligent 'anti-body' operations or discrete, coded counter-measures. The distributed agents turn passive discovery of state into active strategy for 'curing' systems of policy transgressions (28 refs.)

Inspec No.: 8201013

202 Go! A multi-paradigm programming language for implementing multi-threaded agents

Author(s): K.L. Clark (**Affiliation:** Dept. of Comput., Imperial Coll., London, UK), F.G. McCabe

Journal: *Ann. Math. Artif. Intell. (Netherlands)*, vol.41, no.2-4, p.171-206 (Aug. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1012-2443, [Full text](#)

Document type: Journal article

Abstract: Go! is a multi-paradigm programming language that is oriented to the needs of programming secure, production quality, agent based applications. It is multi-threaded, strongly typed and higher order (in the functional programming sense).

It has relation, function and action procedure definitions. Threads execute action procedures, calling functions and querying relations as need be. Threads in different agents communicate and coordinate using asynchronous messages. Threads within the same agent can also use shared dynamic relations acting as Linda-style tuple stores. In this paper we introduce the essential features of Go!. We then illustrate them by programming a simple multi-agent application comprising hybrid reactive/deliberative agents interacting in a simulated ballroom. The dancer agents negotiate to enter into joint commitments to dance a particular dance (e.g., polka) they both desire. When the dance is announced, they dance together. The agents' reactive and deliberative components are concurrently executing threads, which communicate and coordinate using belief, desire and intention memory stores. We believe such a multi-threaded agent architecture represents a powerful and natural style of agent implementation, for which Go! is well suited (39 refs.)

Inspec No.: 8201887

203 Programming rational agents in a modal action logic

Author(s): M. Baldoni, A. Martelli, V. Patti (**Affiliation:** Dipartimento di Informatica, Università degli Studi di Torino, Italy), L. Giordano

Journal: *Ann. Math. Artif. Intell. (Netherlands)*, vol.41, no.2-4, p.207-57 (Aug. 2004) **Publisher:** Kluwer Academic Publishers, Netherlands **Language:** English **ISSN:** 1012-2443, [Full text](#)

Document type: Journal article

Abstract: In this paper we describe a language for reasoning about actions that can be used for modelling and for programming rational agents. We propose a modal approach for reasoning about dynamic domains in a logic programming setting. Agent behavior is specified by means of complex actions, which are defined using modal inclusion axioms. The language is able to handle knowledge producing actions as well as actions, which remove information. The problem of reasoning about complex actions with incomplete knowledge is tackled and the temporal projection and planning problems is addressed; more specifically, a goal directed proof procedure is defined, which allows agents to reason about complex actions and to generate conditional plans. We give a non-monotonic solution for the frame problem by making use of persistency assumptions in the context of an abductive characterization. The language has been used for implementing an adaptive Web-based system (42 refs.)

Inspec No.: 8201888

204 Deriving tabular event-based specifications from goal-oriented requirements models

Author(s): R. De Landtsheer, E. Letier, A. van Lamsweerde (**Affiliation:** Departement d'Ingenierie Informatique, Univ. Catholique de Louvain, Belgium)

Journal: *Requir. Eng. (UK)*, vol.9, no.2, p.104-20 (2004) **Publisher:** Springer-Verlag, UK **Language:** English **ISSN:** 0947-3602, [Full text](#)

Document type: Journal article

Abstract: Goal-oriented methods are increasingly popular for elaborating software requirements. They offer systematic support for incrementally building intentional, structural and operational models of the software and its environment. They also provide various techniques for early analysis, notably, to manage conflicting goals or to anticipate abnormal environment behaviours that prevent goals from being

achieved. On the other hand, tabular event-based methods are well-established for specifying operational requirements for control software. They provide sophisticated techniques and tools for late analysis of software behaviour models through simulation, model checking or table exhaustiveness checks. The paper proposes to take the best out of these two worlds to engineer requirements for control software. It presents a technique for deriving event-based specifications, written in the SCR tabular language, from operational specifications built according to the KAOS goal-oriented method. The technique consists of a series of transformation steps each of which resolves semantic, structural or syntactic differences between the KAOS source language and the SCR target language. Some of these steps need human intervention and illustrate the kind of semantic subtleties that need to be taken into account when integrating multiple formalisms. As a result of our technique SCR specifiers may use upstream goal-based processes a la KAOS for the incremental elaboration, early analysis, organization and documentation of their tables, while KAOS modelers may use downstream tables a la SCR for later analysis of the behaviour models derived from goal specifications (40 refs.)

Inspec No.: 8202057

205 Abduction in well-founded semantics and generalized stable models via tabled dual programs

Author(s): J.J. Alferes, L.M. Pereira (**Affiliation:** Faculdade de Ciencias e Tecnologia, Nova de Lisboa Univ., Lisbon, Portugal), T. Swift

Journal: *Theor. Pract. Log. Program. (UK)*, vol.4, pt.4, p.383-428 (July 2004) **Publisher:** Cambridge University Press, UK **Language:** English **ISSN:** 1471-0684, [Full text](#)

Document type: Journal article

Abstract: Abductive logic programming offers a formalism to declaratively express and solve problems in areas such as diagnosis, planning, belief revision and hypothetical reasoning. Tabled logic programming offers a computational mechanism that provides a level of declarativity superior to that of Prolog, and which has supported successful applications in fields such as parsing, program analysis, and model checking. In this paper we show how to use tabled logic programming to evaluate queries to abductive frameworks with integrity constraints when these frameworks contain both default and explicit negation. The result is the ability to compute abduction over well-founded semantics with explicit negation and answer sets. Our approach consists of a transformation and an evaluation method. The transformation adjoins to each objective literal O in a program, an objective literal $\text{not}(O)$ along with rules that ensure that $\text{not}(O)$ will be true if and only if O is false. We call the resulting program a dual program. The evaluation method, ABDUAL, then operates on the dual program. ABDUAL is sound and complete for evaluating queries to abductive frameworks whose entailment method is based on either the well-founded semantics with explicit negation, or on answer sets. Further, ABDUAL is asymptotically as efficient as any known method for either class of problems. In addition, when abduction is not desired, ABDUAL operating on a dual program provides a tabling method for evaluating queries to ground extended programs whose complexity and termination properties are similar to those of the best tabling methods for the well-founded semantics. A publicly available meta-interpreter has been developed for ABDUAL using the XSB system (34 refs.)

Inspec No.: 8202395

206 Programming with assertions: a prospectus [software development]

Author(s): B.A. Malloy (**Affiliation:** Dept. of Comput. Sci., Clemson Univ., USA), J.M. Voas

Journal: *IT Prof. (USA)*, vol.6, no.5, p.53-9 (Sept.-Oct. 2004)

Publisher: IEEE, USA **Language:** English **ISSN:** 1520-9202, [Full text](#)

Document type: Journal article

Abstract: To address the quality problem for software, developers need a technology that specializes in producing robust software. One widely used process that supports the construction of quality software is testing, which executes the program with input data or test cases, and then compares the output data to expected results. However, the transfer of techniques from research to practice has been slow because many techniques do not scale to real programs. An alternative to testing that has increasing popularity and usage entails the use of assertions to monitor the data attributes of functions or classes. Assertions are formal constraints on the behavior of a software application. Assertions can help improve software quality. To use them effectively, it's important to determine when they are valid

Inspec No.: 8202597

207 Rough set spatial data modeling for data mining

Author(s): T. Beaubouef (**Affiliation:** Comput. Sci. Dept., Southeastern Louisiana Univ., Hammond, LA, USA), R. Ladner, F. Petry

Journal: *Int. J. Intell. Syst. (USA)*, vol.19, no.7, p.567-84 (July 2004) **Publisher:** Wiley, USA **Language:** English **ISSN:** 0884-8173, [Full text](#)

Document type: Journal article

Abstract: Uncertainty management is necessary for real world applications, especially those used with data mining. The region connection calculus (RCC) and egg-yolk methods have proven useful for the representation of vague regions in spatial data. Rough set theory has been shown to be an effective tool for data mining and for uncertainty management in databases. In this study we use a rough set foundation for expressing topological relationships previously defined for the RCC and egg-yolk methods and show that rough sets can improve on the representation of topological relationships and concepts defined with the other models, which leads to improved mining of spatial data. Finally, we provide an extension of spatial association rule generation that will be able to use rough set-modeled spatial data (17 refs.)

Inspec No.: 8203771

208 A methodology for requirements engineering process development

Author(s): L. Jiang (**Affiliation:** Dept. of Electr. & Comput. Eng., Calgary Univ., Alta., Canada), A. Eberlein, B.H. Far

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 263-72

Document type: Conference paper

Abstract: Adoption of the most suitable requirements engineering (RE) process and selection of the most

appropriate RE techniques for a given project is a common challenge faced by industry. This paper presents a methodology for RE process development for a given project. The novelty of this work is four-fold. First, a RE process knowledge base (REPKB) is established and it helps during RE process development. Second, a decision support mechanism is provided during RE process development. Third, this methodology uses three components: process building blocks, standard templates of the RE process and development guidelines, to help process development. Fourth, it explicitly links project characteristics with RE process development so that the most suitable RE process can be developed. The theory behind this methodology is described in the paper. Preliminary results suggest that the methodology is of valuable help to requirements engineers during RE process development and process customization (25 refs.)

Inspec No.: 8205234

209 Verification of emergent behaviors in swarm-based systems

Author(s): C. Rouff, A. Vanderbilt, M. Hinchey, W. Truszkowski, J. Rash

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 443-8

Document type: Conference paper

Abstract: The emergent properties of swarms make swarm-based missions powerful, but at the same time more difficult to design and to assure that the proper behaviors emerge. We are currently investigating formal methods and techniques for verification and validation of swarm-based missions. The Autonomous Nano-Technology Swarm (ANTS) mission is being used as an example and case study for swarm-based missions to experiment and test current formal methods with intelligent swarms. Using the ANTS mission, we have evaluated multiple formal methods to determine their effectiveness in modeling and assuring swarm behavior. This paper introduces how intelligent swarm technology is being proposed for NASA missions, and gives the results of a comparison of several formal methods and approaches for specifying intelligent swarm-based systems and their effectiveness for predicting emergent behavior (14 refs.)

Inspec No.: 8205256

210 Towards a model-driven architecture for autonomic systems

Author(s): D. Gracanin, S.A. Bohner, M. Hinchey (**Affiliation:** Dept. of Comput. Sci., Virginia Tech, Falls Church, VA, USA)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 500-5

Document type: Conference paper

Abstract: Agent based systems and architectures provide a firm foundation for design and development of an autonomic system. The key challenge is the selection and efficient use of effective agent architecture. A model-driven approach accommodates the underlying architecture to automate, as much as possible, the development process. The Cognitive Agent Architecture (COUGAAR) is a distributed agent architecture that provides the primary components and an implementation platform for this research. COUGAAR has been developed primarily for very large-scale, distributed applications that are characterized by hierarchical task decompositions and as such is well suited for autonomic systems. We propose a framework for the agent-based, model-driven architecture for autonomic applications development. The framework consists of two main parts, General COUGAAR application model (GCAM) and general domain application model (GDAM). Some COUGAAR related performance issues are also discussed (10 refs.)

Inspec No.: 8205265

211 Applications for mobile agents in peer-to-peer networks

Author(s): D. Lubke, J.M. Gomez (**Affiliation:** Technische Univ. Clausthal, Germany)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 523-9

Document type: Conference paper

Abstract: Peer-to-peer-networking has lately become one of the most popular applications used in the Internet. Today's applications allow users to share files, offer processing power to other entities and do collaboration by using software which is relatively easy to use. This paper studies possible applications of mobile agents within peer-to-peer-networks to provide the user with more and/or better functionality than the networks in use today are offering (17 refs.)

Inspec No.: 8205268

212 A formal approach for interoperability between mobile agent systems and component based architectures

Author(s): U. Pinsdorf (**Affiliation:** Fraunhofer-Inst. fur Comput. Graphics, Darmstadt, Germany)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 536-42

Document type: Conference paper

Abstract: We present an algorithm that allows to cut a component from a component based system and transplant it into a mobile agent system. We refer to this process as grafting. The overall goal is to run grafted components in a new execution environment. The approach is generic as it

brings the target system into a position to execute any instance of the grafted component type. Moreover it applies for any two systems, not only for component based and agent systems. The approach was tested by grafting OSGi bundles into the mobile agent system SEMOA (33 refs.)

Inspec No.: 8205270

213 An implementation of self-protected mobile agents

Author(s): J. Ametller, S. Robles, J.A. Ortega-Ruiz (**Affiliation:** Dept. of Comput. Sci., Univ. Autonoma de Barcelona, Spain)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 544-9

Document type: Conference paper

Abstract: Traditional approaches to mobile agent code protection rely on platform-based cryptographic services, often based on PKI solutions. In this paper, we discuss some of their shortcomings, and propose a new architecture for secure mobile agents that addresses two outstanding issues: inter-platform portability and agent code protection. Existing solutions often introduce heavyweight frameworks, requiring a major reengineering of legacy systems and severely impeding portability and code reuse. We show how preexisting, as well as new, agent systems can be robustly secured against a wide variety of external attacks, while minimising the impact on the code base of both the agent and its host platform(s). The paper closes with a discussion of our implementation of the proposed mechanisms, as an extension to the well-known JADE platform (9 refs.)

Inspec No.: 8205271

214 A service-oriented software architecture for mobile agent toolkits

Author(s): P. Braun, I. Muller, S. Geisenhainer, V. Schau, W. Rossak (**Affiliation:** Comput. Sci. Dept., Friedrich-Schiller-Univ., Jena, Germany)

Editor(s): V.Dvorak, M.Sveda

Conference: Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems, Brno, Czech Republic **Conference Date:** 24-27 May 2004 **Sponsor(s):** IEEE Comput. Soc.; Brno Univ. of Technol **Publication:** Proceedings. 11th IEEE International Conference and Workshop on the Engineering of Computer-Based Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2125 8 **Page:** 550-6

Document type: Conference paper

Abstract: In this paper we propose a new software architecture for mobile agent toolkits. Agent toolkits mainly consist of a software system that forms an agency, which is responsible to host mobile and stationary software agents. In contrast to most architectures developed so far, which already define a large set of high-level services for agent migration and communication, we propose to employ a kernel-based approach. The kernel only provides fundamental concepts and functions and abstracts from any of these high-level services. We propose a new model for agencies and present our Tracy

mobile agent toolkit acting as a reference implementation of our new model (17 refs.)

Inspec No.: 8205272

215 Improved prediction for Web server delay control

Author(s): D. Henriksson (**Affiliation:** Dept. of Autom. Control, Lund Inst. of Technol., Sweden), Ying Lu, T. Abdelzaher

Conference: Proceedings. 16th Euromicro Conference on Real-Time Systems, Catania, Italy **Conference Date:** 30 June-2 July 2004 **Sponsor(s):** Euromicro **Publication:** Proceedings. 16th Euromicro Conference on Real-Time Systems **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2176 2 **Page:** 61-8

Document type: Conference paper

Abstract: Control methods are being used increasingly for uncertainty management and QoS in modern Web server systems. Previous approaches have suggested combined feedforward and feedback control strategies, using queuing theory for feedforward delay prediction. While queuing theory allows one to predict delay as a function of arrival and service rates, the prediction applies only to long-term averages, and is therefore insensitive to sudden load changes. Unfortunately, Internet load is very bursty, leaving room for predictor improvement. The main contribution of this paper is an extension of the combined feedforward/feedback framework in which the queuing model is replaced with a predictor that instead uses instantaneous measurements to predict future delays. The proposed strategy is evaluated in simulation and by experiments on an Apache Web server. It is shown that the approach performs better than the combined queuing model based feedforward and feedback control presented in earlier papers (27 refs.)

Inspec No.: 8205369

216 Designing Web services with Tropos

Author(s): D. Lau, J. Mylopoulos (**Affiliation:** Dept. of Comput. Sci., Toronto Univ., Ont., Canada)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 306-13

Document type: Conference paper

Abstract: We propose a methodology for designing Web services. The methodology is founded on Tropos (Perini et al., 2001 and Castro, et al., 2002), an agent-oriented software development technique, and supports early and late requirements analysis, as well as architectural and detailed design. An online retailer example is used for illustration of the proposed methodology. We also compare the generated design with a sample design presented in [BPT01] (9 refs.)

Inspec No.: 8205428

217 Model-based discovery of Web services

Author(s): J.H. Hausmann, R. Heckel, M. Lohmann (**Affiliation:** Dept. of Comput. Sci., Paderborn Univ., Germany)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc,

USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 324-31

Document type: Conference paper

Abstract: Web services are software components that can be discovered and employed at runtime using the Internet. Conflicting requirements towards the nature of these services can be identified. From a business perspective, Web services promise to enable the formation of ad-hoc cooperations on a global scale. From a technical perspective, a high degree of standardization and rigorous specifications are required to enable the automated integration of Web services. A suitable technology for Web services has to mediate these needs for flexibility and stability. In this paper a new approach to the description of Web service semantics is introduced. It is a visual approach based on the use of software models and graph transformations and allows for the description of innovative services while providing a precise matching concept. An implementation using current standards and tools is available (23 refs.)

Inspec No.: 8205430

218 Behavior descriptions of mobile agents for Web services integration

Author(s): F. Ishikawa (**Affiliation:** Graduate Sch. of Inf. Sci. & Technol., Tokyo Univ., Japan), Y. Tahara, N. Yoshioka, S. Honiden

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 342-9

Document type: Conference paper

Abstract: Web services technology for interoperability, dynamic discovery and integration of distributed components is to be applied in various environments, such as networks of relatively narrow and unstable wireless connections. However, in such environments it is necessary to deal with resource limitation. Our work adopts mobile agent technology in response to this problem and presents a mobile agent framework for Web Services integration. This framework utilizes BPEL (Business Process Execution Language for Web Services) for description of the integration logic, and provides declarative descriptions for physical behaviors (migration and cloning). This facilitates addition or change of physical behaviors according to the environmental conditions while preserving the integration logic. This paper especially focuses on formal definition and discussion of our framework (15 refs.)

Inspec No.: 8205432

219 A DAML-based repository for QoS-aware semantic Web service selection

Author(s): A. Soydan Bilgin, M.P. Singh (**Affiliation:** Dept. of Comput. Sci., North Carolina State Univ., Raleigh, NC, USA)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 368-75

Document type: Conference paper

Abstract: The Web is moving toward a collection of interoperating Web services. Achieving this interoperability requires dynamic discovery of Web services on the basis of their capabilities. The capability of a service can be properly determined by using not only its functional description (or service interface), but also its quality attributes as judged by previous users of the service. We develop a service repository that extends UDDI registries. This repository combines an ontology of attributes with evaluation data. We base our repository on a new query and manipulation language based on DAML. Our language includes support for a rich set of operations, which are needed to maintain an attribute ontology, publish services, rate services, and select services based on their functional attributes as well as evaluations by others. We have implemented our approach and evaluated its practical completeness via a number of key query and manipulation templates (17 refs.)

Inspec No.: 8205435

220 Services ecosystem: towards a resilient infrastructure for on demand services provisioning in grid

Author(s): Ying Li, Fangyan Rao, Ying Chen, Dong Liu, T. Li
(**Affiliation:** IBM China Res. Lab., Beijing, China)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 394-401

Document type: Conference paper

Abstract: The dynamic nature of the Grid environment brings challenges to the services provisioning technique. A major aspect of Grid is to match demands for services with resources. Since resources and demands are fluctuating, Grid services provisioning need to be adjusted accordingly in responding to changes. In this paper, we propose a novel approach, Grid services migration, to enabling services provisioning on demand. The major motivation is to achieve the desired Quality of Service (QoS) such as performance, reliability, and etc. of Grid services on resources assembled dynamically responding to dynamic demand. In this paper, the technical characteristics of Grid services migration and its taxonomy are introduced. The Grid service migration algorithms are also presented. The Grid services migration framework with MAPE (Monitor, Analyze, Plan, and Execute) loop is proposed based on Open Grid Services Architecture (OGSA) and the prototype named Services Ecosystem, a resilient infrastructure for services provisioning on demand in Grid is developed. Demonstrations and experiments are conducted and results are presented to illustrate the capabilities of Grid services migration (22 refs.)

Inspec No.: 8205438

221 RAWs: reflective engineering for Web services

Author(s): J.P. Fuente, S.S. Alonso, O.S. Martinez, L.J. Aguilar (**Affiliation:** Dept. of Comput. Languages & Syst., Salamanca Pontifical Univ., Madrid, Spain)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc,

USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 488-95

Document type: Conference paper

Abstract: Reflection is a powerful tool for the adaptation of applications at runtime. The modification of Web services is a task that entails the modification and compilation of the source code, as well as the deployment of the new version of the Web service in the application server. In this paper, we introduce RAWs (Reflective and Adaptable Web Service), a Web service design model based on a reflective architecture of two levels. RAWs allows both the dynamic modification of the definition and implementation structure of the Web service, and the dynamic modification of the Web service behavior in order to change the existing code or to add new functionalities. All these dynamic modifications are performed directly on the code during execution, with no need to have the Web service source code. RAWs improves Web services adaptability and maintainability, as well as the ability for authorized clients to remotely modify them (30 refs.)

Inspec No.: 8205449

222 Manners externalize semantics for on-demand composition of context-aware services

Author(s): A. Arsanjani (**Affiliation:** IBM Global Services, Maharishi Univ. of Manage., Fairfield, IA, USA), F. Curbera, N. Mukhi

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 583-90

Document type: Conference paper

Abstract: Architectural mismatch; the semantic and syntactic gap preventing component composition, can be partially alleviated through the ubiquity of the protocols for service composition (syntax). And yet, the semantics of service or component composition have remained elusive. The contribution of this paper builds on the notions introduced by grammar-oriented object design (GOOD) using the notion of manners to specify the semantics of services and drive their automated composition through dynamically reconfigure architectures enabled through GOOD. This leads to the use of manners for explicit service semantics representation in a re-configurable architectural style. We describe the base requirements for automated component and services composition and show how automated assembly of components and services can be accomplished through a dynamically reconfigurable (DyRec) architectural style. In addition, we show how this vision of dynamic reconfiguration is naturally supported by the Web services framework, because it derives from some of the core requirements of the service oriented architectures (SOA) style embodied by Web services. Further, we discuss how such a DyRec architectural style can be implemented using grammar-oriented object design (GOOD) and describe a project that led to a tool called the Business Compiler, to accomplish dynamic composition and collaboration (28 refs.)

Inspec No.: 8205460

223 Semantics based verification and synthesis of BPEL4WS abstract processes

Author(s): Ziyang Duan, A. Bernstein, P. Lewis (**Affiliation:** Dept. of Comput. Sci., SUNY, Stony Brook, NY, USA), Shiyong Lu

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 734-7

Document type: Conference paper

Abstract: We introduce a logic model to formally specify the semantics of workflows and their composite tasks described as BPEL4WS abstract processes. Based on the model, we present a set of inference rules to deduce the strongest postcondition and weakest precondition of a workflow and demonstrate that automatic workflow verification is possible due to the restrictions on data manipulation in an abstract process. We then sketch an algorithm that automatically synthesizes a workflow given its specification and a task library (10 refs.)

Inspec No.: 8205478

224 Location-mediated coordination of Web services in ubiquitous computing

Author(s): A. Sashima, N. Izumi, K. Kurumatani (**Affiliation:** Cyber Assist Res. Center, AIST/CREST, Tokyo, Japan)

Conference: Proceedings. IEEE International Conference on Web Services, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Tech. Committee for Services Comput. (TCSC) **Publication:** Proceedings. IEEE International Conference on Web Services **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2167 3 **Page:** 822-3

Document type: Conference paper

Abstract: A fundamental issue of human-centered coordinating services in ubiquitous computing is concerned with dynamic service coordination according to users' intentions. How can we coordinate the services to assist the user in receiving a coordinated service to maximize the user's satisfaction in an environment? In order to solve this issue, we have been developing a sort of agent-based coordination framework, called "location-mediated agent coordination," that orchestrates Web services embedded in the real world and Web services on the Internet. In this paper, we show a prototype application of the framework, context-aware information assist services in a museum (7 refs.)

Inspec No.: 8205501

225 Optimal resource-aware deployment planning for component-based distributed applications

Author(s): T. Kichkaylo, V. Karamcheti (**Affiliation:** New York Univ., NY, USA)

Conference: Proceedings. 13th IEEE International Symposium on High Performance Distributed Computing, Honolulu, HI, USA **Conference Date:** 4-6 June 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Distributed Comput **Publication:** Proceedings. 13th IEEE International Symposium on High Performance Distributed Computing (IEEE Cat. No.04TH8762) **Publisher:** IEEE, USA, 2004 **Language:** English **ISBN:** 0 7695 2175 4 **Page:** 150-9

Document type: Conference paper

Abstract: Component-based approaches are becoming increasingly popular in the areas of adaptive distributed

systems, Web services, and grid computing. In each case, the underlying infrastructure needs to address a deployment problem involving the placement of application components onto computational, data, and network resources across a wide-area environment subject to a variety of qualitative and quantitative constraints. In general, the deployment needs to also introduce auxiliary components (e.g., to compress/decompress data, or invoke GridFTP sessions to make data available at a remote site), and reuse preexisting components and data. To provide the flexibility required in the latter case, recently proposed systems such as Sekitei and Pegasus have proposed solutions that rely upon AI planning-based techniques. Although promising, the inherent complexity of AI planning and the fact that constraints governing component deployment often involve nonlinear and nonreversible functions have prevented such solutions from generating deployments in resource-constrained situations and achieving optimality in terms of overall resource usage or other cost metrics. We address both of these shortcomings in the context of the Sekitei system. Our extension relies upon information supplied by a domain expert, which classifies component behavior into a discrete set of levels. This discretization, often justified in practice, permits the planner to identify cost-optimal plans (whose quality improves with the level definitions) without restricting the form of the constraint functions. We describe the modified Sekitei algorithm, and characterize, using a media stream delivery application, its scaling behavior when generating optimal deployments for various network configurations (18 refs.)

Inspec No.: 8205620

226 From π -calculus specification to simulation of a mobile agent using Jini

Author(s): A. Barbu, F. Mourlin (**Affiliation:** Paris Univ., Creteil, France)

Editor(s): G.Horton

Conference: 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004, Magdeburg, Germany **Conference Date:** 13-16 June 2004 **Sponsor(s):** SCS **Publication:** 18th European Simulation Multiconference: Networked Simulations and Simulated Networks - ESM'2004 **Publisher:** Gruner Druck GmbH, Germany, 2004 **Language:** English **ISBN:** 3 936150 35 4 **Page:** 77-82

Document type: Conference paper

Abstract: This paper presents a formalism related to mobile agents. It describes the higher order π -calculus, an extension of the π -calculus which was introduced by R. Milner and later developed by Sangiorgi. This formalism defines a mathematical framework that can be used to reason about mobile code; it varies greatly in its expressiveness, in the mechanism it provides to specify mobile code based applications and in its practical usefulness for the validation and the verification of such applications. In this paper we show how this formalism can be used to represent the mobility and communication aspects of a mobile code environment called HoPiTool. We developed the HoPiTool, which is a Jini-based tool to implement mobile agents who respect an initial higher order π -calculus specification. We also introduce the structure of the HoPiTool with its most important elements (7 refs.)

Inspec No.: 8205733

227 Reasoning on temporal conceptual schemas with dynamic constraints

Author(s): A. Artale (**Affiliation:** Dept. of Comput. Sci., Free University of Bozen-Bolzano, Bolzano, Italy)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 79-86

Document type: Conference paper

Abstract: This paper formally clarifies the relevant reasoning problems for temporal EER diagrams. We distinguish between the following reasoning services: (a) entity, relationship and schema satisfiability; (b) liveness and global satisfiability for both entities and relationships; (c) subsumption for either entities or relationships; and (d) logical implication between schemas. We then show that reasoning on temporal models is an undecidable problem as soon as the schema language is able to distinguish between temporal and atemporal constructs, and it has the ability to represent dynamic constraints between entities (16 refs.)

Inspec No.: 8205906

228 Pre-processing time constraints for efficiently mining generalized sequential patterns

Author(s): F. Massegia (**Affiliation:** AxlS Res. Team, INRIA, Sophia Antipolis, France), P. Poncelet, M. Teisseire

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 87-95

Document type: Conference paper

Abstract: In this paper we consider the problem of discovering sequential patterns by handling time constraints. While sequential patterns could be seen as temporal relationships between facts embedded in the database, generalized sequential patterns aim at providing the end user with a more flexible handling of the transactions embedded in the database. We propose a new efficient algorithm, called GTC (graph for time constraints) for mining such patterns in very large databases. It is based on the idea that handling time constraints in the earlier stage of the algorithm can be highly beneficial since it minimizes computational costs by preprocessing data sequences. Our test shows that the proposed algorithm performs significantly faster than a state-of-the-art sequence mining algorithm (6 refs.)

Inspec No.: 8205907

229 Effective data integration in the presence of temporal semantic conflicts

Author(s): Hongwei Zhu, S.E. Madnick, M.D. Siegel (**Affiliation:** MIT Sloan Sch. of Manage., Cambridge, MA, USA)

Editor(s): C.Combi, G.Ligozat

Conference: Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004), Tatiou, Normandie, France **Conference Date:** 1-3 July 2004

Sponsor(s): IEEE; IEEE Comput. Soc.; Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen; Universite de Caen; Conseil Regional de Basse Normandie **Publication:** Proceedings. 11th International Symposium on Temporal Representation and Reasoning (TIME 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2155 X **Page:** 109-14

Document type: Conference paper

Abstract: The change in meaning of data over time poses significant challenges for the use of that data. These challenges exist in the use of an individual data source and are further compounded with the integration of multiple sources. In this paper, we identify three types of temporal semantic heterogeneities. We propose a solution based on extensions to the context interchange framework, which has mechanisms for capturing semantics using ontology and temporal context. It also provides a mediation service that automatically resolves semantic conflicts. We show the feasibility of this approach with a prototype that implements a subset of the proposed extensions (21 refs.)

Inspec No.: 8205910

230 Program comprehension in generative programming: a history of grand challenges

Author(s): D. Batory (**Affiliation:** Dept. of Comput. Sci., Texas Univ. at Austin, TX, USA)

Conference: Proceedings. 12th IEEE International Workshop on Program Comprehension, Bari, Italy **Conference Date:** 24-26 June 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Council on Software Eng **Publication:** Proceedings. 12th IEEE International Workshop on Program Comprehension **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2149 5 **Page:** 2-11

Document type: Conference paper

Abstract: The communities of generative programming (GP) and program comprehension (PC) look at similar problems: GP derives a program from a specification, PC derives a specification from a program. A basic difference between the two is GP's use of specific knowledge representations and mental models that are essential for program synthesis. In this paper, the author presents a historical review of the grand challenges, results, and outlook for GP as they pertain to PC (22 refs.)

Inspec No.: 8205918

231 Program comprehension for the purpose of testing

Author(s): H.M. Sneed (**Affiliation:** ANECON GmbH, Vienna, Austria)

Conference: Proceedings. 12th IEEE International Workshop on Program Comprehension, Bari, Italy **Conference Date:** 24-26 June 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Council on Software Eng **Publication:** Proceedings. 12th IEEE International Workshop on Program Comprehension **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2149 5 **Page:** 162-71

Document type: Conference paper

Abstract: In this paper, program comprehension techniques are examined within the context of testing. First, the tasks of a tester are identified, then the information requirements of a tester to fulfill these tasks. Comprehension is viewed as a

knowledge acquisition process. The knowledge needed depends on the level at which one is testing. For system testing, other knowledge is required than for unit and integration testing. In light of the scope of testing, the paper concludes that it is the tester who needs the broadest knowledge about a software system. Having established the information requirements of testing, a set of tools are presented which help to satisfy these requirements and their practical application discussed (21 refs.)

Inspec No.: 8205926

232 Application of multi-criteria analysis for the creation of a risk assessment knowledgebase for biometric systems

Author(s): C.K. Dimitriadis (**Affiliation:** Expertnet SA, Athens, Greece), D. Polemi

Editor(s): D.Zhang, A.K.Jain

Conference: Biometric Authentication. First International Conference, ICBA 2004. Proceedings, Hong Kong, China

Conference Date: 15-17 July 2004 **Sponsor(s):** Croucher Found.; Int. Assoc. of Pattern Recognition; IEEE Hong Kong Section; Hong Kong Polytechnic Univ.; Nat. Natural Sci. Found in China; Springer-Verlag **Publication:** Biometric Authentication. First International Conference, ICBA 2004. Proceedings (Lecture Notes in Comput. Sci. Vol.3072) **Publisher:** Springer-Verlag, Germany, 2004 **Language:** English **ISBN:** 3 540 22146 8 **Page:** 724-30

Document type: Conference paper

Abstract: This paper presents an enhanced version of a knowledgebase of vulnerabilities, risks and countermeasures for biometric systems. The knowledgebase was created by the application of the multi-criteria analysis methodology to the results of desk research, laboratory testing and interviews. The knowledgebase aims to improve risk assessment procedures, by adding the capability of analyzing the risk of the biometric component of an information system. The application of the knowledgebase is demonstrated for clarifying its functions (22 refs.)

Inspec No.: 8205990

233 State-on-demand execution for adaptive component-based mobile agent systems

Author(s): Yuk Chow, Wenzhang Zhu, Cho-Li Wang, F.C.M. Lau (**Affiliation:** Dept. of Comput. Sci. & Inf. Syst., Hong Kong Univ., China)

Editor(s): N.-F.Tzeng

Conference: Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004), Newport Beach, CA, USA **Conference Date:** 7-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Parallel Processing; IEEE Comput. Soc. Tech. Committee on Distributed Processing **Publication:** Proceedings. Tenth International Conference on Parallel and Distributed Systems (ICPADS 2004) **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2152 5 **Page:** 46-53

Document type: Conference paper

Abstract: The introduction of mobile code in the pervasive computing environment provides a good opportunity for research in ways to enhance execution flexibility. We note that current mobile code is too heavyweight and not adaptive enough to be used in pervasive computing where devices are resource-limited and heterogeneity is the norm. In this paper, we propose a new lightweight, component-based mobile agent system that can adapt to diverse devices and features

resource saving as one of its aims. The system supports strong mobility of mobile code, which is a prerequisite for achieving system flexibility and good performance. The system discretize the transmission of code and execution states and relies on a scheme called state-on-demand (SOD) for the execution of the mobile code. We provide performance results to demonstrate the effectiveness of the SOD scheme (13 refs.)

Inspec No.: 8206055

234 Ontology-driven Web services composition platform

Author(s): I. Budak Arpinar, B. Aleman-Meza, R. Zhang, A. Maduko (**Affiliation:** Dept. of Comput. Sci., Georgia Univ., Athens, GA, USA)

Editor(s): M.Bichler, J.-Y.Chung

Conference: IEEE International Conference on E-Commerce Technology, San Diego, CA, USA **Conference Date:** 6-9 July 2004 **Sponsor(s):** IEEE Comput. Soc. Tech. Committee on Elec. Commerce **Publication:** IEEE International Conference on E-Commerce Technology **Publisher:** IEEE Comput. Soc, USA, 2004 **Language:** English **ISBN:** 0 7695 2098 7 **Page:** 146-52

Document type: Conference paper

Abstract: Discovering and assembling individual Web services into more complex yet new and more useful Web processes is an important challenge. In this paper, we present techniques for (semi) automatically composing Web services into Web processes by using their ontological descriptions and relationships to other services. In interface-matching automatic composition technique, the possible compositions are obtained by checking semantic similarities between interfaces of individual services. Then these compositions are ranked and an optimum composition is selected. In Human-Assisted composition the user selects a service from a ranked list at certain stages. We also address automatic compositions in a peer-to-peer network (17 refs.)

Inspec No.: 8206160

235 Analysis of a free roaming agent result-truncation defense scheme

Author(s): Jianying Zhou, J.A. Onieva, J. Lopez (**Affiliation:** Inst. for Infocomm Res., Heng Mui Keng Terrace, Singapore)

Editor(s): M.Bichler, J.-Y.Chung

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Abstract: Mobile agents play an important role in electronic commerce. Security in free-roaming agents is especially hard to achieve when the mobile code is executed in hosts that may behave maliciously. Some schemes have been proposed to protect agent data (or computation results). However, a known vulnerability of these techniques is the truncation attack where two visited hosts (or one revisited host) can collude to discard the partial results collected between their respective visits. Cheng and Wei proposed a scheme in IC1CS'02 to defense against the truncation of computation results of free-roaming agents (Cheng and Wei, 2002). Cheng-Wei scheme is effective against such an attack in most cases. However, we

demonstrate that it still suffers from the truncation attack when a special loop is established on the path of a free-roaming agent. We further propose two amendments to Cheng-Wei scheme to avoid such an attack (6 refs.)

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236 Reasoning about CBV functional programs in Isabelle/HOL

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Editor(s): K.Slind, A.Bunker, G.Gopalakrishnan

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Abstract: We consider the old problem of proving that a computer program meets some specification. By proving, we mean machine checked proof in some formal logic. The programming language we choose to work with is a call by value functional language, essentially the functional core of standard ML (SML). In future work we hope to add exceptions, then references and I/O to the language (20 refs.)

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237 Formalizing Java dynamic loading in HOL

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Abstract: Dynamic class loading is an important feature of the Java virtual machine. It is the underlying mechanism that supports installing software components at runtime. However, it is also complex. Improperly written class loaders could undermine the type safety of the Java virtual machine. Given the importance of security, the current description provided by the Java virtual machine is deficient. It is ambiguous, imprecise and hard to reason about. In this paper, we suggest a model for the Java virtual machine, which includes the main features of dynamic class loading and linking. We formalize the model and prove its soundness in the HOL system. The soundness theorem demonstrates that our model can preserve types indeed. Based on the model, we can analyze the behaviors of loading in the virtual machine (20 refs.)

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